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Data Compression
Lecture Notes

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1: Lecture 1: Introduction to Data Compression

1.1 Learning Objectives

By the end of this lecture, students will be able to:

- Understand the motivation and benefits of data compression
- Differentiate between lossless and lossy compression techniques
- Compute and interpret common compression performance metrics
- Apply the Huffman coding algorithm step by step
- Analyze real-world compression trade-offs

1.2 Introduction and Motivation: Why Compress Data?

Data compression is the process of representing information using fewer bits than its original form. It is a fundamental component of modern computing systems, enabling efficient storage, faster communication, and reduced operational costs.

Everyday applications of compression include:

- Streaming audio and video
- Image storage and sharing
- File archiving and backups
- Network communication and cloud services

Definition

Data Compression is the process of reducing the number of bits required to represent information, either:

- **Losslessly**: allowing exact reconstruction of the original data
- **Lossily**: allowing controlled loss of information to achieve higher compression

1.2.1 Benefits of Data Compression

Data compression provides three key benefits that are critical in modern computing:

1. **Reduce Storage Space**:
   - Allows more data to be stored in the same physical space
   - Enables archival of historical data that would otherwise be discarded
   - Reduces hardware requirements for storage systems
2. **Reduce Communication Time and Bandwidth**:
   - Enables faster file transfers and downloads
   - Makes high-quality streaming (4K/8K video) practical over limited bandwidth
   - Reduces latency in real-time applications like video conferencing and online gaming
   - Allows IoT devices to transmit data efficiently over wireless networks
3. **Save Money**:
   - Reduces cloud hosting costs (storage and egress fees)
   - Lowers communication costs for data transmission
   - Decreases capital expenditure on storage hardware
   - Reduces energy consumption for data centers and network infrastructure

1.3 Lossless vs. Lossy Compression

1.3.1 Lossless Compression

Lossless compression guarantees perfect reconstruction of the original data. It is essential when accuracy and data integrity are critical.

Typical applications:

- Text files and source code
- Executables and databases
- Medical, scientific, and legal data

1.3.2 Lossy Compression

Lossy compression achieves higher compression ratios by discarding information that is less perceptible or less important.

Typical applications:

- Audio (MP3, AAC)
- Images (JPEG)
- Video (H.264, H.265)

1.3.3 Choosing Between Lossless and Lossy

Factor	**Lossless Compression**	**Lossy Compression**
Reconstruction	Exact	Approximate
Data sensitivity	High	Moderate to low
Typical ratios	Low to moderate	High
Quality impact	None	Controlled degradation

Table 1: Lossless vs. Lossy Compression

1.4 Compression Performance Metrics

1.4.1 Size-Based Metrics

$$\text{Compression Ratio (CR)} = \frac{\text{Original Size}}{\text{Compressed Size}}$$
$$\text{Compression Factor} = \frac{\text{Compressed Size}}{\text{Original Size}}$$
$$\text{Space Savings (\%)} = \left(1 - \frac{\text{Compressed Size}}{\text{Original Size}}\right) \times 100\%$$

Interpretation:

- Larger compression ratios indicate better compression
- Smaller compression factors indicate better compression

1.4.2 Rate-Based Metrics

$$\text{Bits per Sample (bps)} = \frac{\text{Compressed Size (bits)}}{\text{Number of samples}}$$
$$\text{Bit-rate (bps)} = \frac{\text{Compressed Size (bits)}}{\text{Time (seconds)}}$$

These metrics are particularly important in audio and video compression systems.

1.5 Worked Example: Audio Compression Metrics

Example

Uncompressed Audio Properties

- Duration: 180 seconds
- Sampling rate: 44.1 kHz
- Bit depth: 16 bits
- Channels: 2 (stereo)

Original Size Calculation

$$\begin{aligned}\text{Total samples} &= 180 \times 44{,}100 \times 2 = 15{,}876{,}000\\ \text{Size (bits)} &= 15{,}876{,}000 \times 16 = 254{,}016{,}000\\ \text{Size (MB)} &= \frac{254{,}016{,}000}{8 \times 1{,}048{,}576} \approx 30.27\end{aligned}$$

Compression Results

Method	Size (MB)	CR	Savings	Bit-rate
FLAC (lossless)	18.16	1.67:1	40%	807 kbps
MP3 @ 320 kbps	6.75	4.49:1	77.7%	320 kbps
AAC @ 256 kbps	5.40	5.61:1	82.2%	256 kbps

1.6 Huffman Coding

Huffman coding is a widely used **lossless compression algorithm** that assigns variable-length binary codes to symbols based on their frequencies. More frequent symbols receive shorter codes.

1.6.1 Step-by-Step Huffman Coding Example

Example

Message: `MISSISSIPPI RIVER` (17 characters including space)

Symbol Frequencies

Symbol	Frequency
I	5
S	4
P	2
R	2
M	1
V	1
E	1
(space)	1

Tree Construction

1. Combine M(1) + V(1) $\rightarrow$ 2
2. Combine E(1) + (space)(1) $\rightarrow$ 2
3. Combine P(2) + R(2) $\rightarrow$ 4
4. Combine 2 + 2 $\rightarrow$ 4
5. Combine 4 + 4 $\rightarrow$ 8
6. Combine I(5) + S(4) $\rightarrow$ 9
7. Combine 8 + 9 $\rightarrow$ 17

One Possible Code Assignment

Symbol	Code	Length
I	00	2
S	01	2
P	100	3
R	101	3
M	1100	4
V	1101	4
E	1110	4
(space)	1111	4

Compressed Size

$$5(2) + 4(2) + 2(3) + 2(3) + 4(1) = 52 \text{ bits}$$

Original Size (ASCII) $= 17 \times 8 = 136$ bits
Compression Ratio $= 136/52 \approx 2.62 : 1$

1.6.2 Key Properties of Huffman Coding

- Produces prefix-free codes
- Enables instantaneous decoding
- Guarantees minimum average code length among prefix codes
- Widely used in practical compression systems

1.7 End of Chapter Questions

Exercise 1.0

Problem 1: Basic Compression Metrics
An uncompressed grayscale image has the following properties:

- Resolution: 1024×1024 pixels
- Bit depth: 8 bits per pixel

After compression, the image size is 320 KB.
Calculate:

(a) Original image size in KB

(b) Compression ratio

(c) Compression factor

(d) Space savings percentage

Exercise 1.1

Problem 2: Audio Bit-rate and Storage
A mono audio recording has the following parameters:

- Duration: 5 minutes
- Sampling rate: 48 kHz
- Bit depth: 16 bits

The file is compressed using a lossy codec to a constant bit-rate of 192 kbps.
Calculate:

(a) Size of the uncompressed audio file in MB

(b) Size of the compressed file in MB

(c) Compression ratio

(d) Bits per sample after compression

Exercise 1.2

Problem 3: Comparing Compression Options

A video clip has an uncompressed data rate of 120 Mbps. Three compression options are available:

Option	Compressed Bit-rate
A	6 Mbps
B	3 Mbps
C	1.5 Mbps

For each option, calculate:

(a) Compression ratio

(b) Data consumed for a 10-minute video (in MB)

Which option would you choose for:

(i) Live video streaming?

(ii) Archival storage?

Briefly justify your answers.

Exercise 1.3

Problem 4: Huffman Coding Construction

Given the following symbol frequencies:

Symbol	Frequency
A	10
B	8
C	6
D	5
E	4
F	3
G	2
H	2

(a) Construct the Huffman tree step by step

(b) Assign a binary code to each symbol

(c) Compute the total number of bits required to encode the message

(d) Calculate the average number of bits per symbol

Exercise 1.4

Problem 5: Fixed-Length vs. Huffman Coding

Using the symbol set from Problem 4:

(a) Determine the minimum fixed-length code required

(b) Compute the total number of bits using fixed-length coding

(c) Compare the result with Huffman coding

(d) Calculate the percentage reduction in total bits achieved by Huffman coding

Exercise 1.5

Problem 6: Text Compression Scenario

A text file contains 50,000 characters and is stored using 8-bit ASCII encoding. After compression using a lossless algorithm, the file size becomes 18 KB. Calculate:

(a) Original file size in KB

(b) Compression ratio

(c) Compression factor

(d) Space savings percentage

Explain why compression ratios for text files vary significantly depending on content.

Exercise 1.6

Problem 7: Practical Design Question

You are designing a compression system for a wearable health-monitoring device that:

- Records sensor data continuously
- Has limited storage capacity
- Requires exact data reconstruction
- Operates on a low-power processor

(a) Should the system use lossless or lossy compression? Explain.

(b) Which performance metrics are most important in this scenario?

(c) Would a variable-length coding scheme be appropriate? Why or why not?

2: Lecture 2: Theory of Compression — Limits and Optimality

2.1 Types of Codes: From Ambiguous to Instantaneous

Definition

Types of Codes

- **Non-singular Code**: Each source symbol maps to a distinct codeword

$$x_i \neq x_j \Rightarrow C(x_i) \neq C(x_j)$$

- **Uniquely Decodable Code**: Every finite sequence of codewords corresponds to exactly one sequence of source symbols

$$C(x_1)C(x_2)\cdots C(x_n) = C(y_1)C(y_2)\cdots C(y_m) \Rightarrow n = m \text{ and } x_i = y_i$$

- **Prefix Code (Instantaneous Code)**: No codeword is a prefix of another codeword

$$\forall i \neq j: \quad C(x_i) \text{ is not a prefix of } C(x_j)$$

Key Relationships:

$$\text{Prefix Codes} \subset \text{Uniquely Decodable Codes} \subset \text{Non-singular Codes}$$

Important

Why Prefix Codes are Special

- **Instantaneous decoding**: Can decode as soon as codeword ends (no lookahead needed)
- **Tree representation**: Always correspond to leaves of a binary tree
- **Kraft inequality**: Always satisfy $\sum 2^{-\ell_i} \leq 1$
- **Practical**: Used in Huffman coding, many real-world compressors

Example

Example: Comparing Different Code Types

For symbols $\{A, B, C, D\}$ with probabilities $\{0.5, 0.25, 0.125, 0.125\}$:

Code Type	A	B	C	D	Property
Non-singular	0	1	00	11	Distinct but ambiguous: "00" = AA or C?
Uniquely decodable	0	01	011	0111	Unique but need lookahead
Prefix code	0	10	110	111	Instant decoding: "0" = A, stop
Optimal prefix	0	10	110	111	Also Huffman optimal

Decoding examples:

- **Prefix code "010110"**: 0→A, 10→B, 110→C = "ABC" (instant)
- **Uniquely decodable "00111"**: Need to scan ahead to determine split
- **Non-singular "00"**: Ambiguous! Could be "AA" or "C"

Key insight: Prefix codes sacrifice some flexibility in codeword lengths (must satisfy Kraft inequality) for the benefit of instantaneous decoding.

2.2 Basic Terminology and Notation

Definition

Alphabet

An *alphabet* $\mathcal{X}$ is a finite set of possible symbols. Examples:

- Binary alphabet: $\mathcal{X} = \{0, 1\}$
- English letters: $\mathcal{X} = \{\texttt{A}, \ldots, \texttt{Z}\}$
- Bytes: $\mathcal{X} = \{0, 1, \ldots, 255\}$

Definition

Symbol

A *symbol* is a single element drawn from an alphabet. For example, the letter `E` is a symbol from the English alphabet.

Definition

Random Variable

A *random variable* X is a function that assigns a symbol or value to each outcome in a sample space:

$$X : \Omega \to \mathcal{X}$$

- Ω: Sample space (e.g., all possible states of a data source)

- $\mathcal{X}$: Set of possible values (alphabet, e.g., $\{0, 1\}$, ASCII characters)
- For each $\omega \in \Omega$, $X(\omega)$ is the value assigned to outcome ω

Example: Binary Source

- $\Omega = \{\text{emits } 0, \text{emits } 1\}$ (or could be more complex underlying physics)
- $\mathcal{X} = \{0, 1\}$
- $X(\text{emits } 0) = 0$, $X(\text{emits } 1) = 1$
- Probabilities: $P(X = 0) = P(\{\omega : X(\omega) = 0\}) = p$, $P(X = 1) = 1 - p$

Why this matters for compression:

- The entropy $H(X)$ depends on the probability distribution induced by X
- For $x \in \mathcal{X}$: $P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$
- $H(X) = -\sum_{x \in \mathcal{X}} P(X = x) \log_2 P(X = x)$

Definition

Source

A *source* is a process that generates a sequence of symbols $(X_1, X_2, X_3, \dots)$ according to some probability law. In this lecture, we assume discrete sources unless stated otherwise.

Definition

Message (or Sequence)

A *message* is a finite sequence of symbols generated by the source:

$$x^n = (x_1, x_2, \dots, x_n)$$

Compression algorithms operate on messages, not on individual symbols.

Definition

Code and Codewords

A *code* assigns a binary string (codeword) to each symbol or message.

- Source symbols $\to$ codewords (e.g., Huffman coding)
- Messages $\to$ bitstreams (e.g., arithmetic coding)

Definition

Block Length

The *block length* n is the number of source symbols grouped together and encoded as a unit. Larger block lengths generally allow better compression but increase delay and complexity.

Definition

Model

A *model* estimates the probabilities of symbols or sequences. Better models lead to better compression by reducing uncertainty.

2.3 Information and Redundancy: The Core Concepts

2.3.1 Information: A Formal Measure of Uncertainty Reduction

In information theory, information is defined rigorously as a **quantitative measure of the reduction in uncertainty** that results from observing the outcome of a random event.

Definition 2.1. *Let X be a random event that occurs with probability $p = \Pr(X)$. The* ***information content*** *(or* self-information*) $I(X)$ provided by the occurrence of X is defined as:*

$$I(X) = \log_b \left(\frac{1}{p}\right) = -\log_b(p)$$

where:

- $b = 2$ *yields* ***bits*** *(binary digits)*
- $b = e$ *yields* ***nats*** *(natural units)*
- $b = 10$ *yields* ***hartleys*** *or* ***dits***

Example

Predictability vs. Information:

- In a city where it rains every day, the statement "It rained today" conveys almost no information because it was expected
- A file that contains only the bit '1' provides very little information
- A coin that always lands heads produces outcomes, but no information

  Key idea: Perfect predictability implies zero information gain.

Example

Daily Weather Forecast — Information Content:

- Sunny in Phoenix (probability 0.9): $I = -\log_2 0.9 \approx 0.15$ bits
- Snow in Phoenix (probability 0.001): $I = -\log_2 0.001 \approx 9.97$ bits
- Rain in Seattle (probability 0.3): $I = -\log_2 0.3 \approx 1.74$ bits

Interpretation: Rare events carry more information because they reduce uncertainty the most.

2.3.2 Redundancy: The Enemy of Information and the Friend of Compression

Redundancy refers to predictable or repeated structure in data. It is what allows data to be represented using fewer bits.

1. **Spatial Redundancy**: Neighboring data values are highly correlated

   Example

   In a photograph of a clear blue sky, most neighboring pixels have nearly identical color values.

   - **Naive**: Store the RGB value of each pixel independently
   - **Smarter**: Encode repeated pixel values using run-length encoding
   - **Even smarter**: Predict each pixel from its neighbors and encode only the small prediction error

2. **Statistical Redundancy**: Some symbols occur far more frequently than others

   Example

   English letter frequencies:

Letter	Frequency	Letter	Frequency
E	12.7%	Z	0.07%
T	9.1%	Q	0.10%
A	8.2%	J	0.15%

   Frequent letters get shorter codes in variable-length coding schemes.

3. **Knowledge Redundancy**: Information already known to both encoder and decoder

4. **Perceptual Redundancy**: Information that humans cannot perceive

2.4 Entropy: The Fundamental Limit

2.4.1 What is Entropy? Different Perspectives

Definition

Shannon Entropy of a discrete random variable X with possible values $\{x_1, x_2, \ldots, x_n\}$ having probabilities $\{p_1, p_2, \ldots, p_n\}$:

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i \quad \text{bits}$$

Two Complementary Interpretations:

1. **Average Information Content**: Expected value of information content across all symbols
2. **Uncertainty or Surprise**: Measures how uncertain we are about the next symbol

2.4.2 Calculating Entropy: Step by Step

Example

Binary Source Example - Detailed Calculation:
Consider a biased coin: P(Heads) = 0.8, P(Tails) = 0.2

Step 1: **Calculate individual information content**:

$$I_H = -\log_2(0.8) \approx 0.3219 \text{ bits}$$
$$I_T = -\log_2(0.2) \approx 2.3219 \text{ bits}$$

Step 2: **Calculate entropy as expected value**:

$$H = 0.8 \times 0.3219 + 0.2 \times 2.3219 = 0.7219 \text{ bits}$$

Step 3: **Verify using direct formula**:

$$H = -[0.8 \log_2(0.8) + 0.2 \log_2(0.2)] \approx 0.7219 \text{ bits}$$

Key Insights:

- **Extreme cases**:

- Fair coin (P=0.5): $H = 1.0$ bit (maximum uncertainty)
- Always heads (P=1.0): $H = 0$ bits (no uncertainty)
- 90% heads: $H \approx 0.469$ bits

2.4.3 Entropy of English Text: A Practical Case Study

Example

Calculating English Letter Entropy:
Based on letter frequencies in typical English text:

$$H \approx 4.18 \text{ bits/letter}$$

Layered Interpretation:

- **First-order entropy (letters independent)**: 4.18 bits/letter
- **Actual uncertainty is lower**: Letters have dependencies (Q→U)
- **Comparison with encoding schemes**:

Encoding Method	Bits/Letter
Naive (5 bits for 26 letters)	5.00
Huffman (letter-based)	4.30
Using digram frequencies	3.90
Using word frequencies	2.30
Optimal with full context	∼1.50

2.4.4 Beyond First-Order Entropy: The Full Picture

Higher-Order Entropies quantify uncertainty while accounting for increasing context:

- **Zero-order entropy** (H_0): $H_0 = \log_2 |\mathcal{X}|$
- **First-order entropy** (H_1): $H_1 = -\sum p(x) \log_2 p(x)$
- **Second-order entropy** (H_2): $H_2 = -\sum p(x, y) \log_2 p(x|y)$
- **Nth-order entropy** (H_N): $H_N = -\sum p(x_1, \ldots, x_N) \log_2 p(x_N \mid x_1, \ldots, x_{N-1})$

2.4.5 Entropy Rate

The **entropy rate** of a source is defined as the limiting uncertainty per symbol when arbitrarily long contexts are available:

$$H_\infty = \lim_{N\to\infty} H_N$$

2.4.6 The Entropy Theorem: Why It Matters

Definition

Expected Code Length

For a source with symbols $\{x_1, x_2, \ldots, x_n\}$ having probabilities $\{p_1, p_2, \ldots, p_n\}$, and a code that assigns codeword lengths $\{\ell_1, \ell_2, \ldots, \ell_n\}$, the **expected code length** L is:

$$L = \mathbb{E}[\ell(X)] = \sum_{i=1}^{n} p_i \ell_i \quad \text{(bits per symbol)}$$

This measures the average number of bits needed to encode one symbol from the source.

Definition

Compression Ratio and Efficiency

For a source with entropy $H(X)$ and code with expected length L:

- **Compression ratio**: $\rho = \frac{\text{original bits}}{\text{compressed bits}}$
- **Efficiency**: $\eta = \frac{H(X)}{L} \leq 1$
- **Redundancy**: $R = L - H(X) \geq 0$

Perfect compression occurs when $\eta = 1$ (100% efficient) and $R = 0$.

Important

Shannon's Source Coding Theorem (1948)

For a discrete memoryless source with entropy H and any $\epsilon > 0$:

1. **Converse (Impossibility Result)**:
   No lossless coding scheme can achieve expected code length $L < H$.

$$\boxed{L \geq H \quad \text{for any uniquely decodable code}}$$

2. **Achievability (Possibility Result)**:
   There exists a lossless coding scheme (specifically, block coding with suffi-

ciently large block size n) such that:

$$\boxed{H \leq L < H + \epsilon}$$

Equivalently: For any $\epsilon > 0$, $\exists n$ such that:

$$\frac{L_n}{n} < H + \epsilon$$

where L_n is the expected length for blocks of size n.

Interpretation:

- **Entropy is the fundamental limit**: H bits/symbol is the best we can ever do
- **We can get arbitrarily close**: With clever coding, we can approach this limit as closely as desired
- **The gap is achievable**: The "$+\epsilon$" represents practical overhead that can be made arbitrarily small

Example

Understanding the Theorem with Numbers

Consider a binary source with $p(0) = 0.9$, $p(1) = 0.1$:

$$H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.469 \text{ bits/symbol}$$

- **Naive coding**: Use 1 bit per symbol $\rightarrow L = 1.0$, efficiency $\eta = 0.469/1.0 = 46.9\%$
- **Huffman coding**: 0→0, 1→1 (same as naive!) $\rightarrow L = 1.0$, $\eta = 46.9\%$ *Why so bad?* Because we're coding symbols individually.
- **Block coding (n=2)**: Code pairs of symbols:

$$\begin{aligned} 00 &\rightarrow 0 \quad (\ell = 1, p = 0.81) \\ 01 &\rightarrow 10 \quad (\ell = 2, p = 0.09) \\ 10 &\rightarrow 110 \quad (\ell = 3, p = 0.09) \\ 11 &\rightarrow 111 \quad (\ell = 3, p = 0.01) \end{aligned}$$

$L_2 = 0.81 \times 1 + 0.09 \times 2 + 0.09 \times 3 + 0.01 \times 3 = 1.29$ bits/block Per symbol: $L = L_2/2 = 0.645$ bits/symbol, $\eta = 0.469/0.645 \approx 72.7\%$

- **Block coding (n=3)**: Would get even closer to 0.469
- **Theoretical limit**: As $n \to \infty$, $L \to 0.469$

Key insight: The theorem tells us:

1. We can never beat 0.469 bits/symbol (impossibility)
2. We can get as close as we want to 0.469 bits/symbol (achievability)

Important

What the Theorem Does NOT Say

- **It doesn't say how to construct the code** - just that one exists
- **It doesn't guarantee practical implementation** - block size n might need to be huge
- **It doesn't account for computational complexity** - the code might be too complex to implement
- **It assumes we know the true probabilities** - in practice, we estimate them

Yet, this theorem is revolutionary because it:

1. Establishes a **fundamental limit** (like the speed of light in physics)
2. Provides a **benchmark** for evaluating compression algorithms
3. Guides algorithm design toward this limit

2.4.7 Key Takeaways

- Entropy measures both **average information** and **uncertainty**
- Higher-order models reduce entropy by exploiting dependencies
- The entropy rate represents the ultimate compression limit
- Shannon's theorem precisely separates the *possible* from the *impossible*

2.5 Entropy as a Lower Bound

2.5.1 The Fundamental Inequality

Theorem 2.2 (Entropy Lower Bound)**.** *For any* ***uniquely decodable*** *code C for source X:*

$$L(C) \geq H(X)$$

where $L(C) = \mathbb{E}[\ell(X)] = \sum_i p_i \ell_i$ is the expected code length.

Example

Binary Source with $p(0) = 0.9$, $p(1) = 0.1$

$$H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.469 \text{ bits/symbol}$$

Why we can't achieve 0.4 bits/symbol:

1. For 100 symbols, typical sequences: $2^{100 \times 0.469} \approx 2^{46.9}$
2. To encode all uniquely, need at least $2^{46.9}$ codewords
3. At 0.4 bits/symbol, total bits = 40
4. # codewords $\leq 2^{40} < 2^{46.9} \rightarrow$ impossible!

2.6 Kraft-McMillan Inequality: Core Theoretical Tool

2.6.1 Statement and Interpretation

Theorem 2.3 (Kraft-McMillan Inequality (Binary Case))**.** *Let $\ell_1, \ell_2, \ldots, \ell_m$ be the lengths of codewords in a* ***prefix code****. Then:*

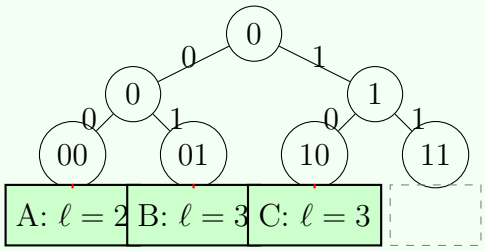
$$\sum_{i=1}^{m} 2^{-\ell_i} \leq 1$$

Conversely, if integers $\ell_1, \ldots, \ell_m$ satisfy this inequality, then there exists a binary prefix code with these lengths.

Example

Tree Visualization of Kraft Inequality

Consider a binary tree of depth $L = \max \ell_i$:



Calculating Kraft sum for $\{\ell_A = 2, \ell_B = 3, \ell_C = 3\}$:

$$\sum 2^{-\ell_i} = 2^{-2} + 2^{-3} + 2^{-3} = 0.25 + 0.125 + 0.125 = 0.5 \leq 1$$

Example

Testing Code Feasibility

1. **Valid lengths**: $\{1, 2, 3, 3\}$

$$\sum = 2^{-1} + 2^{-2} + 2^{-3} + 2^{-3} = 0.5 + 0.25 + 0.125 + 0.125 = 1.0 \quad \text{VALID}$$

2. **Invalid lengths**: $\{1, 1, 2\}$

$$\sum = 2^{-1} + 2^{-1} + 2^{-2} = 0.5 + 0.5 + 0.25 = 1.25 > 1 \quad \text{INVALID}$$

2.7 Optimality of Huffman Codes (Theory Only)

2.7.1 The Optimality Theorem

Theorem 2.4 (Huffman Optimality)**.** *Given a source with symbol probabilities $p_1, p_2, \ldots, p_m$, the Huffman algorithm produces a prefix code that* ***minimizes*** *the expected code length $L = \sum_{i=1}^{m} p_i \ell_i$ among all prefix codes.*

2.7.2 Relation to Entropy

For any Huffman code:

$$H(X) \leq L_{\text{Huffman}} < H(X) + 1$$

Example

Understanding the "+1" Gap

Consider source with probabilities $\{0.6, 0.3, 0.1\}$:

$$H \approx 1.295 \text{ bits}$$

Ideal (non-integer) lengths: $-\log_2 p_i = \{0.737, 1.737, 3.322\}$
Huffman code: 0.6→0, 0.3→10, 0.1→11

$$L = 0.6 \times 1 + 0.3 \times 2 + 0.1 \times 2 = 1.4 \text{ bits}$$

Comparison:

- Entropy: 1.295 bits
- Huffman: 1.400 bits
- Gap: 0.105 bits (much less than 1!)

2.7.3 Why Huffman is Optimal but Not Perfect

Important

Huffman is Optimal Within a Restricted Class
Huffman is optimal among:

- **Symbol-by-symbol** codes
- **Prefix** codes
- **Static** codes

But real optimality might require:

- **Block coding**
- **Fractional bits** (arithmetic coding)
- **Adaptive probabilities**

2.8 Limitations of Symbol-by-Symbol Coding

2.8.1 Three Fundamental Limitations

1. **Cannot Exploit Dependencies**
2. **Integer Length Constraint**: $\ell_i \in \mathbb{Z}^+$ but $-\log_2 p_i \in \mathbb{R}$
3. **Memoryless Assumption**

Example

English Text: The Cost of Symbol-by-Symbol

- **First-order entropy** (ignoring dependencies): 4.0 bits/letter
- **Actual entropy rate** (with dependencies): 1.5 bits/letter
- **Huffman on letters**: 4.0 bits/letter
- **Gap**: 2.5 bits/letter wasted due to ignoring dependencies

2.9 Block Coding and Improved Efficiency

2.9.1 The Block Coding Idea

Instead of coding symbols individually, group them into blocks of length n:

$$\mathbf{X} = (X_1, X_2, \ldots, X_n)$$

Definition

nth Extension of a Source

For a source with alphabet $\mathcal{X}$, the nth extension has alphabet:

$$\mathcal{X}^n = \{(x_1, \ldots, x_n) : x_i \in \mathcal{X}\}$$

with size $|\mathcal{X}|^n$.

2.9.2 Key Mathematical Results

Theorem 2.5 (Entropy of Block Source)**.** *For a discrete memoryless source:*

$$H(X^n) = nH(X)$$

Theorem 2.6 (Block Coding Performance)**.** *There exists a prefix code C_n for X^n such that:*

$$nH(X) \leq L_n < nH(X) + 1$$

Dividing by n:

$$H(X) \leq \frac{L_n}{n} < H(X) + \frac{1}{n}$$

Important

The Magic of Block Coding

As $n \to \infty$:

$$\frac{L_n}{n} \to H(X)$$

We can approach entropy **arbitrarily closely** by making blocks larger!

2.9.3 Step-by-Step Example

Example

Binary Source: $p(0) = 0.9$, $p(1) = 0.1$, $H \approx 0.469$

Step 1: $n = 1$ **(symbol-by-symbol)**

- Huffman: 0→0, 1→1
- $L_1 = 1$ bit/symbol
- Efficiency: $\eta = 0.469/1 = 46.9\%$

Step 2: $n = 2$ **(code pairs)**

- Block probabilities: $P(00) = 0.81$, $P(01) = 0.09$, $P(10) = 0.09$, $P(11) = 0.01$
- Codes: 00→0, 01→10, 10→110, 11→111
- Per symbol: $L_2/2 = 0.645$ bits/symbol
- Efficiency: $\eta = 0.469/0.645 = 72.7\%$

2.10 Trade-Offs in Block Coding

2.10.1 The Engineering Challenges

1. **Exponential Alphabet Growth**: $|\mathcal{X}^n| = |\mathcal{X}|^n$
2. **Memory Requirements**: Huffman tree has $2m^n - 1$ nodes
3. **Computational Complexity**: $O(m^n \log m^n)$
4. **Delay and Latency**: Must wait for n symbols

2.11 From Block Coding to Modern Compression

2.11.1 Arithmetic Coding: Fractional-Bit Block Coding

Important

Arithmetic Coding as "Infinite Block Coding"
Arithmetic coding cleverly avoids the exponential growth problem:

- **Idea**: Encode entire message as a single real number in [0,1)
- **No explicit blocks**: Processes symbols sequentially
- **Fractional bits**: Achieves $L \approx H(X)$ without large n
- **Removes integer constraint**: No "+1" overhead!

2.11.2 Context Modeling: Approximating Large Blocks

Instead of explicit block coding, modern compressors use:

1. **Context Models**: Predict next symbol based on previous k symbols
2. **Prediction + Residual Coding**: Encode only prediction error
3. **Dictionary Methods (LZ family)**: Build dictionary of previously seen phrases

2.12 What Theory Guarantees vs What Practice Achieves

Aspect	Theory Guarantees	Practice Achieves
Optimality	Can approach entropy arbitrarily closely	Gets close, but with practical limits
Block Size	$n \to \infty$ gives optimality	n limited by memory, latency, complexity
Complexity	Ignored, infinite resources allowed	Critical constraint; often dominates design

2.13 Summary and Key Takeaways

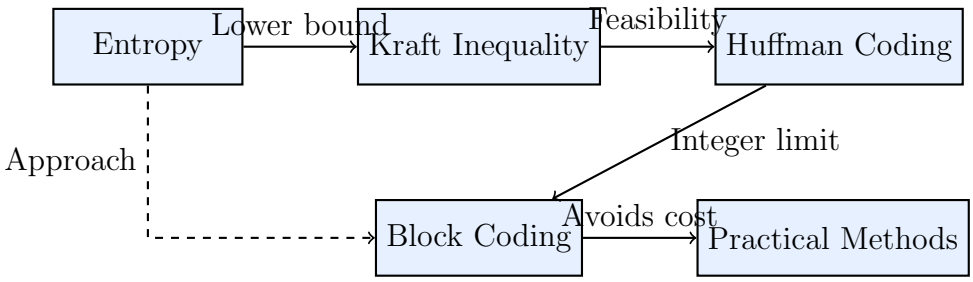
Important

Five Fundamental Lessons

1. **Entropy is the Absolute Limit**: $L \geq H(X)$ for any lossless code

2. **Kraft-McMillan Constrains All Codes**: $\sum 2^{-\ell_i} \leq 1$
3. **Huffman is Optimal Among Prefix Codes**: But limited by integer lengths
4. **Block Coding Allows Approaching Entropy**: $\lim_{n\to\infty} \frac{L_n}{n} = H(X)$
5. **Practical Compression Balances Efficiency and Complexity**

2.13.1 The Big Picture



2.13.2 Looking Forward

- **Next lecture: Arithmetic Coding**: Removes integer constraint
- **Then: Dictionary Methods (LZ family)**: Adaptive to data statistics
- **Finally: Modern Compressors**: Combining multiple techniques

Final Thought

Shannon's 1948 paper told us *exactly how good compression could possibly be.* Every compressor since has been trying to approach that limit while staying within practical constraints.

The gap between theory and practice is where engineering creativity lives!

3: Lecture 3: Advanced Entropy Coding & Extensions

Lecture 3: Beyond Huffman – Advanced Entropy Coding Methods

3.1 Introduction & Motivation

Important

Recall Huffman Coding Limitations:

- **Integer code lengths**: Cannot reach entropy bound for highly skewed distributions
- **Static vs. Adaptive**: Standard Huffman requires prior knowledge of probabilities
- **Codebook overhead**: Need to transmit/store the coding tree
- **Symbol-by-symbol constraint**: Processes one symbol at a time

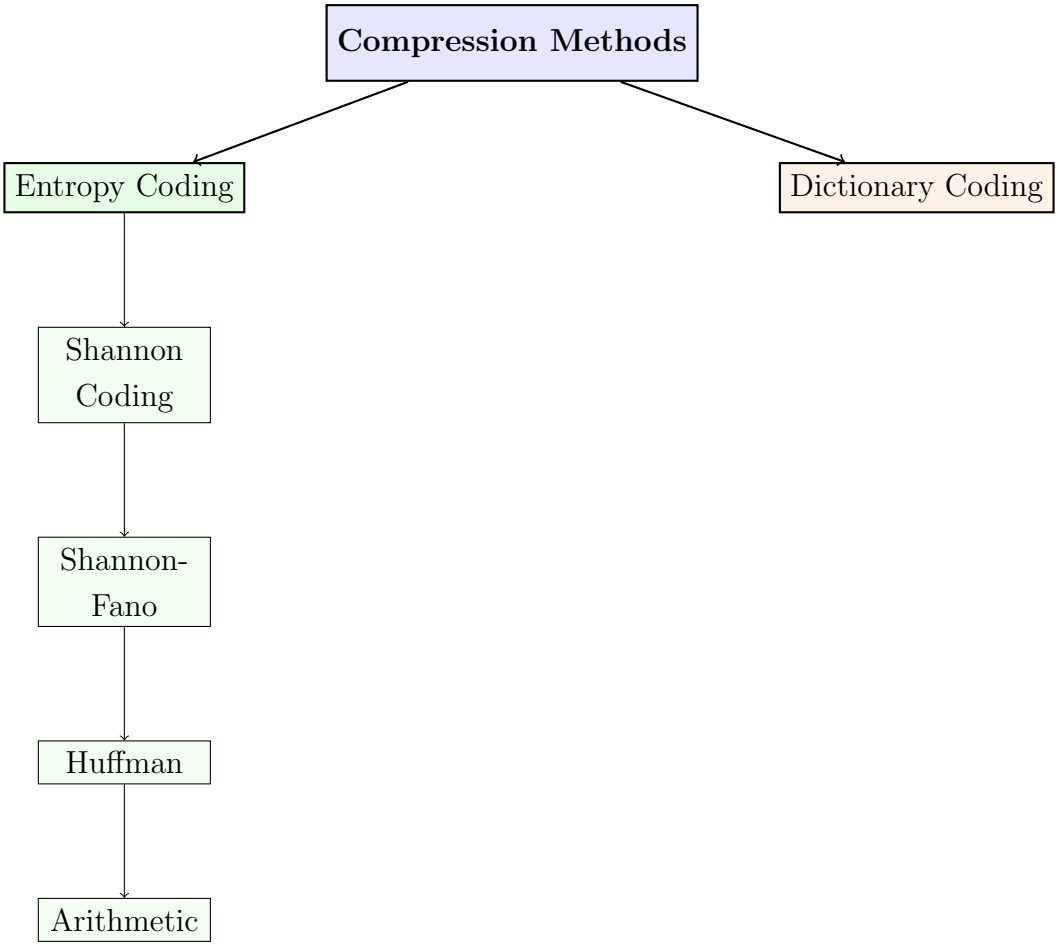
Lecture Roadmap:

1. **Framework**: Coding taxonomy and conceptual organization
2. **Historical methods**: Shannon & Shannon-Fano coding
3. **Practical improvements**: Canonical and Adaptive Huffman
4. **Next generation**: Arithmetic coding paradigm
5. **Synthesis**: Comparison and forward look

3.2 Coding Taxonomy & Framework

Definition

Coding Taxonomy: Classification of compression methods based on key characteristics



Key Dimensions in Compression Algorithm Design

1. **Modeling vs. Coding (Two-Stage View)**:
   - **Modeling Phase**: Analyzes data to estimate symbol probabilities or discover patterns.
     - *Examples*: Frequency counting (Huffman), context modeling (PPM), dictionary construction (LZ77)
   - **Coding Phase**: Converts modeled information into actual bits.
     - *Examples*: Huffman codes, arithmetic codes, LZ77 pointers
   - Some algorithms intertwine both (e.g., LZW builds dictionary while coding).
2. **Knowledge of Source Distribution**:
   - **Static/Fixed**: Uses a predefined model that doesn't change.
     - *Example*: JPEG Huffman tables, known language frequencies
     - Requires prior knowledge of data; fails if distribution differs.
   - **Adaptive**: Learns and updates the model during compression.
     - *Example*: Adaptive Huffman, LZ78 dictionary building

    - No prior knowledge needed; overhead for model transmission.
  - **Universal**: Can compress any source asymptotically optimally.
    - *Theoretical property*: LZ family, arithmetic with adaptive model
    - Note: Most adaptive methods are universal in practice.

3. **Processing Granularity**:
   - **Symbol-by-Symbol**: Each input symbol maps to one codeword.
     - *Example*: Huffman coding
     - Simple but limited to integer bits per symbol.
   - **Block Coding**: Fixed-size groups of symbols coded together.
     - *Example*: Block-sorting (BWT) processes blocks
     - Can capture inter-symbol dependencies within block.
   - **Stream/Incremental**: Continuous processing with immediate output.
     - *Example*: Arithmetic coding, LZ77 sliding window
     - No blocking delay; good for real-time applications.
   - *Note*: "Message-wide" (whole file as one symbol) is theoretical ideal; arithmetic coding approximates it by treating the entire stream as one long fractional code.

4. **Algorithmic Approach (Primary Taxonomy)**:
   - **Statistical Coding**: Uses probability estimates (Huffman, Arithmetic)
   - **Dictionary Coding**: Replaces repeated patterns with references (LZ family)
   - **Transform Coding**: Changes data domain then codes (DCT, wavelet)
   - **Predictive Coding**: Predicts next value, codes difference (DPCM, LPC)

Important Relationships:

- Adaptive methods are usually universal for practical purposes.
- Stream coding is possible with both symbol-by-symbol (Huffman) and message-wide approaches (arithmetic).
- Block processing (like BWT) is often followed by stream coding (like MTF+RLE+arithmetic in bzip2).
- Modeling and coding can be separated (PPM + arithmetic) or combined (LZW).

3.3 Shannon Coding (1948)

Definition

Shannon Coding: A constructive method derived from Shannon's source coding theorem that assigns codewords by taking the binary expansion of cumulative probabilities:

$$l_i = \lceil -\log_2 p_i \rceil \quad \text{and} \quad \text{code}_i = \text{First } l_i \text{ bits of } F_i$$

where p_i is the probability of symbol i, and $F_i = \sum_{j=1}^{i-1} p_j$ is the cumulative probability of symbols ordered by decreasing probability.

Shannon Coding Algorithm

Input: Symbols $S = \{s_1, s_2, \ldots, s_n\}$ with corresponding probabilities $P = \{p_1, p_2, \ldots, p_n\}$
Output: Prefix-free binary code for each symbol

1. **Sort** symbols in non-increasing order of probability: $p_1 \geq p_2 \geq \cdots \geq p_n$
2. **Calculate codeword lengths:** For each symbol i, compute $l_i = \lceil -\log_2 p_i \rceil$
3. **Compute cumulative probabilities:**

$$F_1 = 0, \quad F_i = \sum_{j=1}^{i-1} p_j \quad \text{for } i = 2, \ldots, n$$

4. **Generate codewords:** For each symbol i:
   a) Convert F_i to binary fractional representation (0.xxxx...)
   b) Take the first l_i bits after the binary point as the codeword

Example

Example: Given symbols with probabilities:

Symbol	Probability
A	0.5
B	0.25
C	0.125
D	0.125

Step-by-step construction:

1. **Sort by probability:** Already in non-increasing order ($0.5 \geq 0.25 \geq 0.125 \geq 0.125$)
2. **Calculate lengths:**

- $l_A = \lceil -\log_2 0.5 \rceil = \lceil 1.0 \rceil = 1$
- $l_B = \lceil -\log_2 0.25 \rceil = \lceil 2.0 \rceil = 2$
- $l_C = \lceil -\log_2 0.125 \rceil = \lceil 3.0 \rceil = 3$
- $l_D = \lceil -\log_2 0.125 \rceil = \lceil 3.0 \rceil = 3$

3. **Compute cumulative probabilities:**
   - $F_A = 0$ (first symbol)
   - $F_B = 0.5$ (just A's probability)
   - $F_C = 0.5 + 0.25 = 0.75$ (A + B)
   - $F_D = 0.5 + 0.25 + 0.125 = 0.875$ (A + B + C)
4. **Convert F_i to binary and take first l_i bits:**
   - **A:** $F_A = 0.0_{10}$ in binary is $0.0000\ldots_2$
     - Take first $l_A = 1$ bit: **0**
   - **B:** $F_B = 0.5_{10}$ in binary is $0.1000\ldots_2$
     - Take first $l_B = 2$ bits: **10**
   - **C:** $F_C = 0.75_{10}$ in binary is $0.1100\ldots_2$
     - Take first $l_C = 3$ bits: **110**
   - **D:** $F_D = 0.875_{10}$ in binary is $0.1110\ldots_2$
     - Take first $l_D = 3$ bits: **111**

Resulting code:

Symbol	Probability	l_i	F_i	Shannon Code
A	0.5	1	0.0	0
B	0.25	2	0.5	10
C	0.125	3	0.75	110
D	0.125	3	0.875	111

Expected length: $L = 0.5 \times 1 + 0.25 \times 2 + 0.125 \times 3 + 0.125 \times 3 = 1.75$ bits/symbol

Important

Critical Note on Sorting:
The sorting step is **essential** in Shannon coding because:

- Cumulative probabilities F_i depend on the order of symbols

- Sorting ensures intervals in [0,1) are assigned in decreasing size order
- Without sorting, the resulting code may not be prefix-free
- Sorting guarantees the length condition $\frac{1}{2^{l_i}} \leq p_i < \frac{1}{2^{l_i-1}}$ leads to non-overlapping intervals

Why it works: When probabilities are sorted, each symbol's interval of size p_i starts at F_i and ends at $F_i + p_i$. The codeword length ensures the binary expansion of F_i to l_i bits uniquely identifies the starting point without overlapping with neighboring intervals.

Understanding the Binary Expansion Process

When we write F_i in binary (e.g., $0.5 = 0.1_2$, $0.75 = 0.11_2$), we're essentially:

- Dividing the interval [0,1) into subintervals based on probabilities
- Each symbol gets an interval of size p_i
- The codeword is the **binary fraction** representing the **start** of that interval
- We use exactly $l_i = \lceil -\log_2 p_i \rceil$ bits, which ensures:

$$\frac{1}{2^{l_i}} \leq p_i < \frac{1}{2^{l_i-1}}$$

- This guarantees unique prefixes because intervals don't overlap!

Important

Properties of Shannon Coding:

- **Constructive proof**: Demonstrates that prefix codes exist for any lengths satisfying Kraft inequality
- **Simple to compute**: Direct from probabilities, no tree needed
- **Requires sorting**: Symbols must be ordered by decreasing probability first
- **Not optimal**: Unlike Huffman, doesn't minimize expected length (compare: Huffman would give A=0, B=10, C=110, D=111 **same in this case!**)
- **Theoretical importance**: Foundation for Shannon's source coding theorem
- **Efficiency bound**: $H(X) \leq L < H(X) + 1$ (like Shannon's theorem says)

Example

Counterexample: What happens if we don't sort?
Consider the same symbols but in different order: C (0.125), A (0.5), D (0.125), B (0.25)
Without sorting:

- $F_C = 0$, $l_C = 3 \rightarrow$ codeword: 000
- $F_A = 0.125$, $l_A = 1 \rightarrow$ codeword: 0 (binary: 0.001... $\rightarrow$ first bit is 0)
- $F_D = 0.625$, $l_D = 3 \rightarrow$ codeword: 101
- $F_B = 0.75$, $l_B = 2 \rightarrow$ codeword: 11

Problem: Codeword "0" (for A) is a prefix of "000" (for C) $\rightarrow$ **not prefix-free!**
This demonstrates why sorting is essential in Shannon coding.

3.4 Shannon–Fano Coding (1949)

Definition

Shannon–Fano Coding: A top-down, recursive source coding technique that assigns binary codewords by repeatedly partitioning a set of symbols into two subsets whose total probabilities are as close as possible. The method was developed independently by *Claude Shannon* and *Robert Fano* in 1949.

Algorithm Description

High-level idea: Symbols with higher probabilities should receive shorter codewords. This is achieved by repeatedly splitting the symbol set into two probability-balanced groups and assigning binary prefixes.

Step-by-step procedure:

1. **Sort** the symbols in decreasing order of probability:

$$p_1 \geq p_2 \geq \cdots \geq p_n.$$

2. **Recursive partitioning:**

   - If the current set contains only one symbol, stop (this is the base case).

- Find an index k that minimizes

$$\left|\sum_{i=1}^{k} p_i - \sum_{i=k+1}^{n} p_i\right|.$$

- This divides the symbols into two subsets:

$$S_1 = \{1, \ldots, k\}, \qquad S_2 = \{k+1, \ldots, n\}.$$

- Append bit **0** to the codewords of all symbols in S_1.
- Append bit **1** to the codewords of all symbols in S_2.
- Apply the same procedure recursively to S_1 and S_2.

Example with Six Symbols

Example

Example: Consider six symbols with the following probabilities.

Symbol	Probability	$-\log_2 p_i$
A	0.30	1.74
B	0.25	2.00
C	0.20	2.32
D	0.10	3.32
E	0.10	3.32
F	0.05	4.32

Construction process

Step 1: First split (balance 0.55 vs. 0.45)

- Sorted symbols: A(0.30), B(0.25), C(0.20), D(0.10), E(0.10), F(0.05)
- Best split: $\{A, B\}(0.55)$ and $\{C, D, E, F\}(0.45)$
- Prefix assignment: $\{A, B\} \to 0$, $\{C, D, E, F\} \to 1$

Step 2: Split $\{A, B\}$

- A: **00**, B: **01**

Step 3: Split $\{C, D, E, F\}$

- Best split: $\{C\}(0.20)$ and $\{D, E, F\}(0.25)$
- C: **10**, $\{D, E, F\} \to 11$

Step 4: Split $\{D, E, F\}$

- Best split: $\{D\}(0.10)$ and $\{E, F\}(0.15)$
- D: **110**, $\{E, F\} \to 111$

Step 5: Split $\{E, F\}$

- E: **1110**, F: **1111**

Final codes:

Symbol	Probability	Code	Length
A	0.30	00	2
B	0.25	01	2
C	0.20	10	2
D	0.10	110	3
E	0.10	1110	4
F	0.05	1111	4

Expected code length:

$$L = 2.40 \text{ bits/symbol.}$$

Entropy:

$$H(X) \approx 2.25 \text{ bits/symbol.}$$

Efficiency:

$$\frac{H(X)}{L} \approx 93.8\%.$$

Visual Tree Representation

```
                    ROOT
                   /    \
                 0/      \1
                 /        \
        {A,B} 0.55    {C,D,E,F} 0.45
           / \            / \
         0/   \1        0/   \1
         /     \        /     \
{A} 0.30  {B} 0.25  {C} 0.20  {D,E,F} 0.25
  00        01        10         / \
                                0/   \1
                                /     \
                            {D} 0.10  {E,F} 0.15
                              110        / \
                                       0/   \1
                                       /     \
                                   {E} 0.10  {F} 0.05
                                     1110      1111
```

Key Observations

- **Probability-balanced splits:** The algorithm focuses on balancing probabilities rather than the number of symbols.
- **Variable code lengths:** More probable symbols receive shorter codewords.
- **Prefix-free property:** No codeword is a prefix of another.
- **Near-optimal performance:** The efficiency is high but not guaranteed to be optimal.

Important

Limitations and Historical Context

- Shannon–Fano coding does *not* always produce the optimal code.
- Huffman coding (1952) guarantees the minimum average code length.
- Historically important as a precursor to Huffman coding.

3.5 Canonical Huffman Codes

Definition

Canonical Huffman Code: A standardized representation of a Huffman code in which:

- Codes are assigned in lexicographic (binary) order
- All codewords of the same length are consecutive binary numbers
- The first codeword of each length is the smallest possible binary value
- Only the code lengths are required to reconstruct the entire code

This enables compact storage and fast table-based decoding.

Why Canonical Huffman Codes?

A standard Huffman algorithm produces *optimal code lengths*, but the exact bit patterns depend on implementation details such as tie-breaking and tree construction:

- Different trees can yield the same optimal set of code lengths
- Different bit assignments, but identical compression performance

Canonical Huffman coding fixes **one unique and deterministic assignment** for a given set of code lengths so that:

1. Only code lengths need to be transmitted
2. The decoder can reconstruct the codes without ambiguity
3. Decoding can be implemented efficiently using lookup tables

Two-Step Process

Complete workflow:

1. **Run the standard Huffman algorithm** to obtain optimal code lengths l_i
2. **Apply the canonical transformation** to convert lengths into standardized codes

$$\boxed{\text{Symbol probabilities} \xrightarrow{\text{Huffman}} \text{Code lengths} \xrightarrow{\text{Canonical}} \text{Canonical codes}}$$

Canonical Huffman Construction Algorithm

Input: Code lengths l_i obtained from a standard Huffman algorithm **Output:** Canonical Huffman codes

1. **Sort symbols** in ascending order of code length l_i (primary key), and by symbol order or value (secondary key for tie-breaking)
2. **Count symbols per length:**
   - Let $l_{\min}$ and $l_{\max}$ be the minimum and maximum code lengths
   - For each length l, compute $\text{count}[l]$ = number of symbols with length l
3. **Compute starting codes:**
   - $\text{start_code}[l_{\min}] = 0$
   - For $l = l_{\min} + 1$ to $l_{\max}$:

$$\text{start_code}[l] = (\text{start_code}[l-1] + \text{count}[l-1]) \times 2$$

4. **Initialize assignment pointers:**

$$\text{next_code}[l] = \text{start_code}[l] \quad \text{for all } l$$

5. **Assign codes to symbols** (in sorted order):
   - For each symbol i with length l_i:

- Assign next_code[l_i] as its code (converted to l_i-bit binary)
- Increment next_code[l_i] by 1

Understanding the Shift Operation: Why Multiply by 2?

The key step start_code[l] = (start_code[$l-1$] + count[$l-1$]) $\times$ 2 ensures:

Important

Visual Intuition: Imagine all codes of length $l-1$ are written as binary numbers. When we multiply by 2 (left shift by 1 bit), we:

- Append a 0 to each existing code
- Create room for new codes that are one bit longer
- Ensure the new codes maintain prefix-free property

Example: If the last code of length 2 is "11" (binary 3), then:

$$(\text{last code} + 1) \times 2 = (3 + 1) \times 2 = 8 = 1000_{\text{binary}}$$

The first code of length 3 is "100" (3 bits of 1000), which follows "11" properly.

Mathematical Explanation:

- start_code[$l-1$] + count[$l-1$] gives the **first unused code value** after all codes of length $l-1$
- Multiplying by 2 (left shift by 1 bit) makes room for an extra bit
- This ensures all codes of length l start with a 0 in their new bit position
- The result is the smallest valid starting point for the next length group

Bit Pattern Visualization:

```
Length 2 codes: 00, 01, 10, 11
Length 3 codes start at: (11 + 1) × 2 = 1000_binary
So length 3 codes are: 100, 101, 110, 111
Notice: 11 is NOT a prefix of 100!
```

Complete Worked Example

Example

Example: Suppose the Huffman algorithm produces the following code lengths:

Symbol	Length (l_i)
A	2
B	3
C	3
D	3
E	4
F	4

Key Point: The code lengths (l_i) are already determined by the Huffman tree construction. These lengths tell us *exactly* how many bits each symbol's code will use.

Step 1: Sort symbols by length (primary), then symbol order (secondary)

- **Critical:** Sorting ensures consistent code assignment
- Order: A (2), B (3), C (3), D (3), E (4), F (4)
- Within same length group, maintain original symbol order

Step 2: Count symbols per length

$$\text{count}[2] = 1, \quad \text{count}[3] = 3, \quad \text{count}[4] = 2$$

Step 3: Compute starting codes (with shift explanation)

$\text{start_code}[2] = 0$ (represented as 00 in binary, using **2 bits** because length=2)

$\text{start_code}[3] = (\text{start_code}[2] + \text{count}[2]) \times 2$

$= (0 + 1) \times 2 = 2$ (represented as 010 in binary, using **3 bits** because length=3)

Interpretation: After 1 code of length 2, shift left (add 0 bit) for length 3

$\text{start_code}[4] = (\text{start_code}[3] + \text{count}[3]) \times 2$

$= (2 + 3) \times 2 = 10$ (represented as 1010 in binary, using **4 bits** because length=4)

Interpretation: After 3 codes of length 3, shift left (add 0 bit) for length 4

Important: The multiplication by 2 is a **left shift** operation. It adds one more bit to the code while maintaining the prefix-free property.

Step 4: Initialize assignment pointers

$$\text{next_code}[2] = 0, \quad \text{next_code}[3] = 2, \quad \text{next_code}[4] = 10$$

Step 5: Assign codes (in sorted order)

Symbol	Length	Decimal	Binary Code	next_code after
A	2	0	`00`	1
B	3	2	`010`	3
C	3	3	`011`	4
D	3	4	`100`	5
E	4	10	`1010`	11
F	4	11	`1011`	12

How binary conversion works:

- For symbol A (length=2, decimal=0): Binary: 00 (padded to 2 bits)
- For symbol B (length=3, decimal=2): Binary: 010 (2 in binary is 10, padded to 3 bits: 010)
- For symbol E (length=4, decimal=10): Binary: 1010 (10 in binary is 1010, already 4 bits)

Algorithm Summary:

1. **Input**: Code lengths l_i for each symbol (from Huffman tree)
2. **For each length** l:
   - First code of length l: start_code[l]
   - Subsequent codes: increment by 1
   - All codes use exactly l bits
3. **Key formula**: start_code[$l + 1$] = (start_code[l] + count[l]) $\times$ 2

Final canonical codes (verify properties):

Symbol	Length	Canonical Code	Check
A	2	00	✓ First length 2 code
B	3	010	✓ First length 3 code
C	3	011	✓ Consecutive with B
D	3	100	✓ Consecutive with C
E	4	1010	✓ First length 4 code
F	4	1011	✓ Consecutive with E

Verification:

- All codes of same length are consecutive binary numbers ✓
- Codes are lexicographically ordered ✓
- No code is a prefix of another ✓

What If We Don't Sort? A Counterexample

Example

Problem without sorting: Suppose we have the same code lengths but assign codes without proper sorting:

Symbol	Length
D	3
A	2
E	4
B	3
F	4
C	3

If we assign codes in this random order using the same algorithm:

- D (length 3) gets code 010
- A (length 2) gets code 00
- E (length 4) gets code 1000
- **Problem:** Code "00" (A) is a prefix of "1000" (E)!

Conclusion: Sorting ensures that shorter codes are assigned first, preventing prefix conflicts between codes of different lengths.

Transmission and Decoding

Transmission format (very compact):

- Transmit only the sequence of code lengths in symbol order:

$$\langle l_1, l_2, \ldots, l_n \rangle$$

- Example for our symbols: $\langle 2, 3, 3, 3, 4, 4 \rangle$
- Each length can be represented using $\lceil \log_2 l_{\max} \rceil$ bits

Decoder reconstruction:

1. Read code lengths for all symbols (in original symbol order)
2. Sort symbols by (l_i, index) to match encoder's order
3. Reconstruct canonical codes using the same algorithm
4. Build decoding tables for fast lookup

Comparison with Standard Huffman

Standard Huffman (tree-based)	**Canonical Huffman (length-based)**
Output: Huffman tree structure	Output: Code lengths only
Must transmit tree (complex)	Transmit only lengths (simple)
Decoding by tree traversal (slower)	Decoding by table lookup (faster)
Multiple equivalent trees possible	Single deterministic assignment
Same optimal compression	Same optimal compression
More complex implementation	Simple, robust implementation
No sorting required	**Requires sorting by length**

Key Properties and Applications

- **Optimality:** Identical compression ratio to standard Huffman
- **Compactness:** Only code lengths stored/transmitted
- **Speed:** Table-based decoding is significantly faster
- **Standardization:** Used in DEFLATE (gzip/ZIP), JPEG, PNG, MPEG
- **Deterministic:** Same lengths always produce same codes

Important

Critical Insights:

1. Canonical Huffman is **not** a different compression algorithm from Huffman
2. The Huffman algorithm determines *how long* each code should be
3. The canonical transformation determines the *exact bit patterns* consistently
4. **Sorting by length** is essential to ensure prefix-free property
5. The $\times 2$ operation (left shift) ensures proper separation between different length groups

3.6 Adaptive Huffman Coding

Overview

Definition

Adaptive Huffman Coding is a single-pass, lossless data compression technique in which the Huffman model is updated dynamically as symbols are encoded and decoded. The Huffman tree evolves on-the-fly, allowing both the encoder and decoder to adapt to changing symbol statistics without any prior knowledge of the source distribution.

Unlike static Huffman coding, adaptive Huffman coding does not require a preprocessing phase to compute symbol frequencies. Instead, symbol statistics are learned incrementally as the data stream is processed.

Limitations of Static Huffman Coding

Static Huffman coding assumes that symbol statistics are known in advance:

- Requires two passes over the data:
  1. Collect symbol frequencies
  2. Encode using the constructed Huffman tree
- The Huffman tree (or code lengths) must be transmitted as side information
- Cannot adapt to non-stationary or evolving symbol distributions

These limitations motivate adaptive schemes when symbol statistics are unknown or change over time.

Key Principle of Adaptive Huffman Coding

Adaptive Huffman coding maintains a Huffman tree that is updated after encoding or decoding each symbol.

Key invariant: After processing each symbol, the encoder and decoder maintain *identical Huffman trees* by applying the same updates in the same order. This ensures correct decoding without transmitting the tree explicitly.

Initialization

The algorithm begins with a special symbol:

- A single **NYT** (Not Yet Transmitted) node

When a symbol appears for the first time:

- The code for the NYT node is output
- The raw symbol value is transmitted
- The NYT node is expanded into:
  - A new NYT node
  - A leaf node representing the new symbol

Note: Some variants preinitialize the tree if the alphabet is fixed, but the standard adaptive Huffman algorithm begins with only the NYT node.

Tree Update Mechanism

After each symbol is processed:

- The frequency (weight) of the corresponding leaf node is incremented
- The tree is updated to preserve the **sibling property**

Definition

Sibling Property: Nodes are numbered such that nodes with higher weights have higher numbers. For any given weight, all nodes with that weight form a contiguous block. Each internal node has exactly two children.

To restore this property, nodes may be swapped with others of equal weight, followed by incrementing parent node weights. This update process proceeds bottom-up toward the root.

Encoding and Decoding Process

- If the symbol has appeared before:
  - Output its current Huffman code
- If the symbol is new:
  - Output the code for the NYT node
  - Output the symbol in raw (fixed-length) form
- Update the tree identically at both encoder and decoder

Note: Exact bit patterns depend on the update order and implementation. Adaptive Huffman coding guarantees prefix-free optimality but not unique codes.

Algorithms

Two classical adaptive Huffman algorithms are widely used:

- **FGK Algorithm** (Faller–Gallager–Knuth)
- **Vitter's Algorithm (Algorithm V)**, which improves worst-case performance and is commonly preferred

Both algorithms maintain the sibling property while ensuring efficient updates.

Performance Characteristics

- Update cost is **amortized** $O(1)$ per symbol
- Worst-case update time is proportional to the tree height
- Compression efficiency:
  - Initially suboptimal
  - Converges toward entropy-optimal coding as statistics stabilize

Comparison with Static Huffman Coding

Feature	**Static Huffman**	**Adaptive Huffman**
Number of passes	Two	One
Prior statistics required	Yes	No
Model transmission	Required	Not required
Adaptation to changes	No	Yes
Initial efficiency	High	Low
Asymptotic efficiency	Optimal	Near-optimal

Applications

Adaptive Huffman coding is well suited for:

- Streaming data sources
- Real-time communication
- Situations where full statistics cannot be collected in advance

However, it may be less suitable when symbol distributions are known and stable, or when computational simplicity is critical.

Summary

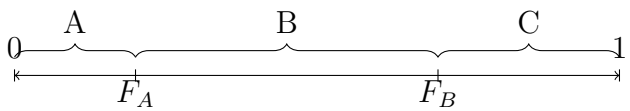
Adaptive Huffman coding extends classical Huffman coding to dynamic environments. By updating the coding model incrementally and maintaining strict structural invariants, it enables efficient, single-pass compression without prior knowledge of source statistics.

3.7 Arithmetic Coding: The Paradigm Shift

Definition

Arithmetic Coding: Encodes an entire message into a single fractional number in the interval $[0, 1)$, approaching the entropy bound very closely.

Core Idea: Represent messages as subintervals of $[0, 1)$:



Algorithm 1 Arithmetic Encoding Algorithm

Require: Message $m = s_1 s_2 \ldots s_k$, symbol probabilities p_i
Ensure: Final interval $[low, high)$

```
low ← 0.0, high ← 1.0
for each symbol s in m do
    range ← high − low
    high ← low + range × F_s          ▷ F_s: cumulative prob up to s
    low ← low + range × F_{s−1}       ▷ F_{s−1}: cumulative prob before s
end forreturn any number in [low, high)
```

Example

Example: Encode message "CAB" with probabilities: A(0.5), B(0.25), C(0.25)

Symbol	Probability	Cumulative
A	0.5	0.5
B	0.25	0.75
C	0.25	1.0

Encoding:

1. Start: $[0, 1)$
2. Process 'C': $[0.75, 1.0)$ (C occupies $[0.75, 1.0)$)
3. Process 'A': $range = 0.25$
   - $low = 0.75 + 0.25 \times 0.0 = 0.75$

- $high = 0.75 + 0.25 \times 0.5 = 0.875$
- New interval: $[0.75, 0.875)$

4. Process 'B': $range = 0.125$
   - $low = 0.75 + 0.125 \times 0.5 = 0.8125$
   - $high = 0.75 + 0.125 \times 0.75 = 0.84375$
   - Final interval: $[0.8125, 0.84375)$

Output: Any number in $[0.8125, 0.84375)$, e.g., 0.8125 in binary

Important

Practical Implementation Issues:

- **Finite precision**: Use integer arithmetic with scaling
- **Renormalization**: Output bits when interval confined to one half
- **Carry-over**: Handle when interval spans midpoint
- **Termination**: Need special end-of-message symbol

Adaptive Arithmetic Coding: Easier than adaptive Huffman - just update probabilities as you go!

3.8 Comparison & Synthesis

Method	**Optimal?**	**Adaptive?**	**Complexity**	**Near Entropy?**	**Key Applications**
Shannon Coding	No	No	Low	No	Theoretical proofs
Shannon-Fano	No	No	Low	Sometimes	Historical
Huffman	Yes*	No	Low	Moderate	General purpose
Canonical Huffman	Yes*	No	Low	Moderate	DEFLATE, JPEG, PNG
Adaptive Huffman	Yes*	Yes	Medium	Moderate	Early compressors
Arithmetic Coding	**Near-opt**	**Yes**	**High**	**Yes (close)**	**JPEG2000, H.264, HEVC**

*Optimal for symbol-by-symbol coding given probabilities

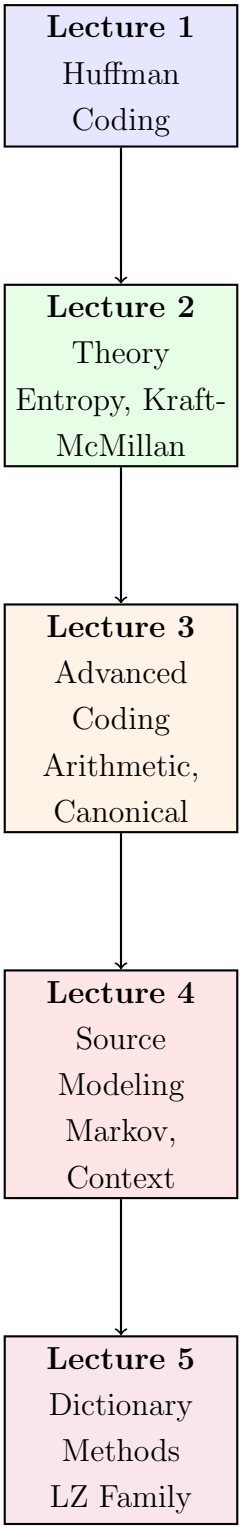
Important

Key Insights:

- **Huffman vs. Arithmetic**: Huffman is simpler but has an "integer penalty"; Arithmetic approaches entropy bound
- **Modern standards**: Arithmetic coding (CABAC) used in video compression for 10-20% better compression
- **Practical choice**: For general compression, Canonical Huffman (DEFLATE); for media, Arithmetic coding
- **The missing piece**: All these methods assume we have good probability estimates. Where do those come from?

3.9 Forward Look

The Complete Picture: What Comes Next?



Next Lecture: Source Modeling and Statistical Dependence

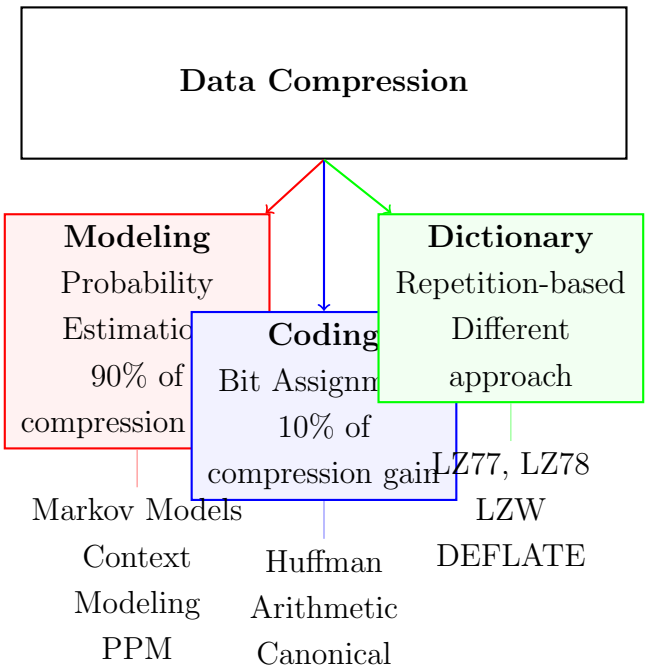
- **The missing half**: We now know how to code efficiently, but where do the probabilities come from?
- **Real data has memory**: 'Q' is usually followed by 'U' in English text
- **Markov models**: Capturing dependencies between symbols

- **Context modeling**: Using past symbols to predict future ones
- **The modeling-coding separation**: Modern compressors separate these two tasks

The Big Question for Next Time:

> **If Arithmetic coding can get within 0.01 bits of entropy, what's the real limit to compression?**
> **The answer: It's not the coding, it's the *modeling*!**

The Two Pillars of Compression (Revised View):



Exercise 3.0

Exercise 3.1: Given symbols with probabilities: A(0.4), B(0.3), C(0.2), D(0.1)

(a) Construct a Shannon code and compute its expected length

(b) Construct a Shannon-Fano code

(c) Compare with Huffman code from Lecture 2

Exercise 3.1

Exercise 3.2: Convert the following Huffman code to canonical form:

Symbol	Huffman Code
A	0
B	100
C	101
D	110
E	1110
F	1111

Exercise 3.2

Exercise 3.3: Encode the message "ABAC" using arithmetic coding with probabilities: A(0.6), B(0.3), C(0.1). Show each step.

Exercise 3.3

Exercise 3.4: Thinking Ahead: Consider the English phrase "THE QUICK BROWN FOX"

(a) If we use Huffman coding with letter frequencies, what's wrong with this approach?

(b) How might knowing that 'Q' is usually followed by 'U' help compression?

(c) Why would arithmetic coding be better than Huffman for this kind of data?

Exercise 3.4

Exercise: 3.5 Given code lengths: A=2, B=2, C=3, D=3, E=3, F=4

1. Sort the symbols properly
2. Compute the canonical codes
3. Verify no code is a prefix of another
4. What happens if you don't sort by length?

End of Lecture 3 – Advanced Entropy Coding Methods

Next: Lecture 4 – Source Modeling and Statistical Dependence

We now have efficient coding methods. Next: Where do the probabilities come from?

4: Lecture 4: Source Modeling and Statistical Dependence

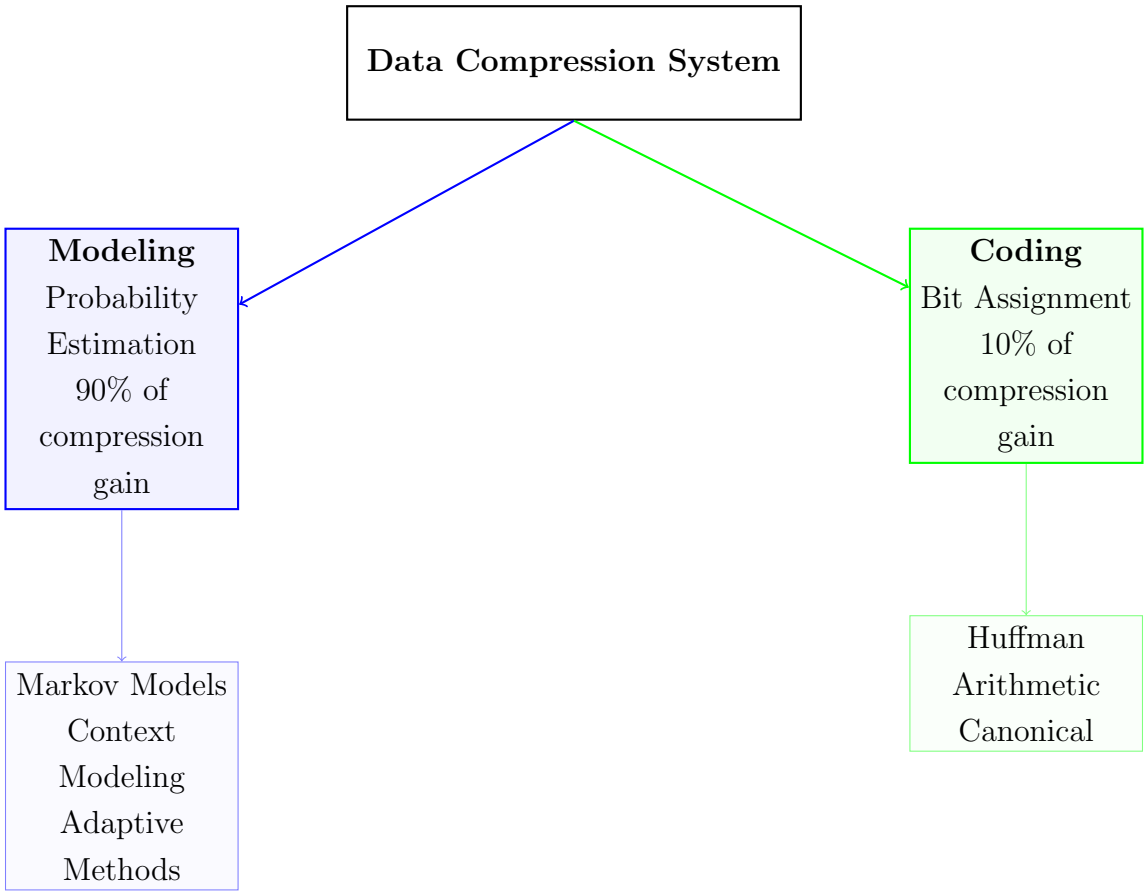
Lecture 4: Beyond Coding – The Power of Source Modeling

4.1 Introduction: Beyond Coding

Important

Key Insight: In the first three lectures, we've focused on **coding** - efficient ways to represent symbols given their probabilities. Now we address the other half: **modeling** - how to get good probability estimates in the first place.

The Big Picture:



Why Real Data Defies IID Assumptions:

- **IID (Independent Identically Distributed)**: Assumption behind simple Huffman
- **Reality**: Data has **memory** and **dependencies**
- Example: In English text, 'Q' is almost always followed by 'U'

- Example: In images, neighboring pixels are highly correlated

Today's Roadmap:

1. Understand statistical dependence in data
2. Learn Markov models for capturing memory
3. Explore context modeling techniques
4. See practical examples with real data
5. Connect modeling to coding (Lecture 3)

4.2 Memoryless vs. Sources with Memory

Definition

Memoryless Source (IID): Each symbol is generated independently of all previous symbols. Probability distribution: $P(X_n = x) = p(x)$ for all n.

Definition

Source with Memory: The probability of a symbol depends on previous symbols. Example: $P(X_n = x | X_{n-1} = y, X_{n-2} = z, \dots)$.

Example

Examples of Dependence in Real Data:

- **Text**: 'TH' is common, 'TQ' is rare
- **Images**: Neighboring pixels have similar colors
- **Audio**: Sound waves have temporal continuity
- **Video**: Consecutive frames are nearly identical
- **Source code**: Keywords, variable names repeat

Important

Measuring Dependence: Autocorrelation

$$\rho(k) = \frac{\mathbb{E}[(X_t - \mu)(X_{t+k} - \mu)]}{\sigma^2}$$

where k is the lag. High $\rho(k)$ means strong dependence at distance k.

4.3 Conditional Entropy and Mutual Information

Definition

Conditional Entropy: The average uncertainty about X given knowledge of Y:

$$H(X|Y) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log_2 p(x|y)$$

Example

Intuition: "Knowing the Past Helps Predict the Future"
Consider English letters:

- Unconditional: $H(\text{letter}) \approx 4.07$ bits
- Given previous letter: $H(\text{letter}|\text{previous}) \approx 3.36$ bits
- Given previous 2 letters: $H(\text{letter}|\text{previous } 2) \approx 2.77$ bits

Each additional letter of context reduces uncertainty!

Definition

Mutual Information: Measures how much knowing Y reduces uncertainty about X:

$$I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

Example

Worked Example: English Letter Dependence
Let X = current letter, Y = previous letter.

$$\begin{aligned} H(X) &= 4.07 \text{ bits} \\ H(X|Y) &= 3.36 \text{ bits} \\ I(X;Y) &= 4.07 - 3.36 = 0.71 \text{ bits} \end{aligned}$$

This means knowing the previous letter gives us 0.71 bits of information about the current letter.

4.4 Markov Sources

Definition

Markov Property (Memory-m): The future depends only on the last m symbols:

$$P(X_n = x | X_{n-1}, X_{n-2}, \ldots, X_1) = P(X_n = x | X_{n-1}, \ldots, X_{n-m})$$

First-Order Markov Model ($m = 1$):

- Only the immediately previous symbol matters
- Represented by transition probabilities $p_{ij} = P(X_n = j | X_{n-1} = i)$
- Can be shown as a state diagram or transition matrix

Example

Example: Weather Prediction Markov Chain

States: $\{\text{Sunny (S)}, \text{Rainy (R)}\}$

Transition probabilities:

	S	R
S	0.8	0.2
R	0.3	0.7

Interpretation: If today is sunny, 80% chance tomorrow is sunny, 20% chance rainy.

Higher-Order Markov Models:

- Order-k: Depends on last k symbols
- More accurate but exponentially more parameters
- Number of parameters grows as $|\mathcal{A}|^{k+1}$ where $\mathcal{A}$ is alphabet size

Important

Memory-Complexity Trade-off:

- **Order 0**: 26 parameters for English (simple but weak)
- **Order 1**: $26 \times 26 = 676$ parameters
- **Order 2**: $26 \times 26 \times 26 = 17,576$ parameters
- **Order 5**: $26^6 \approx 308$ million parameters!

This is the **context explosion problem**.

4.5 Entropy Rate Revisited

Definition

Entropy Rate of a Stationary Source:
For a stationary stochastic process $\mathcal{X} = (X_1, X_2, \dots)$, the entropy rate is:

$$H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} H(X_1, X_2, \dots, X_n)$$

Equivalently (and equal for stationary processes):

$$H(\mathcal{X}) = \lim_{n\to\infty} H(X_n \mid X_1, \dots, X_{n-1})$$

Special case: Stationary Markov chain of order m
If $\mathcal{X}$ is a stationary m-th order Markov chain, then for all $t > m$:

$$H(X_t \mid X_1, \dots, X_{t-1}) = H(X_t \mid X_{t-m}, \dots, X_{t-1})$$

and this conditional entropy is constant over time. Therefore:

$$H(\mathcal{X}) = H(X_{m+1} \mid X_1, \dots, X_m)$$

No limit needed — the conditional entropy stabilizes immediately.

Example

Example: First-Order Markov Chain ($m = 1$)
Step 1: Define the model Two weather states: S (sunny) and R (rainy).
Transition probabilities (tomorrow's weather depends only on today's):
Let X_t be the weather on day t. Then:

$$P(X_{t+1} = S \mid X_t = S) = 0.8, \qquad P(X_{t+1} = R \mid X_t = S) = 0.2$$
$$P(X_{t+1} = S \mid X_t = R) = 0.3, \qquad P(X_{t+1} = R \mid X_t = R) = 0.7$$

In matrix form (rows = current state, columns = next state):

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} \qquad \begin{array}{l} \text{Row 1: } X_t = S \\ \text{Row 2: } X_t = R \end{array}$$

Step 2: Find stationary distribution π Stationary distribution $\pi = [\pi_S, \pi_R]$ satisfies $\pi P = \pi$, where $\pi_S = P(X_t = S)$ and $\pi_R = P(X_t = R)$ in the long run.

$$\begin{cases} \pi_S = 0.8\pi_S + 0.3\pi_R \\ \pi_R = 0.2\pi_S + 0.7\pi_R \\ \pi_S + \pi_R = 1 \end{cases}$$

Solving (from first equation):

$$\pi_S - 0.8\pi_S = 0.3\pi_R \Rightarrow 0.2\pi_S = 0.3\pi_R \Rightarrow \pi_S = 1.5\pi_R$$

$$1.5\pi_R + \pi_R = 1 \Rightarrow 2.5\pi_R = 1 \Rightarrow \pi_R = 0.4$$

$$\pi_S = 1.5 \times 0.4 = 0.6$$

$$\pi = [\pi_S, \pi_R] = [0.6, 0.4]$$

Step 3: Compute conditional entropies *Uncertainty about tomorrow given today's weather:*
Given $X_t = S$: Tomorrow's distribution is $[0.8, 0.2]$

$$H(X_{t+1} \mid X_t = S) = -0.8 \log_2 0.8 - 0.2 \log_2 0.2 \approx 0.7219 \text{ bits}$$

Given $X_t = R$: Tomorrow's distribution is $[0.3, 0.7]$

$$H(X_{t+1} \mid X_t = R) = -0.3 \log_2 0.3 - 0.7 \log_2 0.7 \approx 0.8813 \text{ bits}$$

Step 4: Compute entropy rate For a stationary first-order Markov chain:

$$H(\mathcal{X}) = H(X_{t+1} \mid X_t) = \sum_{s \in \{S,R\}} P(X_t = s) \cdot H(X_{t+1} \mid X_t = s)$$

With $P(X_t = s) = \pi_s$ (stationarity):

$$H(\mathcal{X}) = \pi_S \cdot H(X_{t+1} \mid X_t = S) + \pi_R \cdot H(X_{t+1} \mid X_t = R)$$

$$= 0.6 \times 0.7219 + 0.4 \times 0.8813 = 0.43314 + 0.35252 = 0.78566 \text{ bits}$$

$$\approx 0.786 \text{ bits per day}$$

Important

What This Means for Data Compression:
Two approaches to compression:

1. **IID (Independent Identically Distributed) assumption:**

- Treat each symbol as independent of previous ones
- Best achievable rate: $H(X)$ (marginal entropy)
- Example: English letters ≈ 4.07 bits/letter

2. **With modeling (using context):**
   - Account for dependencies between symbols (Markov structure)
   - Use conditional probabilities based on previous symbols
   - Best achievable rate: $H(\mathcal{X})$ (entropy rate)
   - Example: English text ≈ 2.3 bits/letter

Compression gain:

$$\frac{H(X) - H(\mathcal{X})}{H(X)} \approx \frac{4.07 - 2.3}{4.07} \approx 43\%$$

Up to $\sim 45\%$ better compression by exploiting dependencies!

4.6 Context Modeling in Practice

Definition

Context Modeling: Maintain separate probability distributions for each possible context (history).

Fixed-Length vs. Variable-Length Contexts:

- **Fixed-length**: Always use last k symbols as context
- **Variable-length**: Use longest matching context in database
- Example: PPM (Prediction by Partial Matching) uses variable-length

Example

The Context Explosion Problem:
For English text (26 letters + space):

Order	Contexts	Parameters
0	1	27
1	27	729
2	729	19,683
3	19,683	531,441
4	531,441	14,348,907
5	14,348,907	387,420,489

By order 5: 387 million parameters need estimation!

Solutions to Context Explosion:

1. **Escaping**: Fall back to lower-order model when context unseen
2. **Blending**: Combine predictions from different order models
3. **Pruning**: Remove low-frequency contexts
4. **Adaptive methods**: Update probabilities as data arrives

4.7 Case Study: Text Compression Modeling

Example

1. Entropy Estimates for English Text (Theoretical Limits)

Model/Estimation Method	Bits/Letter	Year/Source
Letter frequencies only (Order-0)	4.07	Shannon 1948
+ 1st order Markov (bigrams)	3.36	
+ 2nd order (trigrams)	2.77	
+ 3rd order	2.43	
Human prediction experiment	1.3	Shannon 1951
Modern best estimates	1.0–1.5	Various studies

2. Practical Compression Performance

Compressor/Method	Bits/Letter	Notes
gzip (LZ77 + Huffman)	2.5–3.0	Fast, widely used
bzip2 (BWT + MTF + Huffman)	2.3–2.8	Better but slower
PPMd	2.1–2.4	Context modeling
PAQ variants	1.5–1.8	State of the art, very slow

Key insight: Better models (using more context) $\rightarrow$ lower entropy $\rightarrow$ better compression.
But practical compressors face tradeoffs: memory, speed, and model complexity.

PPM (Prediction by Partial Matching):

- Uses **variable-length** contexts
- Tries highest-order model first
- Escapes to lower order if context unseen
- Blends probabilities from different orders
- State-of-the-art for text compression in 1990s

Important

Practical Entropy Reduction:

$$\begin{aligned} \text{IID model (Huffman)} &: 4.07 \text{ bits/letter} \\ \text{With context modeling} &: 2.23 \text{ bits/letter} \\ \text{Improvement} &: 45\% \text{ better compression!} \end{aligned}$$

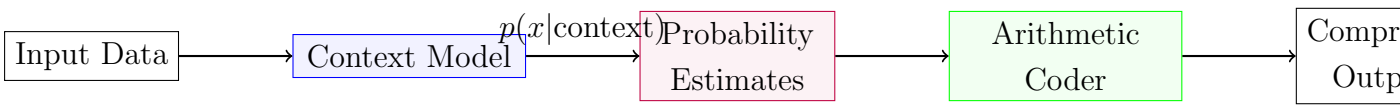
4.8 The Modeling–Coding Separation Principle

Definition

Modeling–Coding Separation: Modern compressors separate probability estimation (modeling) from bit assignment (coding).

Historical Evolution:

- **Early**: Integrated (Huffman builds tree from frequencies)
- **Modern**: Separated (Model → Probabilities → Arithmetic Coder)



Example

How PPM + Arithmetic Beats Huffman:

1. **PPM**: Sees context "TH" → predicts E with 80% probability
2. **Arithmetic**: Encodes E using $p = 0.8 \rightarrow$ 0.32 bits
3. **Huffman**: Would need at least 1 bit for any symbol

4. **Gain**: 0.32 bits vs 1+ bits = 3× better for this symbol!

Important

Why Arithmetic Coding is the Perfect Backend:

- Can handle **fractional bits** per symbol
- Accepts **changing probabilities** symbol by symbol
- Works with **adaptive models** naturally
- Achieves entropy bound for good models

4.9 Adaptive vs. Static Modeling

Definition

Static Models: Train once on representative data, use fixed model for all files.

- **Pros**: Fast encoding/decoding
- **Cons**: Model may not match specific file
- **Example**: Early text compressors using English statistics

Definition

Semi-Adaptive (Two-Pass): First pass: collect statistics; Second pass: encode.

- **Pros**: Tailored to specific file
- **Cons**: Need to transmit model (overhead)
- **Example**: Standard Huffman with tree transmission

Definition

Fully Adaptive (One-Pass): Update model while encoding/decoding.

- **Pros**: No model transmission, adapts to local changes
- **Cons**: Slower, initial poor compression
- **Example**: Adaptive Huffman, PPM with update

Example

Comparison in Practice:

Type	Compression	Speed	Memory
Static	Medium	Fast	Low
Semi-Adaptive	Good	Medium	Medium
Fully Adaptive	Best	Slow	High

Choice depends on application constraints!

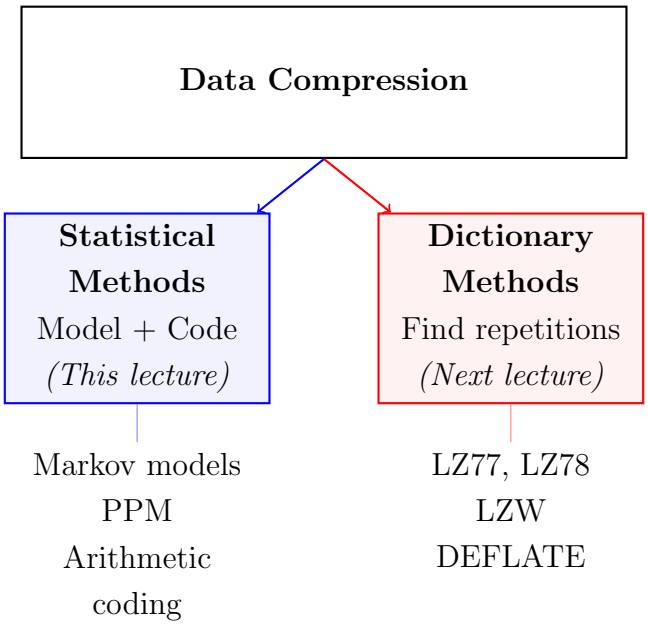
4.10 Summary and Forward Look

Important

Key Takeaways:

1. **The real compression is in modeling**, not just coding
2. **Context matters**: Using past symbols reduces uncertainty
3. **Markov models** capture memory in data
4. **Context explosion** limits practical model order
5. **Arithmetic coding** enables efficient use of good models
6. **Adaptive methods** avoid model transmission overhead

The Two Pillars Revisited:



Preview: Next Lecture on LZ Family (Dictionary Methods):

- A completely different approach: find repeating patterns
- No probability estimation needed
- Works well for files with exact repetitions
- Used in ZIP, GIF, PDF, and many modern formats
- Often combined with statistical methods in practice

Exercise 4.0

Exercise 4.1: Given the first-order Markov chain for weather:

	S	R
S	0.7	0.3
R	0.4	0.6

a) Find the stationary distribution $\pi = [\pi_S, \pi_R]$ b) Calculate the entropy rate $H(\mathcal{X})$ c) How does this compare to an IID source with $P(S) = 0.55, P(R) = 0.45$?

Exercise 4.1

Exercise 4.2: Consider the text fragment: "THE CAT SAT ON THE MAT" a) Build an order-1 (bigram) model for this text b) Calculate $H(X)$ (order-0 entropy) c) Calculate $H(X|Y)$ where Y is previous letter (order-1 conditional entropy) d) How much mutual information exists between consecutive letters?

Exercise 4.2

Exercise 4.3: Explain why arithmetic coding is better than Huffman coding when used with: a) A high-order Markov model b) An adaptive context model c) A model that gives very skewed probabilities (e.g., $p = 0.99$ for one symbol)

End of Lecture 4 – Source Modeling and Statistical Dependence

Next: Dictionary-based Compression (LZ Family)

5: Lecture 5: Dictionary-Based Compression: The Lempel–Ziv Revolution

5.1 Motivation: Hitting the Limits of Statistical Coding

Statistical coding methods such as Huffman and arithmetic coding achieve near-optimal compression *when the source statistics are known or can be accurately estimated.* However, these methods fundamentally rely on estimating probabilities over symbols or blocks of symbols.

As discussed earlier, block-based statistical coding faces a fundamental dilemma.

5.1.1 Recap: The Block Coding Dilemma

Increasing block length allows a coder to better capture dependencies between symbols and approach the entropy rate of the source. However, this comes at a steep cost:

- The number of possible blocks grows exponentially with block length.
- Estimating probabilities reliably requires exponentially more data.
- Memory and computational complexity quickly become impractical.

As a result, practical statistical coders are constrained to small contexts and local dependencies.

5.1.2 The Promise of Exploiting Long-Range Repetition

Real-world data—text, executable files, logs, DNA sequences—often contains *long-range repetition.* Patterns may repeat far apart, well beyond the reach of fixed-size statistical contexts.

Important

Statistical coding models *how often* symbols occur, but does not directly model *where long repeated patterns occur.*

This observation motivates a radically different approach to compression.

5.2 Paradigm Shift: From Statistics to Dictionaries

Instead of estimating probabilities, dictionary-based compression learns the source *by example.*

5.2.1 Core Philosophy of Dictionary Coding

Dictionary-based coders operate on a simple idea:

> *Replace repeated substrings by references to earlier occurrences.*

As the input is processed sequentially, a dictionary of previously seen substrings is constructed. Future occurrences are encoded by references into this dictionary.

Important

You can think of Lempel–Ziv methods as *learning the source structure rather than estimating probabilities.*

Crucially, the encoder and decoder process the input in exactly the same order and therefore build identical dictionaries *without transmitting the dictionary explicitly.*

5.2.2 Explicit vs. Implicit (Adaptive) Dictionaries

Dictionary-based methods fall into two categories:

- **Implicit dictionaries**: The dictionary is defined by a sliding window of recent output (e.g., LZ77, LZSS).
- **Explicit dictionaries**: The dictionary is stored and indexed explicitly (e.g., LZ78).

5.3 The Lempel-Ziv Algorithms

5.3.1 LZ77: The Sliding Window Algorithm

Published in 1977 by Abraham Lempel and Jacob Ziv, LZ77 introduced the concept of using recent history as a dictionary. It forms the foundation for many practical compression algorithms including gzip, PNG, and ZIP.

The Sliding Window Model: LZ77 maintains a sliding window divided into two parts:

- **Search buffer**: Contains the N most recently processed symbols (the "dictionary").
- **Look-ahead buffer**: Contains the next L symbols to be encoded.

The Encoding Tuple: (Offset, Length, Next Symbol): At each step, the encoder searches for the longest string in the search buffer that matches the beginning of the look-ahead buffer. It outputs a triple:

$$(\text{offset}, \text{length}, \text{next symbol})$$

5.3.2 LZSS: Improving Practical Efficiency

Published in 1982 by James Storer and Thomas Szymanski, LZSS addresses a key inefficiency in LZ77.

Flag Bits: Literal vs. Match: LZSS introduces a **flag bit** preceding each encoded element:

- **Flag = 0**: Next item is a literal symbol (8 bits for ASCII).
- **Flag = 1**: Next item is a match pair (offset, length).

Match Threshold: LZSS introduces a **match threshold**: only encode matches that are *long enough* to actually save bits.

5.3.3 LZ78: The Dictionary Growth Algorithm

Published in 1978, LZ78 takes a different approach: it builds an explicit dictionary that grows as encoding proceeds.

Encoding Pairs: (Dictionary Index, New Symbol): LZ78 outputs pairs:

$$(\text{index}, \text{symbol})$$

where the new phrase (dictionary[index] + symbol) is added to the dictionary.

5.3.4 LZW (Lempel-Ziv-Welch)

LZW (1984) is a popular LZ78 variant used in GIF, TIFF, and UNIX compress.

LZW Key Features:

- **No explicit next symbol**: Dictionary entries are added *before* they're used.
- **Fixed-width codes**: Typically 12-bit codes (4096 dictionary entries).
- **Adaptive dictionary**: Starts with all single symbols (256 entries for bytes).

5.4 Theoretical Properties: Why Lempel–Ziv Works

5.4.1 The Universality Principle

Lempel–Ziv algorithms have a remarkable theoretical property:

Definition

Universal Compression: An algorithm is universal if it can compress any stationary ergodic source to its entropy rate *without prior knowledge* of the source statistics.

Both LZ77 and LZ78 are universal compressors.

5.4.2 Asymptotic Optimality

Theorem 5.1 (Ziv & Lempel, 1977-1978)**.** *For any stationary ergodic source with entropy rate H, let L_n be the length of the LZ77 or LZ78 encoding of n source symbols. Then:*

$$\lim_{n\to\infty} \frac{\mathbb{E}[L_n]}{n} = H$$

5.5 Bridging Paradigms: Dictionary vs. Entropy Coding

5.5.1 Complementary Approaches

Dictionary coding and entropy coding address different aspects of compression:

- **Dictionary coding**: Exploits *structural redundancy* (repetitions).
- **Entropy coding**: Exploits *statistical redundancy* (skewed symbol frequencies).

5.5.2 The Hybrid Approach

Modern compressors combine both approaches. The DEFLATE algorithm (used in ZIP, PNG, gzip) works this way:

1. LZSS finds repeated strings, outputs literals and matches.
2. Huffman coding compresses:
    - Literal symbols (0-255)
    - Match lengths (257-285)
    - Match distance codes (0-29)

5.6 Summary and Forward Look

5.6.1 Key Takeaways

- **Paradigm shift**: Dictionary coding exploits repetition rather than probabilities.
- **LZ77**: Sliding window approach, encodes matches as triples.
- **LZSS**: Practical improvement with flag bits and match thresholds.
- **LZ78**: Explicit dictionary growth, encodes pairs.
- **Universality**: LZ methods work on any stationary source without prior knowledge.
- **Modern practice**: Hybrid systems (LZSS + Huffman) dominate.

End of Chapter Exercises

Exercise 5.0

Exercise 5.1 (LZ77 Encoding)
Encode the string `TOBEORNOTTOBEORTOBEORNOT` using LZ77 with a search buffer of size 12. Show your work step by step with the search buffer, look-ahead buffer, and output tuples.

Exercise 5.1

Exercise 5.2 (LZSS Efficiency Analysis)
Compare LZ77 and LZSS for encoding a sequence with parameters: 8-bit symbols, 12-bit offsets, 4-bit lengths.

(a) How many bits does LZ77 use for a non-match?

(b) How many bits does LZSS use for a literal?

(c) What is the minimum match length that saves bits in LZSS?

(d) When would LZ77 be more efficient than LZSS?

Exercise 5.2

Exercise 5.3 (LZ78 Dictionary Growth)
Encode the string `ABABABABA` using LZ78. Show the dictionary after each step and the complete output sequence. How many dictionary entries are created?

Exercise 5.3

Exercise 5.4 (Algorithm Comparison)
For each string below, predict which would perform better: LZ77/LZSS or LZ78? Explain why.

(a) `AAAAAAAAAAAAAAAAAAAA` (20 A's)

(b) `ABCDEFGHIJKLMNOPQRSTUVWXYZ`

(c) `ABABABABABABABABABAB`

(d) A file containing the same 1KB block repeated 100 times

Exercise 5.4

Exercise 5.5 (Decoding Challenge)
Decode the following LZSS encoded sequence (match threshold = 3):
`0 T`

```
0 H
0 E
1 (3,5)
0 _
0 _
1 (5,4)
0 .
```

Where: - Each line represents one encoded element - Flag 0 indicates a literal character - Flag 1 indicates a match (offset, length) - _ represents space character - . represents period

What is the original string? Show your decoding step by step.

Exercise 5.5

Exercise 5.6 (Window Size Analysis)

Consider LZ77 with search buffer size N.

(a) How many bits are needed to encode an offset value?

(b) If we increase N from 4096 to 65536, how many more bits are needed for offsets?

(c) What is the trade-off in choosing N?

(d) In practice, why might we choose $N = 32768$ over $N = 65536$ even if memory is available?

Exercise 5.6

Exercise 5.7 (Universal Compression Intuition)

Explain in your own words why Lempel–Ziv algorithms are universal. Why don't they need to know source statistics in advance? How do they "learn" the source structure?

Exercise 5.7

Exercise 5.8 (Hybrid Coding Design)

Design a simple hybrid compressor that:

(a) Uses LZSS with match threshold 3

(b) Collects statistics on literals, lengths, and offsets

(c) Applies Huffman coding to each alphabet separately

How would the decoder know which Huffman trees to use?

Exercise 5.8

Exercise 5.9 (Search Efficiency)
A naive LZ77 implementation searches for matches by comparing the look-ahead buffer against every position in the search buffer.

(a) What is the time complexity of this approach?

(b) Describe how a hash table can improve search efficiency.

(c) What information would you store in the hash table?

(d) How would you handle hash collisions?

Exercise 5.9

Exercise 5.10 (Theoretical Limits)
Prove or provide intuition for:

(a) LZ77 cannot achieve compression ratio better than the entropy rate for i.i.d. sources.

(b) LZ78 may perform poorly on very short inputs.

(c) For sources with long-range dependencies, LZ methods can outperform block Huffman coding even with large blocks.

6: Lecture 6: DEFLATE Algorithm: The Standard in Practice

6.1 Introduction: The Universal Compression Standard

The DEFLATE compression algorithm, specified in RFC 1951 (1996), is arguably the most widely deployed compression algorithm in history. Its success lies not in being the theoretically optimal algorithm, but in striking an exceptional balance between compression ratio, speed, and implementation simplicity.

Definition

DEFLATE: A lossless data compression algorithm combining LZ77-style string matching with Huffman coding. It is the compression engine behind gzip, ZIP, PNG, and HTTP compression.

6.1.1 Where DEFLATE Is Used: gzip, ZIP, PNG, HTTP

- **gzip**: The Unix compression utility (RFC 1952)
- **ZIP file format**: Phil Katz's PKZIP format (1989), later standardized
- **PNG image format**: Replaced GIF, uses DEFLATE on filtered scanlines
- **HTTP compression**: Content-Encoding: gzip/deflate in web traffic
- **PDF documents**: FlateDecode filter for stream compression
- **Java JAR files**: Based on ZIP format

6.1.2 Historical Context: Phil Katz, ZIP Format, and Open Standards

The story of DEFLATE is intertwined with the "compression wars" of the late 1980s-1990s:

- **1985**: Terry Welch's LZW algorithm patented (used in GIF)
- **1986**: Phil Katz creates PKARC (based on ARC format)
- **1989**: Katz releases PKZIP with proprietary compression
- **1991**: gzip created as free alternative to UNIX compress (which used LZW)
- **1993**: Unisys enforces LZW patents, creating demand for patent-free alternatives
- **1996**: DEFLATE specification published as RFC 1951

The key innovation was making DEFLATE specification **open** and **royalty-free**, leading to widespread adoption.

6.2 High-Level Architecture: A Two-Stage Pipeline

DEFLATE employs a classic transform-coding approach:

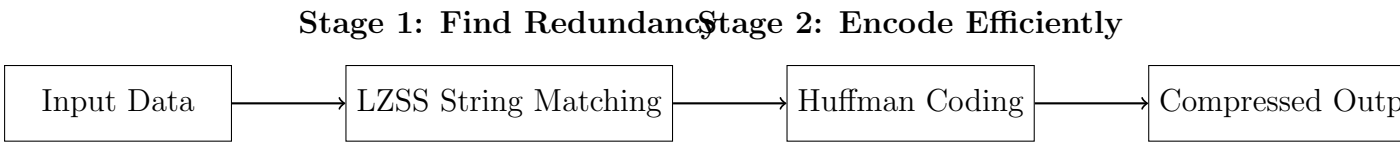


Figure 1: DEFLATE's two-stage compression pipeline

6.2.1 Stage 1: LZSS String Matching (Finding Redundancy)

The LZSS (Lempel-Ziv-Storer-Szymanski) stage identifies repeated sequences in the data:

- Scans input with a **sliding window** (32KB history buffer)
- For each position, looks for longest match in recent history
- Outputs either:
  - **Literal byte**: When no good match found
  - **(Length, Distance) pair**: When match found

Example: For the string "the cat sat on the mat":

- First occurrence of "the": output as literals 't','h','e'
- Second occurrence of "the": output as (length=3, distance=15)

6.2.2 Stage 2: Huffman Coding (Encoding the Reduced Data)

The Huffman stage compresses the output from Stage 1:

- Takes stream of literals and (length,distance) pairs
- Builds **two Huffman trees**:
  - One for literals and lengths (combined alphabet)
  - One for distances
- Uses **canonical Huffman codes** for compact representation

6.2.3 Why the Hybrid Approach Works: Transforming Redundancy

The genius of DEFLATE is in how it **transforms redundancy**:

1. **LZSS finds structural redundancy**: Repeated sequences become (length,distance) pairs
2. **Huffman finds statistical redundancy**: Frequent symbols get shorter codes
3. Together they exploit **both types** of redundancy in data

Important

Key Insight: LZSS is excellent at finding *long-range* repetition (phrases, patterns). Huffman is excellent at compressing *skewed distributions*. Together they handle both English text (skewed letter frequencies + repeated words) effectively.

6.2.4 Design Philosophy: Fast Decoding Over Maximum Compression

DEFLATE was designed with clear priorities:

Priority	Implementation Choice
Fast decoding	Canonical Huffman (O(1) decode)
Memory efficiency	32KB sliding window (tiny by today's standards)
Implementation simplicity	Two independent stages
Streaming capability	Independent compression blocks
Acceptable compression	"Good enough" ratio for most data

This "fast decode" philosophy made DEFLATE ideal for distribution formats (ZIP, PNG) where files are compressed once but decompressed many times.

6.3 LZSS in DEFLATE: Detailed Implementation

6.3.1 Sliding Window: 32KB History and Lookahead Buffer

DEFLATE's LZSS uses specific, fixed parameters:

- **History buffer**: 32,768 bytes (2^{15}) of recently processed data
- **Lookahead buffer**: 258 bytes maximum match length
- **Minimum match length**: 3 bytes (shorter matches not worth encoding)

Example

Why 32KB?

- 32KB was a practical limit in 1990s memory-constrained systems
- Larger windows provide diminishing returns for most data
- Cache-friendly size for modern processors
- Still adequate for most short-to-medium range repetitions

6.3.2 Hash Chains for Fast Match Finding

Instead of brute-force searching 32KB for matches, DEFLATE uses a hash table:

Algorithm 2 DEFLATE Match Finding with Hash Chains

```
procedure FindBestMatch(current_position, data)
    hash ← hash(data[current_position : current_position + 3])
    best_length ← 0
    best_distance ← 0
    for each prev_pos in hash_table[hash] do
        if current_pos − prev_pos > 32768 then          ▷ Outside window
            continue
        end if
        length ← MatchLength(prev_pos, current_pos)
        if length > best_length then
            best_length ← length
            best_distance ← current_pos − prev_pos
        end if
    end for
    return (best_length, best_distance)
end procedure
```

The hash is typically computed from the first 3 bytes of the potential match. Hash chains link all positions with the same 3-byte prefix.

6.3.3 Lazy Matching and Greedy Heuristics

DEFLATE uses sophisticated heuristics to find better matches within the sliding window constraints:

- **Greedy parsing**: Immediately take the longest match found at the current position

- **Lazy matching**: Consider outputting a literal instead, hoping for a better match starting at the next position

Important

Window Constraints in Lazy Matching:

- All matches must reference data within the 32KB search buffer
- The lookahead buffer contains data to be encoded (maximum 258 bytes)
- Lazy matching decides: "match now" vs. "literal now + potentially better match next"

Example

Lazy Matching Example Where Lazy Parsing Wins

Consider the input data:

`aaaaabaaaaa`

Indexing the data:

0	1	2	3	4	5	6	7	8	9	10
a	a	a	a	a	b	a	a	a	a	a

Assume that the first six bytes have already been processed. The sliding window state is:

a	a	a	a	a	b	a	a	a	a	a
Search Buffer						Lookahead Buffer				

Search buffer: ``aaaaab'' (positions 0–5)
Lookahead buffer: ``aaaaa'' (positions 6–10)
Decision point: position 6 (first 'a' in lookahead)

- **Greedy parsing**:
  - Find longest match ≥ 3 starting at position 6
  - ``aaaaa'' matches ``aaaaa'' starting at position 1 in search buffer
  - Match length = 5, distance = 5
  - Greedy immediately emits: (distance=5, length=5)
  - Covers 5 bytes
- **Lazy parsing**:
  - At position 6: Finds match length = 5, distance = 5

- Checks next position (position 7)
- Starting at position 7: ``aaaa'' (4 bytes in lookahead)
- Cannot find match longer than 5 bytes (only 4 bytes remain)
- Greedy wins - lazy doesn't help here

Example where lazy actually wins:
Consider: ``abcabcdabcde''
After processing first 3 bytes:

<table>
<tr><td>a</td><td>b</td><td>c</td><td>a</td><td>b</td><td>c</td><td>d</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td></td></tr>
<tr><td colspan="3">Search Buffer</td><td colspan="10">Lookahead Buffer</td></tr>
</table>

Search buffer: ``abc'' (positions 0-2)
Lookahead: ``abcdabcde'' (positions 3-11)
At position 4 ('a'):

- **Greedy**:
  - ``abc'' matches at position 0 (distance=3, length=3)
  - Encodes: (3,3)
  - Advances to position 6
  - Position 6: ``dabcde'' - no match ≥ 3
  - Output literals: 'd','a','b','c','d','e'
- **Lazy**:
  - At position 3: Finds ``abc'' (length=3, distance=3)
  - Checks position 4: ``bcdabcde''
  - ``bcd'' doesn't match (only ``bc'' in search buffer)
  - Length=2 < 3
  - Greedy wins

The reality: Simple examples showing dramatic lazy advantage are mathematically rare. Lazy matching typically provides small gains (1-2 bytes) in specific edge cases.

Algorithm 3 Lazy Matching with Sliding Window Constraints

```
procedure LazyMatch(search_buffer, lookahead)
    pos ← 0                                          ▷ Position within lookahead buffer
    best_match_here ← FindBestMatch(search_buffer, lookahead, pos)
    if best_match_here.length ≥ 3 then
        best_match_next ← FindBestMatch(search_buffer, lookahead, pos + 1)
        if best_match_next.length > best_match_here.length then
            Output lookahead[pos] as literal
            return pos + 1                           ▷ Move to next position
        else
            Output match (best_match_here.distance, best_match_here.length)
            return pos + best_match_here.length
        end if
    else
        Output lookahead[pos] as literal
        return pos + 1
    end if
end procedure
```

Important

Why Lazy Matching Works with Sliding Windows:

- By outputting a literal, we shift the window by 1 byte
- This adds that literal to the search buffer for future matches
- Sometimes the literal + new search buffer position enables a longer match
- The decision is: "Is sacrificing 1 byte now worth gaining several bytes later?"

Typical lazy matching optimization:

- Only consider lazy if current match length < 8 (configurable threshold)
- Check only next 1-2 positions (not all possible positions)
- Stop lazy checking if a very long match (e.g., > 128 bytes) is found

6.3.4 Output Format: (Length, Distance) Pairs vs. Literal Bytes

The LZSS stage produces a stream of symbols from two alphabets:

Type	Range	Encoding Details
Literal bytes	0–255	Direct byte values
Length codes	3–258	Base code + 0–5 extra bits for longer lengths
Distance codes	1–32768	Base code + 0–13 extra bits in logarithmic ranges
End-of-block	256	Special marker (end of DEFLATE block)

Important encoding details:

- **Lengths 3–258**: Encoded using symbols 257–285
  - For example: Length 3 = symbol 257 (no extra bits)
  - Length 10 = symbol 264 (no extra bits)
  - Length 258 = symbol 285 (no extra bits)
  - Lengths 11–258 use 1–5 extra bits for precise encoding
- **Distances 1–32768**: Encoded using symbols 0–29 with logarithmic ranges
  - Example ranges:
    * Symbol 0: Distance 1 (no extra bits)
    * Symbol 4: Distances 5–6 (1 extra bit)
    * Symbol 29: Distances 24577–32768 (13 extra bits)
  - Distance 100 would use symbol for range 97–128 with extra bits

Example

Encoding Distance 100:

1. Distance 100 falls in range 97–128
2. According to DEFLATE specification:
   - Range 97–128 corresponds to distance symbol 18
   - Base distance for this symbol: 97
   - Extra bits needed: 2 bits (since 128-97+1 = 32 possibilities)
3. Compute extra bits: 100 - 97 = 3
4. Encode as: symbol 18 followed by binary '11' (3 in 2 bits)

This logarithmic encoding allows representing 32768 possible distances with only 30 symbols plus extra bits, significantly reducing the Huffman alphabet size.

6.4 DEFLATE's Unique Huffman Coding Scheme

This is where DEFLATE shows its brilliance. Instead of storing full Huffman trees, it stores only **code lengths**.

6.4.1 Two Alphabets: Literals (0-285) and Distances (0-29)

DEFLATE uses separate Huffman trees for different symbol types:

- **Literals/Lengths alphabet**: 286 symbols
  - 0-255: Literal bytes
  - 256: End-of-block marker
  - 257-285: Length codes (with extra bits)
- **Distances alphabet**: 30 symbols
  - 0-3: Direct distances 1-4
  - 4-29: Distance codes with extra bits

6.4.2 Canonical Huffman via Code Lengths: The Brilliant Compression

Instead of storing:

- Full tree structure (large), or
- Symbol-to-code mapping (variable size)

DEFLATE stores only:

- **Code length for each symbol** (4 bits each, 19 maximum length)
- **Canonical reconstruction rules**:
  1. Sort symbols by code length
  2. Assign first code of length k as 0
  3. Each subsequent code = previous code + 1
  4. When increasing length, left-shift and add zeros

Example

Canonical Huffman Example:
Given code lengths: A=2, B=2, C=3, D=3, E=3

1. Sort by length: A(2), B(2), C(3), D(3), E(3)
2. Length 2: Start with 00

3. Assign: A=00, B=01 (00+1)
4. Length 3: Start with 100 (01¡¡1)
5. Assign: C=100, D=101, E=110

Decoder only needs code lengths to reconstruct this!

6.4.3 Run-Length Encoding of Code Lengths (19-Symbol Alphabet)

Code lengths themselves are compressed using RLE on a 19-symbol alphabet:

- 0-15: Actual code length 0-15
- 16: Repeat previous length 3-6 times (2 extra bits)
- 17: Repeat zero 3-10 times (3 extra bits)
- 18: Repeat zero 11-138 times (7 extra bits)

This compression means the Huffman tree description is itself heavily compressed!

6.4.4 Why Canonical Codes Enable O(1) Decoding

With canonical codes, decoding becomes a table lookup:

```
// Precomputed tables from code lengths
min_code[length] = smallest code of this length
max_code[length] = largest code of this length
first_symbol[length] = first symbol with this length

// Decoding loop
while (need more symbols) {
    code = peek_bits(MAX_BITS);  // Look at next bits
    for (len = 1 to MAX_BITS) {
        if (code <= max_code[len]) {
            symbol = first_symbol[len] + (code - min_code[len]);
            consume_bits(len);
            break;
        }
    }
}
```

Listing 1: Canonical Huffman Decoding (Pseudocode)

This is **O(1)** per symbol (constant-time lookup), crucial for fast decompression.

6.5 Block Structure and Streaming Format

DEFLATE supports streaming through independent compression blocks.

6.5.1 Three Block Types: Stored, Fixed Huffman, Dynamic Huffman

Type	Bits	Use Case
Stored	00	Uncompressible data
Fixed Huffman	01	Default trees (predefined)
Dynamic Huffman	10	Custom trees per block
Reserved	11	(Error)

- **Stored blocks**: Raw data with length header (for already-compressed data)
- **Fixed blocks**: Use predefined Huffman trees (no tree description overhead)
- **Dynamic blocks**: Custom optimized trees (best compression)

6.5.2 Bit-Level Layout: BFINAL, BTYPE, and Data

Each block has precise bit-level layout:

```
// Block header
BFINAL: 1 bit  (1 = last block in stream)
BTYPE:  2 bits (00, 01, or 10)

// If BTYPE == 00 (Stored block)
//    Skip to next byte boundary
//    LEN: 16 bits (block length)
//    NLEN: 16 bits (~LEN for error checking)
//    Data: LEN bytes of raw data

// If BTYPE == 01 (Fixed Huffman)
//    Compressed data using fixed trees

// If BTYPE == 10 (Dynamic Huffman)
//    HLIT: 5 bits (257-286 literal/length codes)
//    HDIST: 5 bits (1-30 distance codes)
//    HCLEN: 4 bits (4-19 code length codes)
//    Code lengths for code length alphabet (HCLEN+4 times)
//    Compressed literal/length and distance code lengths
//    Compressed data
```

Listing 2: DEFLATE Block Structure

6.5.3 Streaming Support: Independent Blocks and Reset Points

- Each block compressed independently
- History buffer **does not reset** between blocks (maintains 32KB window)
- Allows streaming compression of unlimited data
- Enables random access in some formats (ZIP seeks to file boundaries)

6.6 Step-by-Step Compression Walkthrough

Let's trace through compressing a simple English phrase.

6.6.1 Input: "The cat sat on the mat"

Assume ASCII encoding (simplified for example):

```
Position: 0123456789012345678901234 56
Text:     The cat sat on the mat
Indices:  0         1         2
```

6.6.2 LZSS Processing: Finding Matches

1. Position 0-2: "The" - No match in empty history → Output literals 'T','h','e'
2. Position 3: ' ' (space) - No match → Output literal ' '
3. Position 4-6: "cat" - No match → Output 'c','a','t'
4. Continue... Position 18-20: "the" matches position 0-2!
5. Match length = 3, distance = 18
6. Encode as (length=3, distance=18)

Final LZSS output (simplified):

```
Literals: T h e   c a t   s a t   o n   (length,distance=3,18) m a t
Symbols: 84 104 101 32 99 97 116 32 115 97 116 32 111 110 (257,18) 109 97 116
```

(Note: 257 is the symbol for length=3 in DEFLATE's encoding)

6.6.3 Building Canonical Huffman Tables

From the symbol stream, compute frequencies:

Symbol	Frequency
32 (space)	4
97 ('a')	3
116 ('t')	3
Others	1 each
257 (length=3)	1
18 (distance)	1

Build Huffman trees, then extract code lengths. For example, space (32) gets shortest code.

6.6.4 Bitstream Assembly: Final Compressed Output

1. Write block header: BFINAL=1, BTYPE=10 (dynamic)
2. Write HLIT, HDIST, HCLEN values
3. Write compressed code lengths for the 19 code-length symbols
4. Write compressed literal/length and distance code lengths
5. Write compressed data using the Huffman codes
6. Pad final byte with zeros if needed

6.7 Compression Levels and Performance Trade-offs

zlib (the reference implementation) offers 9 compression levels.

6.7.1 zlib's 9 Levels: From "fast" to "best"

Level	Strategy	Use Case
0	No compression (store only)	Testing/disabled
1	Fastest LZ77, lazy matching off	Real-time compression
6	Default: moderate search, lazy matching	General purpose
9	Full search, optimal parsing	Archival storage

Levels mainly differ in:

- Hash table size and chain length
- Lazy matching on/off
- Match selection heuristics

6.7.2 Speed vs. Ratio: Why Level 6 is the Default

The default level 6 represents the "knee" in the speed/ratio curve:

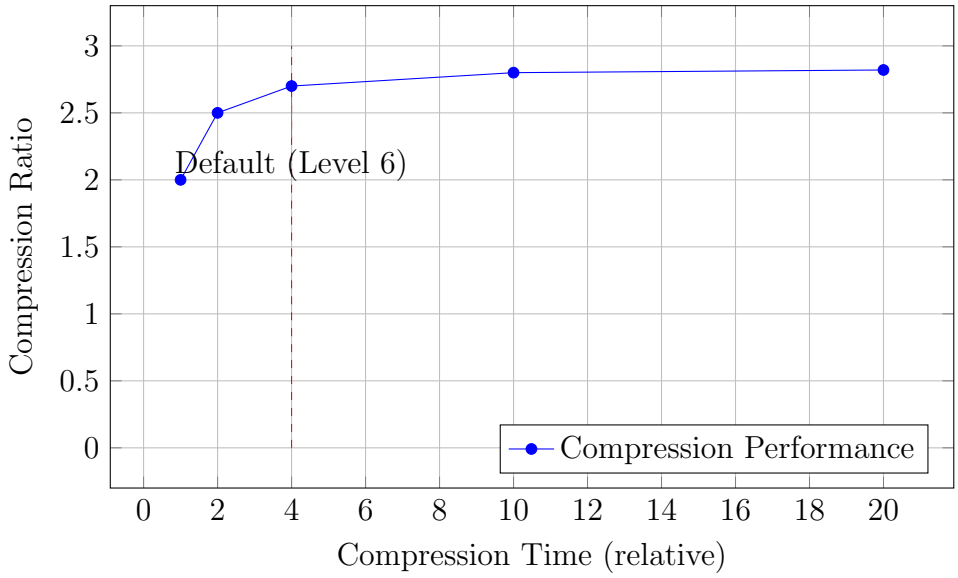


Figure 2: Trade-off between compression time and ratio in zlib

Level 9 is only ∼5% better than Level 6 but takes 2-5x longer.

6.7.3 Memory Footprint: 32KB to 256KB Windows

- **Default**: 32KB history buffer + 16KB hash table ≈ 48KB
- **Maximum**: Some implementations allow 256KB windows
- **Modern systems**: Typically use 32KB-64KB for cache efficiency

6.8 Practical Applications and File Formats

6.8.1 gzip Format: Headers, Footers, and CRC32

A gzip file contains:

```
+--------+------+------+-----+------+------+-------+------+
| Header | Extra| Name |Comment|HCRC | DEFLATE | CRC32 | ISIZE |
+--------+------+------+-----+------+------+-------+------+
  10 bytes optional optional 2    compressed  4      4
```

- **Header**: Magic bytes (0x1F, 0x8B), compression method (8=DEFLATE)
- **CRC32**: 32-bit checksum of uncompressed data
- **ISIZE**: Uncompressed size modulo 2^{32}

6.8.2 ZIP Format: Multiple Files and DEFLATE

ZIP is a container format supporting multiple compression methods:

- Each file compressed independently with DEFLATE
- Central directory at end enables random access
- Supports encryption, comments, extra fields

6.8.3 PNG: DEFLATE on Filtered Scanlines

PNG applies a **filter** to each scanline before DEFLATE:

- Subtract pixel from left pixel (Sub filter)
- Subtract pixel from above pixel (Up filter)
- Various predictors to create more compressible data
- Then DEFLATE compresses the filtered bytes

6.8.4 HTTP Compression: Accept-Encoding Header

Web servers compress responses on-the-fly:

```
Request:  GET /page.html HTTP/1.1
          Accept-Encoding: gzip, deflate

Response: HTTP/1.1 200 OK
          Content-Encoding: gzip
          [gzip-compressed body]
```

Saves 60-80% bandwidth for text content (HTML, CSS, JS).

6.9 Limitations and When DEFLATE Fails

6.9.1 Poor Performance on Already-Compressed or Random Data

- **Already compressed**: JPEG, MP3, ZIP files → DEFLATE can't compress further
- **Encrypted data**: Random-looking → No patterns to find
- **Result**: May actually *increase* size due to overhead

Good compressors detect this and use "stored" (uncompressed) blocks.

6.9.2 The "Shakespeare Problem": Global vs. Local Repetition

DEFLATE's 32KB window can't capture **global repetition**:

Example

Shakespeare's Complete Works:

- Word "the" appears thousands of times
- But only instances within 32KB of each other are linked
- Distant repetitions (across chapters) aren't captured
- Result: Compresses to ∼2 bits/byte instead of potential ∼1.5

Modern compressors (LZMA, zstd) use larger dictionaries (MBs) for this case.

6.9.3 Why Not Arithmetic Coding? Patents, Speed, and Complexity

In the 1990s, arithmetic coding was:

- **Patented**: IBM held key patents
- **Slower**: 2-10x slower than Huffman
- **Complex**: Harder to implement correctly
- **Bit-level operations**: Unfriendly to byte-oriented systems

Today, arithmetic coding is patent-free but DEFLATE's ecosystem is entrenched.

6.9.4 Security Issues: CRIME, BREACH, and Compression Side-Channels

Compression can leak information:

- **CRIME (2012)**: HTTPS compression reveals session cookies
- **BREACH (2013)**: HTTP compression reveals CSRF tokens
- **Principle**: If secret affects compressed size, attacker can deduce it
- **Solution**: Disable compression on sensitive data or use random padding

6.10 Modern Alternatives and Successors

6.10.1 LZMA and 7-Zip: Better Compression, Slower Speed

LZMA (Lempel-Ziv-Markov chain Algorithm):

- **Larger window**: Up to 4GB vs DEFLATE's 32KB
- **Range coding**: Arithmetic coding variant

- **Better compression**: 30-50% smaller than DEFLATE
- **Slower**: 5-10x slower compression
- **Used in**: 7z format, xz utilities, software installers

6.10.2 Zstandard (zstd): Facebook's Balanced Alternative

zstd (2016) aims to replace DEFLATE:

- **Faster**: 5-10x faster decompression than zlib
- **Better compression**: 10-20% better than zlib at same speed
- **Modern features**: Dictionary training, reproducible builds
- **Adoption**: Linux kernel, SquashFS, Facebook, Netflix

6.10.3 Brotli: Google's Web-Optimized Successor

Brotli (2013) targets web compression:

- **Static dictionary**: 120KB of common web strings
- **Better text compression**: 20-26% better than gzip
- **Slower compression**: But fast decompression
- **Adoption**: CDNs, web servers, browsers

6.10.4 Why DEFLATE Still Dominates: The Legacy Effect

Despite better alternatives, DEFLATE persists because:

- **Ubiquitous support**: Every system has zlib
- **Standardized**: RFC 1951, no licensing issues
- **Good enough**: For most use cases
- **Network effects**: Hard to change entrenched standards
- **Hardware support**: Some processors have DEFLATE acceleration

6.11 Hands-On Analysis and Debugging

6.11.1 Using gzip -v -l and infgen for Inspection

```
# Basic compression
gzip -9 file.txt              # Maximum compression
gzip -1 file.txt              # Fastest compression

# Analyze compressed file
gzip -l file.txt.gz           # List compression ratio
gzip -v -l file.txt.gz        # Verbose listing

# Decompress with verbose output
gzip -v -d file.txt.gz        # Show compression ratio

# Use infgen for detailed DEFLATE analysis
infgen -d file.gz             # Disassemble DEFLATE stream
```

6.11.2 Visualizing Matches and Huffman Trees

Tools for deeper analysis:

- **puff**: Educational DEFLATE implementation (readable C)
- **infgen**: DEFLATE stream disassembler
- **Custom scripts**: Parse zlib output with ZLIB_DEBUG

6.11.3 Benchmarking: Compression Ratio vs. Speed

```
# Time compression
time gzip -9 large_file.txt

# Compare compression levels
for level in {1..9}; do
    echo -n "Level $level: "
    gzip -c -$level file.txt | wc -c
done

# Benchmark suite
pigz                  # Parallel gzip implementation
zstd -b               # zstd benchmark mode
```

6.12 Summary and Key Insights

6.12.1 Why DEFLATE Dominated for 30+ Years

1. **Right trade-offs**: Fast decode over maximum compression

2. **Clever engineering**: Canonical Huffman, compact tree representation
3. **Open standard**: RFC 1951, no patents, reference implementation
4. **Good enough**: Adequate compression for most real-world data
5. **Ecosystem**: Tools, libraries, hardware support

6.12.2 Lessons for Compression Engineers: Practical Beats Perfect

- **Decode speed matters more than encode speed** for distribution formats
- **Implementation simplicity** leads to wider adoption
- **Open standards** defeat technically superior proprietary formats
- **Backward compatibility** is a powerful force
- Sometimes **"good enough" wins over "best"**

6.12.3 Looking Forward: The Future of Compression Standards

- **DEFLATE will persist** in legacy systems for decades
- **zstd and Brotli** gaining ground in new applications
- **Hardware acceleration** becoming common (Intel QAT, AWS Nitro)
- **AI/ML approaches** showing promise but not yet practical
- The next "DEFLATE" will need to be **10x better** to displace it

Important

Final Thought: DEFLATE represents a peak in practical algorithm design. It shows how understanding both theory (information theory) and practice (hardware constraints, use cases) leads to enduring solutions. Study DEFLATE not just to use it, but to learn how to design algorithms that stand the test of time.

End of Chapter Exercises

Exercise 6.0

Exercise 6.1: DEFLATE Architecture Understanding

Explain in your own words why DEFLATE uses a two-stage approach (LZSS + Huffman) instead of just one of these techniques. Consider:

(a) What types of redundancy does LZSS exploit that Huffman cannot?

(b) What types of redundancy does Huffman exploit that LZSS cannot?

(c) Why can't Huffman coding alone achieve good compression on typical text data?

(d) Why can't LZSS alone achieve optimal compression?

Exercise 6.1

Exercise 6.2: LZSS Match Analysis

Given the input string: "abracadabraabracadabra"

(a) Using a sliding window of size 12, trace through LZSS processing and show the output as a sequence of literals and (length, distance) pairs.

(b) Calculate the compression ratio if literals are 8 bits and each (length, distance) pair requires 24 bits (8 for length, 16 for distance).

(c) What would be the output if the sliding window was only size 6 instead of 12?

Exercise 6.2

Exercise 6.3: Distance Encoding Practice

Using DEFLATE's distance encoding scheme (logarithmic ranges):

(a) Encode the following distances: 1, 5, 10, 100, 1000, 20000

(b) For each, specify: symbol number, base distance, extra bits required, and final binary encoding.

(c) What is the maximum distance that can be encoded without extra bits?

(d) Why does DEFLATE use logarithmic encoding instead of direct binary representation?

Hint: Refer to RFC 1951 Section 3.2.5 for the exact distance code table.

Exercise 6.3

Exercise 6.4: Canonical Huffman Construction

Given the following symbol frequencies for a small alphabet:

Symbol	Frequency
A	40
B	30
C	15
D	10
E	5

(a) Build a Huffman tree and determine code lengths for each symbol.

(b) Generate the canonical Huffman codes from these lengths.

(c) Show the exact bit sequence DEFLATE would store to represent these code lengths.

(d) Calculate the average code length and compare it to the entropy.

Exercise 6.4

Exercise 6.5: DEFLATE Block Analysis

Consider the following DEFLATE block header (in binary):

$$1\ 10\ 01101\ 00101\ 0100$$

Where the bits represent: BFINAL (1), BTYPE (10), HLIT (5 bits), HDIST (5 bits), HCLEN (4 bits).

(a) Decode each field and explain what it means.

(b) What type of block is this?

(c) How many literal/length codes will follow?

(d) How many distance codes will follow?

(e) How many code-length codes will follow?

Exercise 6.5

Exercise 6.6: Compression Level Trade-offs

You are designing a compression system for:

(a) A web server compressing HTML pages in real-time

(b) An archival system storing historical documents

(c) A mobile app with limited CPU and battery

For each scenario:

- Recommend a DEFLATE compression level (0-9) with justification
- Explain whether you would use fixed or dynamic Huffman blocks
- Discuss any modifications you might make to the standard DEFLATE parameters

Exercise 6.6

Exercise 6.7: DEFLATE vs. Modern Alternatives

Compare DEFLATE with two modern alternatives (choose from LZMA, zstd, or Brotli):

(a) Create a comparison table covering: compression ratio, speed (encode/decode), memory usage, and typical use cases.

(b) For each of the following, recommend which compressor to use and why:

- Compressing software updates for download
- Real-time compression in a database
- Web assets on a content delivery network

(c) Explain why DEFLATE is still widely used despite newer, better algorithms.

Exercise 6.7

Exercise 6.8: Implementation Challenge

Write pseudocode for a DEFLATE decoder that:

(a) Reads the block header and identifies block type

(b) For dynamic blocks, reconstructs the Huffman trees from code lengths

(c) Decodes the compressed data using canonical Huffman decoding

(d) Handles both literals and (length, distance) pairs

Focus on the key algorithms, not edge cases or optimizations. Your pseudocode should show the overall structure and critical steps.

Exercise 6.8

Exercise 6.9: Security Analysis

The CRIME and BREACH attacks exploit compression side-channels:

(a) Explain the fundamental vulnerability that enables these attacks.

(b) Why does DEFLATE's LZSS stage make this vulnerability particularly dangerous?

(c) Propose three different strategies to mitigate this risk while still using compression.

(d) Would switching to a different compression algorithm (like zstd or Brotli) solve the problem? Why or why not?

Exercise 6.9

Exercise 6.10: Research and Reflection

(a) DEFLATE was standardized in 1996. Identify three technological constraints from that era that influenced DEFLATE's design decisions.

(b) If you were designing DEFLATE today, what would you change given modern hardware (multi-core CPUs, gigabytes of RAM, SSD storage)?

(c) Research the current state of hardware-accelerated compression (Intel QAT, AWS Nitro, etc.). How might these technologies change the optimal compression algorithm design?

(d) Predict: Will DEFLATE still be widely used in 2030? Justify your answer.

Important

Exercise Guidelines:

- **Exercises 6.1-6.5**: Core concepts – recommended for all students
- **Exercises 6.6-6.8**: Intermediate application – for deeper understanding
- **Exercises 6.9-6.10**: Advanced topics – for interested students or projects
- **Time estimates**: Basic exercises (30 min each), advanced (60-90 min each)
- **Resources**: Refer to RFC 1951, zlib source code, and infgen tool for hands-on exploration

Optional Project Challenge

Build a Simple DEFLATE Analyzer: Write a program in your preferred language that:

- Takes a gzip file as input
- Parses the gzip header and trailer
- Extracts the DEFLATE stream
- Identifies block boundaries and types
- Reports basic statistics (compression ratio, block types used, etc.)
- **Bonus**: Visualize the Huffman trees or match distances

Use existing libraries for decompression, but implement the analysis logic yourself. This project will deepen your understanding of the actual byte-level format.

7: Lecture 7: Run-Length Encoding: Exploiting Equality Redundancy

Important

Learning Objectives:

- Understand the concept of equality redundancy and why it's the simplest form of compressible pattern
- Master the basic RLE algorithm and its variants for different data types
- Analyze compression ratios, best/worst cases, and theoretical limits of RLE
- Learn practical RLE implementations used in real-world file formats
- Understand why RLE fails on some data types and how to combine it with other techniques
- Build the conceptual bridge to transform-based compression, especially BWT

7.1 Introduction and Motivation

7.1.1 The Simplest Form of Redundancy: Repetition

When we look at data through the lens of compression, we're constantly asking: "What patterns repeat?" Among all possible patterns, the absolute simplest is when the *same symbol repeats consecutively*. We call this **equality redundancy**.

Definition

Equality Redundancy: A form of data redundancy where the same symbol appears multiple times in consecutive positions within a sequence. Each such group of identical consecutive symbols is called a **run**.

Consider these examples:

- Text: `AAAABBBCCCC` contains three runs: `AAAA`, `BBB`, and `CCCC`
- Images: A blue sky photo has long runs of similar blue pixel values
- Signals: Silence in audio contains runs of zero amplitude
- Documents: White space in scanned text creates runs of white pixels

The key insight is elegant in its simplicity: instead of storing each symbol individually, we can store each symbol once along with a count of how many times it repeats. For `AAAA`, instead of storing 4 bytes, we store (`A`,4).

Example

Intuition Building: Raw data: `WWWWWWWWWWWWWBBWWWWWWWWWWWWWBBBWWWWWWWWW`
With runs identified:

- 13 W's
- 2 B's
- 13 W's
- 3 B's
- 9 W's

Storing as pairs: (W,13)(B,2)(W,13)(B,3)(W,9)
Original: 40 characters Encoded: 10 values (5 symbols + 5 counts) → significant savings!

Important

Why Study RLE Today? Despite its age and simplicity, RLE remains relevant because:

1. It's optimal for certain data types (binary images, sparse data)
2. It's extremely fast—often the fastest compression method available
3. It's patent-free and simple to implement correctly
4. It's a building block for more complex algorithms (BWT, JPEG)
5. Understanding RLE teaches fundamental compression concepts applicable to all methods

7.2 Basic Run-Length Encoding

7.2.1 Formal Definition of a Run

Before we can encode runs, we need a precise mathematical definition.

Definition

Run: Given a sequence $S = s_1, s_2, s_3, \ldots, s_n$ over an alphabet Σ, a run is a maximal consecutive subsequence $s_i, s_{i+1}, \ldots, s_j$ such that $s_i = s_{i+1} = \ldots = s_j$, and either $i = 1$ or $s_{i-1} \neq s_i$, and either $j = n$ or $s_{j+1} \neq s_j$.
In simpler terms: a run is the longest possible sequence of identical symbols that appears consecutively.

Properties of Runs:

1. **Maximality:** A run cannot be extended—the symbol before it (if any) is different, and the symbol after it (if any) is different.
2. **Disjointness:** Runs partition the sequence; every symbol belongs to exactly one run.
3. **Alternation:** Adjacent runs always contain different symbols.

Example

Run Decomposition Examples:

Sequence	Runs	Notes
AAAAA	{AAAAA}	Single run of length 5
ABABAB	{A,B,A,B,A,B}	Six runs of length 1
AAABBBCCC	{AAA},{BBB},{CCC}	Three runs
ABBBCCDD	{A},{BBB},{CC},{DD}	Mixed lengths
AAAAAAABBBBBBB	{AAAAAAA},{BBBBBBB}	Two runs, long

Run Statistics:

- Minimum run length: 1
- Maximum run length: n (the entire sequence)
- Number of runs: Between 1 and n

7.2.2 Encoding Algorithm (Symbol, Count)

The basic RLE encoding algorithm transforms an input sequence into a sequence of (symbol, count) pairs. The algorithm is elegantly simple.

Algorithm 4 Basic RLE Encoding

```
procedure RLE-Encode(S[1..n])
    result ← empty list
    if n = 0 then
        return result
    end if
    current_symbol ← S[1]
    run_length ← 1
    for i ← 2 to n do
        if S[i] = current_symbol then
            run_length ← run_length + 1
        else
            append (current_symbol, run_length) to result
            current_symbol ← S[i]
            run_length ← 1
        end if
    end for
    append (current_symbol, run_length) to result        ▷ Don't forget the last run!
    return result
end procedure
```

Important

Key Implementation Considerations:

- **First symbol initialization:** The algorithm must handle the first symbol specially
- **Last run handling:** Always append the final run after the loop
- **Empty input:** Edge case that must be handled gracefully
- **Count representation:** How to store the count (fixed width, variable width, etc.)
- **Symbol representation:** Whether symbols are bytes, bits, or larger units

7.2.3 Decoding Algorithm

Decoding is equally straightforward: for each (symbol, count) pair, output the symbol count times.

Algorithm 5 Basic RLE Decoding

```
procedure RLE-Decode(pairs[1..m])
    result ← empty list
    for each (symbol, count) in pairs do
        for i ← 1 to count do
            append symbol to result
        end for
    end for
    return result
end procedure
```

7.2.4 Worked Examples (Text and Binary Data)

Let's work through several examples to build intuition.

Example

Example 1: Simple Text

Input: `WWWWWWWWWWWWWBBWWWWWWWWWWWWWBBBWWWWWWWW`
Step-by-step encoding:

1. Start at position 1: `W`, length = 13 (positions 1-13)
2. Position 14: `B` (different) → output (W,13), start new run
3. New run: `B`, length = 2 (positions 14-15)
4. Position 16: `W` (different) → output (B,2), start new run
5. New run: `W`, length = 13 (positions 16-28)
6. Position 29: `B` (different) → output (W,13), start new run
7. New run: `B`, length = 3 (positions 29-31)
8. Position 32: `W` (different) → output (B,3), start new run
9. Final run: `W`, length = 9 (positions 32-40) → output (W,9)

Output: (W,13)(B,2)(W,13)(B,3)(W,9)
Compression Analysis:

- Original size: 40 bytes
- Encoded size: 5 symbols + 5 counts = 10 values
- If each value is 1 byte: 10 bytes vs 40 bytes → 75% reduction!
- Compression ratio: 4:1

Example

Example 2: Binary Image Data

Consider a scan line from a fax document (1 = black, 0 = white):
Input: `0000000000111110000011111100000000000`
Let's count runs:

- Run 1: zeros, length = 10
- Run 2: ones, length = 5
- Run 3: zeros, length = 5
- Run 4: ones, length = 6
- Run 5: zeros, length = 11

Output: (0,10)(1,5)(0,5)(1,6)(0,11)
Comparison with original: Original: 37 bits Encoded: 10 values (5 symbols + 5 counts) If counts fit in 4 bits each: 5×1 + 5×0.5 = 7.5 bytes → 80% reduction!

Example

Example 3: Worst Case - No Runs

Input: `ABCDEFGHIJKLMNOPQRSTUVWXYZ`
Run decomposition: 26 runs, each of length 1
Output: (A,1)(B,1)(C,1)...(Z,1)
Compression Analysis:

- Original: 26 bytes
- Encoded: 52 values (26 symbols + 26 counts)
- At 1 byte per value: 52 bytes vs 26 bytes → 100% expansion!
- At 4 bits per count: 26 + 13 = 39 bytes → still 50% expansion

This illustrates RLE's critical weakness: data without runs actually expands.

7.2.5 Time and Space Complexity

Understanding the computational efficiency of RLE helps in deciding when to use it.

Important

Time Complexity Analysis:

- **Encoding:** $O(n)$ where n is input length
  - Single pass through the data
  - Constant work per symbol (comparison and counter increment)
  - No backtracking or lookahead needed
  - Optimal for any one-pass algorithm
- **Decoding:** $O(m)$ where m is output length
  - Single pass through the encoded pairs
  - For each pair, output loop runs *count* times
  - Total output size equals original input size
  - Also optimal (must produce all output symbols)
- **Overall:** Linear time in both input and output size
  - Cannot be faster than linear (must examine each input symbol)
  - Cannot decode faster than output size (must produce each output)
  - RLE achieves the theoretical minimum time complexity

Important

Space Complexity Analysis:

- **Encoding (in-place):** $O(1)$ auxiliary space
  - Only need current symbol and counter
  - Can output as runs are discovered (streaming)
  - No need to store entire input in memory
- **Encoding (with buffering):** $O(r)$ where r is number of runs
  - If storing all pairs before output
  - $r \leq n$ (at most n runs)
  - Typically much smaller than n for compressible data
- **Decoding:** $O(1)$ auxiliary space
  - Process one pair at a time

- Output as we go
- Perfect for streaming applications

Complexity Summary Table

Metric	Encoding	Decoding
Time Complexity	$O(n)$	$O(n)$
Auxiliary Space (streaming)	$O(1)$	$O(1)$
Auxiliary Space (buffered)	$O(r)$	$O(1)$
Optimal?	Yes (linear lower bound)	Yes (output size lower bound)

7.3 Compression Analysis and Theoretical Properties

Understanding when RLE works well and when it fails requires careful analysis of its compression behavior.

7.3.1 Best-Case Compression Ratio

The best case for RLE occurs when the entire input consists of a single long run.

Definition

Compression Ratio:

$$\text{Ratio} = \frac{\text{Original Size}}{\text{Compressed Size}}$$

Higher ratio means better compression.

Theorem 7.1. *For a sequence of length n consisting of a single symbol, the best-case compression ratio achievable with basic RLE is approximately $\frac{n}{c}$, where c is the size (in bytes) of one (symbol, count) pair.*

Example

Best-Case Calculation:
Consider a 1 MB file (1,048,576 bytes) of all 'A's.

- **Original:** 1,048,576 bytes
- **Encoded:** (A, 1048576)
- **With 4-byte counts (typical):**
  - Symbol: 1 byte ('A')

- Count: 4 bytes (32-bit integer)
- Total: 5 bytes

- **Compression ratio:** 1,048,576 / 5 ≈ 209,715 : 1
- **Space savings:** 99.9995%

Impact of count field size:

Count field size	Pair size	Compressed size	Ratio
1 byte	2 bytes	2 bytes	524,288:1
2 bytes	3 bytes	3 bytes	349,525:1
4 bytes	5 bytes	5 bytes	209,715:1
8 bytes	9 bytes	9 bytes	116,508:1

Key insight: Smaller count fields give better ratios for long runs, but limit maximum run length.

7.3.2 Worst-Case Expansion

The worst case for basic RLE is when the data has no runs at all—every symbol differs from its predecessor.

Theorem 7.2. *For a sequence of length n with no repeated consecutive symbols (all runs of length 1), basic RLE achieves a compression ratio of $\frac{n}{2n} = 0.5$ when using fixed-size counts equal to symbol size, resulting in 100% expansion.*

Proof. If every symbol is different from its predecessor:

- Number of runs $= n$ (each symbol is its own run)
- Encoded output $= n$ (symbol, count) pairs
- If count field size equals symbol size (e.g., 1 byte each):
  - Original size $= n$ bytes
  - Compressed size $= n$ symbols $+ n$ counts $= 2n$ bytes
  - Ratio $= n/(2n) = 0.5$
  - Expansion $= (2n - n)/n = 1 = 100\%$

□

Example

Worst-Case Demonstration:
Input: `ABABABABAB` (10 bytes, alternating symbols)

Run decomposition: (A,1)(B,1)(A,1)(B,1)(A,1)(B,1)(A,1)(B,1)(A,1)(B,1)

With 1-byte counts:

- Original: 10 bytes
- Encoded: 20 bytes
- Expansion: 100%

With 4-bit counts (0.5 bytes):

- Original: 10 bytes
- Encoded: 10 symbols + 5 bytes of counts = 15 bytes
- Expansion: 50%

With clever encoding (PackBits style):

- Can mark some runs as literals to avoid counts
- But adds overhead for escape symbols
- Trade-off: complexity vs. worst-case behavior

7.3.3 Expected Performance on Random Data

Understanding how RLE performs on random data is crucial because:

1. Many real data sources have random components
2. It establishes a baseline for "no compression possible"
3. It helps identify when data is truly compressible

Theorem 7.3. *For an i.i.d. (independent and identically distributed) random sequence over an alphabet of size k, the expected run length is:*

$$E[\textit{run length}] = \frac{1}{1-p} = \frac{k}{k-1}$$

where $p = 1/k$ is the probability of a symbol repeating.

Proof. For a random sequence with equiprobable symbols:

- Probability that next symbol equals current $= 1/k$
- Run length follows geometric distribution: $P(\text{run length} = L) = (1/k)^{L-1} \cdot (1-1/k)$

- Expected value: $\sum_{L=1}^{\infty} L \cdot (1/k)^{L-1} \cdot (1 - 1/k) = \frac{1}{1-1/k} = \frac{k}{k-1}$

□

Example

Expected Run Lengths for Different Alphabets:

Alphabet	Size (k)	Expected run length
Binary	2	$2/(2-1) = 2$
DNA bases	4	$4/3 \approx 1.333$
Lowercase letters	26	$26/25 = 1.04$
ASCII	256	$256/255 \approx 1.0039$

Implications:

- Binary data has the longest expected runs (2)
- ASCII text has runs barely longer than 1
- For large alphabets, runs of length > 1 are rare events

Important

Expected Compression Ratio for Random Data:
For an alphabet of size k with symbols and counts each stored in 1 byte:

- Expected number of runs $= n/E[\text{run length}] = n \cdot (1 - 1/k)$
- Compressed size $= 2 \cdot \#\text{runs} = 2n(1 - 1/k)$ bytes
- Original size $= n$ bytes
- Expected ratio $= n/[2n(1 - 1/k)] = 1/[2(1 - 1/k)]$

Alphabet	Expected runs	Compressed size	Ratio
Binary	$0.5n$	n	$1:1$
4 symbols	$0.75n$	$1.5n$	$0.667:1$
ASCII	$0.996n$	$1.992n$	$0.502:1$

Key observation: For alphabets larger than binary, RLE on random data causes expansion! Binary random data breaks even (no compression, no expansion).

7.3.4 Relation to Entropy

Understanding the relationship between RLE and information theory helps place it in the broader compression landscape.

Definition

Shannon Entropy: For a source with probability distribution P, the entropy $H(P) = -\sum p_i \log_2 p_i$ bits per symbol represents the theoretical minimum average code length.

Theorem 7.4. *For a memoryless source with symbol probabilities p_i, the entropy of run lengths is given by:*

$$H_{run} = H(P) + (1 - H(P)) \cdot \frac{\sum p_i^2}{1 - \sum p_i^2} \text{ bits per run}$$

This is generally higher than the per-symbol entropy, explaining why RLE alone cannot achieve optimal compression.

Important

Information-Theoretic Perspective:

1. **RLE as a transform:** RLE transforms a sequence into a sequence of (symbol, length) pairs. This is a reversible transformation that doesn't lose information.

2. **Redundancy remaining:** After RLE, the symbol and length sequences still contain structure:

   - Symbols alternate (never two same symbols in a row)
   - Run lengths have their own distribution
   - There may be correlation between symbol and run length

3. **Optimal compression:** To achieve entropy limits, we must further compress the RLE output using statistical methods (Huffman, arithmetic coding).

4. **Markov property:** RLE effectively models the source as a simple Markov chain where state = current symbol, and run length determines transition time.

Example

Binary Source Example:
Consider a binary source with $p(0) = 0.9$, $p(1) = 0.1$.

- **Source entropy:** $H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.469$ bits/symbol
- **Theoretical minimum:** 0.469 bits per symbol on average
- **Run length distribution:**

- Runs of 0: geometric with mean $1/0.1 = 10$
- Runs of 1: geometric with mean $1/0.9 \approx 1.111$

- **RLE output:** (symbol, length) pairs
- **If we store lengths in 8 bits:**
  - Each run: 1 bit for symbol + 8 bits for length = 9 bits
  - Average run length: $0.9 \times 10 + 0.1 \times 1.111 \approx 9.111$ symbols
  - Average bits per symbol $= 9/9.111 \approx 0.988$ bits/symbol
  - Far from entropy limit (0.469)!
- **If we entropy-code the lengths:**
  - Symbol: still 1 bit (or entropy code symbols too)
  - Lengths: entropy $H_{\text{lengths}} \approx 3.4$ bits (for this distribution)
  - Average bits per symbol $= (1 + 3.4)/9.111 \approx 0.483$ bits/symbol
  - Much closer to entropy limit!

Lesson: RLE is most effective when combined with entropy coding of the run lengths.

7.4 Engineering Variants of RLE

Basic RLE works well in theory, but practical implementations require engineering choices to handle real-world constraints and improve performance.

7.4.1 Count-After-Symbol vs. Count-Before-Symbol

The ordering of symbol and count in the output stream affects decoding and has subtle implications.

Important

Two Common Conventions:

1. Count-After-Symbol (Symbol then Count):

- Format: `[symbol][count][symbol][count]...`
- Example: `A5B3C2` means "5 A's, 3 B's, 2 C's"
- **Advantages:**
  - Natural reading order: "Here's what to output, then how many"

- Easier for streaming: see symbol first, can prepare output
- Common in many implementations

- **Disadvantages:**
  - Decoder must read count before knowing how many to output
  - Slightly less intuitive for some applications

2. **Count-Before-Symbol (Count then Symbol):**
   - Format: `[count][symbol][count][symbol]...`
   - Example: `5A3B2C` means "5 A's, 3 B's, 2 C's"
   - **Advantages:**
     - Decoder knows length before seeing symbol
     - Can allocate buffer space if needed
     - Used in some formats (e.g., PCX)
   - **Disadvantages:**
     - Slightly less natural for human reading
     - Streaming requires looking ahead

Example

Implementation Comparison:

```python
def decode_after(encoded):
    result = []
    i = 0
    while i < len(encoded):
        symbol = encoded[i]
        count = encoded[i + 1]
        result.extend([symbol] * count)
        i += 2
    return result
```

Listing 3: Count-After-Symbol Decoding (Python)

```python
def decode_before(encoded):
    result = []
    i = 0
    while i < len(encoded):
        count = encoded[i]
        symbol = encoded[i + 1]
```

```
        result.extend([symbol] * count)
        i += 2
    return result
```

Listing 4: Count-Before-Symbol Decoding (Python)

The code is nearly identical—the difference is mainly conceptual and affects how we think about the data.

7.4.2 Escape-Symbol Schemes

One of the biggest problems with basic RLE is expansion on non-repetitive data. Escape-symbol schemes address this by treating runs and non-repetitive data differently.

Definition

Escape Symbol: A special marker that indicates whether the following data represents a run or a sequence of literal (uncompressed) symbols.

Important

Common Escape Scheme Design:

1. Choose an escape symbol (often a value not likely to appear in data, or one that can be escaped itself)
2. When encountering a run of length ≥ 3:
   - Output escape symbol
   - Output the repeated symbol
   - Output the run length
3. For non-repetitive data:
   - Output symbols literally
   - If the literal data contains the escape symbol, need a way to represent it (e.g., double escape)

Example

Escape Scheme Example:
Let's design a simple escape scheme with escape marker = `0xFF` (255 in byte values).
Encoding rules:

- For run of length ≥ 3 of symbol S: output `[0xFF][S][length]`
- For single symbols or pairs: output as-is

- If data contains `0xFF`, output it twice: `[0xFF][0xFF]` means literal 0xFF

Example: Input: `AAAAABCDEFFFFGG`
Step-by-step encoding:

- `AAAAA` (5 A's): `FF A 05`
- `B`: `B` (literal)
- `C`: `C` (literal)
- `D`: `D` (literal)
- `E`: `E` (literal)
- `FFFF` (4 F's): `FF F 04`
- `GG` (2 G's): `GG` (pairs output as literals)

Encoded: `FF A 05 B C D E FF F 04 G G`
Analysis:

- Original: 15 bytes
- Encoded: 11 bytes
- Savings: 4 bytes (27%)
- Without escape scheme (basic RLE): would be 16 bytes (expansion!)

7.4.3 Threshold-Based RLE (Minimum Run Length)

Threshold-based RLE only encodes runs that exceed a minimum length, leaving shorter sequences as literals.

Definition

Threshold T: The minimum run length that will be encoded as a run. Runs of length $< T$ are treated as literals.

Important

Design Trade-offs:

- **Low threshold (e.g., T=2):**
  - Catches more runs
  - But overhead for many short runs

  - May still cause expansion on some data

- **High threshold (e.g., T=4):**
  - Only encodes long, beneficial runs
  - Less overhead
  - Misses opportunities for moderate compression
- **Optimal threshold:** Depends on data characteristics and count field size
  - With 1-byte counts, break-even is usually 3-4
  - With smaller counts (e.g., 4 bits), threshold can be lower

Example

Threshold Comparison:
Input: `AAABBBBCDDDDD` (A:3, B:4, C:1, D:5)
With T=2 (encode runs of length ≥ 2):

- Run A(3): encode as run $\rightarrow$ (A,3)
- Run B(4): encode as run $\rightarrow$ (B,4)
- Run C(1): literal $\rightarrow$ C
- Run D(5): encode as run $\rightarrow$ (D,5)
- Total: 3 runs + 1 literal = $3\times2 + 1 = 7$ values

With T=3 (encode runs of length ≥ 3):

- Run A(3): encode $\rightarrow$ (A,3)
- Run B(4): encode $\rightarrow$ (B,4)
- Run C(1): literal $\rightarrow$ C
- Run D(5): encode $\rightarrow$ (D,5)
- Same as T=2 in this case (no runs of length 2)

With T=4 (encode runs of length ≥ 4):

- Run A(3): literal $\rightarrow$ A A A (3 literals)
- Run B(4): encode $\rightarrow$ (B,4)
- Run C(1): literal $\rightarrow$ C

- Run D(5): encode → (D,5)
- Total: 2 runs + 4 literals = 2×2 + 4 = 8 values

With T=5 (encode runs of length ≥ 5):

- Run A(3): literals (3)
- Run B(4): literals (4)
- Run C(1): literal (1)
- Run D(5): encode → (D,5)
- Total: 1 run + 8 literals = 1×2 + 8 = 10 values

Comparison:

Threshold	Encoded size	Notes
T=2	7 values	Best for this data
T=3	7 values	Same as T=2
T=4	8 values	Worse
T=5	10 values	Much worse

7.4.4 PackBits-Style Mixed Literal/Run Encoding

PackBits is a classic RLE variant used in many file formats (TIFF, Macintosh resource forks). It elegantly handles both runs and literals in a single byte header.

Definition

PackBits Encoding: Each chunk begins with a 1-byte header h:

- If $0 \leq h \leq 127$: copy the next $h + 1$ bytes literally
- If $-127 \leq h \leq -1$: repeat the next byte $1 - h$ times
- $h = -128$: reserved (no operation, sometimes used for special purposes)

Important

PackBits Algorithm Details:
Header byte interpretation as signed 8-bit integer:

- **Positive (0 to 127):** Literal run
  - Value n means "copy next $n + 1$ bytes as-is"
  - Range: 1 to 128 bytes per literal chunk
- **Negative (-1 to -127):** Repeated run

- Value n means "repeat next byte $1 - n$ times"
- Since n is negative, $1 - n$ is positive
- Range: 2 to 128 repetitions
- Example: $h = -3$ means repeat next byte $1 - (-3) = 4$ times

- **-128:** Reserved (some implementations ignore, others use as EOF marker)

Example

PackBits Encoding Examples:
Example 1: Mixed data Input: `AAAAABCDEFFFFGG`
Scanning the data:

- Run of 5 A's → header = -4 (repeat 1-(-4)=5 times), then byte 'A'
- Literals B,C,D,E (4 bytes) → header = 3 (copy 3+1=4 bytes), then B C D E
- Run of 4 F's → header = -3 (repeat 4 times), then byte 'F'
- Literals G,G (2 bytes) → header = 1 (copy 2 bytes), then G G

Encoded stream (as bytes):

```
[-4, 'A', 3, 'B','C','D','E', -3, 'F', 1, 'G','G']
```

Example 2: Long literal run Input: `ABCDEFGHIJKLMNOP` (16 distinct letters)
Since PackBits literal chunks max at 128 bytes, we can encode as one chunk: Header = 15 (copy 16 bytes), then all 16 letters.
Example 3: Long repeated run Input: 150 bytes of 'X'
We need multiple chunks:

- First chunk: header = -127 (repeat 128 times), byte 'X'
- Second chunk: header = -21 (repeat 22 times), byte 'X'
- Total: 4 bytes + 2 bytes overhead = 6 bytes for 150 bytes of data

Algorithm 6 PackBits Decoding

```
procedure PackBits-Decode(data[1..n])
    i ← 1
    result ← empty list
    while i ≤ n do
        h ← data[i]                                   ▷ header byte
        i ← i + 1
        if h ≥ 0 then                                 ▷ Literal run
            len ← h + 1
            for j ← 1 to len do
                append data[i] to result
                i ← i + 1
            end for
        else if h = −128 then                         ▷ Reserved, usually skip
                                                      ▷ No operation, sometimes used as marker
        else                                          ▷ Repeated run
            len ← 1 − h                               ▷ h is negative
            value ← data[i]
            i ← i + 1
            for j ← 1 to len do
                append value to result
            end for
        end if
    end while
    return result
end procedure
```

Important

PackBits Advantages and Disadvantages:

Advantages:

- Simple, elegant design
- No expansion on non-repetitive data (literal chunks are efficient)
- Handles both runs and literals seamlessly
- Widely supported in file formats

Disadvantages:

- Limited to 128-byte chunks (max run length 128)

- 1-byte overhead per chunk (minimal, but adds up)
- Not optimal for very long runs (requires splitting)
- Signed byte arithmetic can be confusing

7.4.5 Fixed-Width vs. Variable-Length Count Fields

The representation of run lengths significantly impacts compression efficiency.

Important

Fixed-Width Count Fields:
Design: Each run length stored in a fixed number of bits (e.g., 8 bits, 16 bits, 32 bits)
Advantages:

- Simple to implement
- Fast encoding/decoding (no bit-level parsing)
- Random access possible (know where each run ends)
- Good for hardware implementation

Disadvantages:

- Wasted space for short runs
- Limited maximum run length (e.g., 255 for 8 bits)
- Need to split longer runs into multiple chunks
- Not adaptive to run length distribution

Typical trade-offs:

Count size	Max run	Overhead/run	Best for
4 bits (nibble)	15	0.5 bytes	Very short runs
8 bits (byte)	255	1 byte	General purpose
16 bits	65,535	2 bytes	Long runs
24 bits	16.7M	3 bytes	Images
32 bits	4.3B	4 bytes	Very large files

Important

Variable-Length Count Fields:
Design: Use a variable-length code (like Elias gamma coding, Fibonacci coding) to represent run lengths
Common approaches:

1. **Gamma/Elias codes:** Good for small numbers, inefficient for large
2. **Fibonacci codes:** Better balance, no need for separator
3. **Exponential-Golomb:** Used in H.264/AVC video coding
4. **Adaptive Huffman:** Best compression, but complex

Advantages:

- Optimal for the run length distribution
- No arbitrary maximum run length
- No wasted bits for short runs
- Can achieve near-entropy compression of run lengths

Disadvantages:

- Complex implementation (bit-level I/O)
- Slower encoding/decoding
- No random access without decoding entire stream
- Error propagation if bit errors occur

Understanding Variable-Length Codes Through Elias Gamma

To truly appreciate the trade-off between fixed and variable-length encoding, we need to understand how variable-length codes actually work. Elias gamma coding provides the clearest illustration of the fundamental principle: small numbers should use fewer bits than large numbers.

Elias Gamma Coding: The Core Idea

Elias gamma coding builds numbers in two parts:

1. **The size indicator:** A sequence of zeros that tells us "how many bits the number will need," terminated by a 1

2. **The remainder:** The actual value in binary, using exactly the number of bits indicated

More formally, for any positive integer N:

- Find the largest power of two 2^k that is less than or equal to N
- Let $M = N - 2^k$ (the remainder)
- Write k zeros, then a 1 (the size indicator)
- Append M in binary using exactly k bits

Building Elias Gamma Codes: Step by Step

Let's construct codes systematically to see the pattern:

1. **For N = 1:**
   - Largest power of two ≤ 1 is $2^0 = 1$, so $k = 0$
   - Remainder $M = 1 - 1 = 0$
   - Size indicator: k zeros followed by 1 = "1" (just the 1, no zeros)
   - Remainder in k bits: $k = 0$ means no remainder bits
   - Final code: "1"
2. **For N = 2:**
   - Largest power of two ≤ 2 is $2^1 = 2$, so $k = 1$
   - Remainder $M = 2 - 2 = 0$
   - Size indicator: "01" (one zero, then a 1)
   - Remainder in $k = 1$ bits: "0"
   - Final code: "010"
3. **For N = 3:**
   - Largest power of two ≤ 3 is $2^1 = 2$, so $k = 1$
   - Remainder $M = 3 - 2 = 1$
   - Size indicator: "01" (one zero, then a 1)
   - Remainder in $k = 1$ bits: "1"
   - Final code: "011"
4. **For N = 4:**

- Largest power of two ≤ 4 is $2^2 = 4$, so $k = 2$
- Remainder $M = 4 - 4 = 0$
- Size indicator: "001" (two zeros, then a 1)
- Remainder in $k = 2$ bits: "00"
- Final code: "00100"

5. **For N = 5:**
   - Largest power of two ≤ 5 is $2^2 = 4$, so $k = 2$
   - Remainder $M = 5 - 4 = 1$
   - Size indicator: "001" (two zeros, then a 1)
   - Remainder in $k = 2$ bits: "01"
   - Final code: "00101"

Complete Reference Table

N	2^k	k	M	Elias Gamma Code
1	1	0	0	1
2	2	1	0	010
3	2	1	1	011
4	4	2	0	00100
5	4	2	1	00101
6	4	2	2	00110
7	4	2	3	00111
8	8	3	0	0001000
9	8	3	1	0001001
10	8	3	2	0001010
15	8	3	7	0001111
16	16	4	0	000010000

How Decoding Works

The decoding process is symmetric and equally straightforward:

1. Read bits until you encounter a 1. Count how many zeros came before it—this count is k
2. The 1 you just read is the termination of the size indicator
3. Read the next k bits—this is the remainder M in binary
4. Compute $N = 2^k + M$

Decoding Example: Given bitstream "001011..."

1. Read zeros: "00" then a 1 → found "001", so $k = 2$
2. Next $k = 2$ bits: "01" → this binary is $M = 1$
3. $N = 2^2 + 1 = 4 + 1 = 5$
4. First number decoded is 5, remaining bits "...", continue decoding

Fixed-Width vs. Variable-Length: A Detailed Comparison

Now that we understand how variable-length coding works, let's compare the two approaches with concrete examples.

Example

Count Field Comparison:
Consider a file with runs of lengths: 3, 5, 2, 100, 4, 200
Fixed 8-bit counts:

- Each run: 1 byte for count
- Total count overhead: 6 bytes = 48 bits
- Can represent all runs (max 200 < 255)
- Simple, fast, randomly accessible

Fixed 4-bit counts:

- Cannot represent runs ¿15
- Would need to split run 100 into: 15,15,15,15,15,15,10 (7 runs!)
- Run 200: even worse, would need 14 runs (13×15 + 5)
- Total overhead: 6 runs become 6+7+14 = 27 runs!
- 27 runs × 0.5 bytes = 13.5 bytes (worse than 8-bit!)

Variable-length (Elias gamma):

- 3 → "011" (3 bits)
- 5 → "00101" (5 bits)
- 2 → "010" (3 bits)
- 100 → "0000001100100" (13 bits)

- 4 → "00100" (5 bits)
- 200 → "000000011001000" (15 bits)
- Total: 44 bits ≈ 5.5 bytes

Comparison Table:

Method	Total Bits	Characteristics
Fixed 4-bit	Not possible (runs too long)	Would need run splitting
Fixed 8-bit	48 bits	Simple, fast, random access
Elias gamma	44 bits	8.3% savings, complex, no random access

Bit-by-bit breakdown:

Method	3	5	2	100	4	200	Total
Fixed 8-bit	8	8	8	8	8	8	48
Elias gamma	3	5	3	13	5	15	44
Savings	+5	+3	+5	-5	+3	-7	+4

Notice the economic trade-off: small numbers (2,3,4,5) use fewer bits than fixed-width, which subsidizes the extra bits needed for large numbers (100,200). In typical data where short runs are much more common than long runs, this trade-off yields net compression.

When to Use Each Approach

The choice between fixed and variable-width count fields depends on your priorities:

Consideration	Fixed-Width Better	Variable-Width Better
Implementation complexity	Simple hardware/software	Complex bit-level I/O
Speed	Fast encoding/decoding	Slower parsing
Random access	Yes—know where each run ends	No—must decode sequentially
Run length distribution	Uniform or bounded	Highly skewed toward small values
Maximum run length	Limited by width	Unlimited
Error resilience	Errors isolated to one run	Errors propagate

Real-World Hybrid Approaches

Many practical formats use hybrid strategies:

- **JPEG:** Uses different fixed-width tables for DC and AC coefficients, adapting to their different distributions

- **PNG:** Uses a combination of fixed and variable (LZ77) with Huffman coding
- **H.264:** Uses Exponential-Golomb (variable) for syntax elements
- **FAX compression:** Uses modified Huffman (variable) for run lengths

The key insight is that variable-length coding, while more complex, can significantly improve compression when run length distributions are skewed. Elias gamma provides the simplest introduction to this fundamental concept in data compression.

7.5 Bit-Level and Binary RLE

When working with binary data, RLE takes on special characteristics and applications.

7.5.1 Run-Length Encoding of Bits (0/1 Streams)

Binary RLE operates on bit streams rather than byte streams, which changes the dynamics significantly.

Definition

Binary RLE: A variant of run-length encoding that operates on a stream of bits (0s and 1s), encoding each run of consecutive identical bits as a length.

Important

Key Characteristics of Binary RLE:

1. **Alphabet size = 2:** Only two symbols, so runs alternate strictly
2. **No need to store symbol:** Since runs alternate, we only need to store the first symbol once, then just lengths
3. **Run lengths dominate:** With only two symbols, the entire encoding can be just a sequence of lengths
4. **Boundary handling:** Need to know where the stream ends

Algorithm 7 Binary RLE Encoding

```
procedure BinaryRLE-Encode(bits[1..n])
    result ← empty list
    if n = 0 then
        return result
    end if
    current_bit ← bits[1]
    append current_bit to result                    ▷ Store first bit
    run_length ← 1
    for i ← 2 to n do
        if bits[i] = current_bit then
            run_length ← run_length + 1
        else
            append run_length to result
            current_bit ← bits[i]
            run_length ← 1
        end if
    end for
    append run_length to result
    return result
end procedure
```

Example

Binary RLE Examples:

Example 1: Simple alternating pattern Input bits: `000111000111`

- First bit: 0
- Runs: 3 zeros, 3 ones, 3 zeros, 3 ones
- Encoded: [0, 3, 3, 3, 3]
- Original: 12 bits
- Encoded: 1 bit + 4 runs × (length field)
- If length fields are 4 bits: total = 1 + 16 = 17 bits (expansion!)

Example 2: Long runs (fax-like) Input bits: `0000000000111111111110000000000`

- First bit: 0
- Runs: zeros(10), ones(10), zeros(8)

- Encoded: [0, 10, 10, 8]
- Original: 28 bits
- With 4-bit lengths: 1 + 12 = 13 bits (53% compression!)
- With 8-bit lengths: 1 + 24 = 25 bits (still compression!)

Note: Binary RLE only makes sense when runs are long enough to offset length field overhead.

7.5.2 Applications in Fax and Document Compression

The most successful application of binary RLE has been in facsimile (fax) transmission and document compression.

Important

Fax Compression Overview:

- **CCITT Group 3 Fax Standard (1980):** The first widely adopted digital fax standard
- **Modified Huffman (MH) coding:** Combines RLE with Huffman coding for run lengths
- **Modified READ (MR) coding:** 2D variant that uses previous line for prediction
- **Modified Modified READ (MMR) coding:** Used in Group 4 fax, no errors allowed

How Group 3 Fax Works:

1. **Scan line resolution:** Typical fax: 1728 pixels per line (standard resolution)
2. **Binary representation:** White = 0, Black = 1 (or vice versa, depending on standard)
3. **Run-length extraction:** Each line becomes alternating runs of white and black
4. **Run-length coding:** Each run length is encoded using a Huffman code
5. **Special codes:** End-of-line (EOL) markers for synchronization

Example

Modified Huffman (MH) Coding Tables:

The Group 3 standard defines Huffman codes for run lengths:

Terminating codes (run lengths 0-63):

Run length	White code	Black code
0	00110101	0000110111
1	000111	010
2	0111	11
3	1000	10
4	1011	011
...	...	...
63	00110110	000001100110

Make-up codes (run lengths multiples of 64):

- 64: White 11011, Black 0000001111
- 128: White 10010, Black 000011001000
- ... up to 2560

Encoding process:

1. For run length L, if $L \leq 63$: use terminating code
2. If $L > 63$: output make-up code for $64 \times \lfloor L/64 \rfloor$, then terminating code for remainder

Example

Fax Encoding Example:

Scan line with runs: white(150), black(3), white(20), black(1500)

White run of 150:

- Make-up for 128: "10010"
- Terminating for 22: white code for 22 = "0001100"
- Combined: "100100001100"

Black run of 3:

- Terminating code for 3 black: "10"

White run of 20:

- Terminating code for 20 white: "0001000"

Black run of 1500:

- 1500/64 = 23 remainder 28
- Make-up for 23×64 = 1472: black make-up code
- Terminating for 28: black terminating code

Result: A compact bitstream that can be transmitted over phone lines.

7.5.3 Two-Dimensional Run-Length Encoding

While 1D RLE works on scan lines independently, 2D RLE exploits correlation between adjacent lines.

Definition

2D Run-Length Encoding: A technique that encodes runs of pixels by referencing the previous line, exploiting vertical redundancy in document images.

Important

CCITT Group 3 2D (MR) and Group 4 (MMR):
Key concepts:

- **Reference line:** Previously encoded line used as prediction
- **Changing elements:** Pixels where color changes (black to white or vice versa)
- **Modes of encoding:**
  - **Pass mode:** A run on current line corresponds to a gap on reference line
  - **Vertical mode:** Run boundary aligns with reference line boundary (±3 pixels)
  - **Horizontal mode:** Encode current run directly when not correlated

Example

2D RLE Encoding Modes:
Consider two scan lines (a0 = current position, a1 = next changing element):
Reference line: `---XXXX------XXXXXX----` **Current line:** `---XXXXXXX---XXXXXX----`
Pass mode: When a run on current line "passes over" a run on reference line

- Efficient for slanted lines or shifted content

Vertical mode: When a1 aligns with a changing element on reference line ±3

- Encode as vertical(offset) where offset = difference in positions
- Very efficient for aligned text

Horizontal mode: When no correlation, encode a0a1 and a1a2 lengths directly

- Fallback when 2D prediction fails

Compression gain: 2D typically achieves 2-3× better compression than 1D alone.

7.6 Zero Run-Length Encoding in Transform Coding

One of the most important modern applications of RLE is in transform-based compression systems.

7.6.1 Sparse Data and Long Zero Sequences

After applying transforms like DCT (Discrete Cosine Transform) or wavelet transforms, coefficients often contain long runs of zeros.

Important

Why Zero Runs Occur in Transform Coding:

1. **Energy compaction:** Transforms concentrate signal energy into few coefficients
2. **Quantization:** Small coefficients become zero after quantization
3. **High-frequency coefficients:** Often near-zero in natural images
4. **Zigzag scanning:** Orders coefficients from low to high frequency, creating long zero runs at the end

Definition

Zero Run-Length Encoding: A specialized form of RLE that focuses exclusively on runs of zero values, often combined with encoding the non-zero values that interrupt them.

Example

Zigzag Scan in JPEG:
After 8×8 DCT and quantization, coefficients are scanned in zigzag order:

```
Original 8×8 block (quantized coefficients):
```

```
[150, -25, 0,   0,   0,   0,   12,  0
 0,   0,   0,   0,   0,   0,   0,   0
 0,   0,   0,   0,   0,   0,   0,   0
 0,   0,   0,   0,   0,   0,   0,   0
 0,   0,   0,   0,   0,   0,   0,   0
 0,   0,   0,   0,   0,   0,   0,   0
 0,   0,   0,   0,   0,   0,   0,   0
 0,   0,   0,   0,   0,   0,   0,   0]
```

Zigzag scan order (simplified): Position 1: 150 (DC) Positions 2-10: -25, 0, 0, 0, 0, 12, 0, 0, 0 Remaining 54 positions: all zeros
Zero-run encoding: (0,4) meaning 4 zeros, then 12, then (0,54) for end-of-block
JPEG representation: (0,4,12) followed by End-of-Block marker.

7.6.2 Zero-Run Encoding in Image Compression Pipelines

Modern image and video codecs integrate zero-run encoding as a standard component.

Important

Zero-Run Encoding Pipeline:

```
Raw Image → Transform → Quantization → Zigzag Scan →
Zero-Run Encoding → Entropy Coding → Compressed Bitstream
```

Key components:

- **Transform:** DCT (JPEG, MPEG), DWT (JPEG 2000), or other
- **Quantization:** Reduces precision, creates zeros
- **Zigzag scan:** Reorders to maximize zero runs
- **Zero-run encoding:** (run, level) pairs
- **Entropy coding:** Huffman (JPEG) or arithmetic (JPEG 2000)

Example

JPEG's Zero-Run Encoding:
JPEG encodes AC coefficients as pairs (RUNLENGTH, LEVEL):

- **RUNLENGTH:** Number of consecutive zeros preceding this coefficient (0-15)
- **LEVEL:** Non-zero coefficient value

- Special case: (0,0) = End-of-Block (EOB)
- Special case: (15,0) = ZRL (16 zeros, handled specially)

Example coefficient sequence: 0, 0, 0, 5, 0, 0, -3, 0, 0, 0, 0, 2, [rest zeros]
Encoded as:

- (3,5) → 3 zeros then 5
- (2,-3) → 2 zeros then -3
- (4,2) → 4 zeros then 2
- (0,0) → EOB (rest all zeros)

Huffman coding: Each (RUNLENGTH, LEVEL) pair is Huffman coded using tables optimized for typical distributions.

7.6.3 Interaction with Entropy Coding

Zero-run encoding and entropy coding work together synergistically.

Important

Why Combine Zero-Run and Entropy Coding:

1. **Dimensionality reduction:** Zero-run encoding converts 2D position information into 1D run lengths
2. **Probability modeling:** Run lengths have predictable distributions (short runs common, long runs rare)
3. **Context modeling:** Can use different entropy codes for different coefficient types
4. **Efficiency:** Run-length encoding handles the zeros; entropy coding handles the values

Example

JPEG vs. JPEG 2000 Approach:
JPEG (DCT-based):

- Fixed 8×8 blocks
- Zigzag scan per block
- (run, level) pairs

- Huffman coding per pair
- Separate DC prediction

JPEG 2000 (Wavelet-based):

- Multi-resolution decomposition
- Embedded block coding with optimized truncation (EBCOT)
- Fractional bit-plane coding
- Arithmetic coding with context modeling
- Zero-run encoding implicit in bit-plane representation

Compression comparison:

- JPEG: Good at medium-high bitrates
- JPEG 2000: Better at low bitrates, no blocking artifacts
- Both rely heavily on zero-run redundancy

7.7 Limitations of Run-Length Encoding

Understanding RLE's limitations is as important as understanding its strengths.

7.7.1 Why RLE Fails on Natural Language Text

Natural language text represents RLE's worst nightmare.

Important

Characteristics of Natural Language Text:

1. **Large alphabet:** 26 letters + punctuation + case + spaces
2. **Rare repeats:** In English, repeated letters are uncommon
   - Double letters: "book", "little", "success" — relatively rare
   - Triple letters: Almost nonexistent (except "beeeeep" in comics)
3. **Statistical distribution:** Letter frequencies follow Zipf's law
4. **Context structure:** Redundancy is in patterns, not runs

Example

RLE on Shakespeare:
Consider the opening of Hamlet's soliloquy: `To be, or not to be, that is the question:`
Run analysis:

- Total characters (including spaces): 41
- Runs of length 1: 39 runs
- Runs of length 2: 1 run (double space after colon? No, single space)
- Actually let's count properly:
  - "T" (1), "o" (1), " " (1), "b" (1), "e" (1), "," (1), " " (1), "o" (1), "r" (1), " " (1), "n" (1), "o" (1), "t" (1), " " (1), "b" (1), "e" (1), "," (1), " " (1), "t" (1), "h" (1), "a" (1), "t" (1), " " (1), "i" (1), "s" (1), " " (1), "t" (1), "h" (1), "e" (1), " " (1), "q" (1), "u" (1), "e" (1), "s" (1), "t" (1), "i" (1), "o" (1), "n" (1), ":" (1)
- Every run is length 1!

RLE encoding: 41 runs $\times$ 2 values each = 82 values **At 1 byte per value:** 82 bytes vs 41 original $\rightarrow$ 100% expansion

Important

Why Text Has No Runs:
From information theory perspective:

- English entropy: 1-1.5 bits per character (for ASCII text)
- But this entropy comes from word structure, grammar, semantics
- Equality redundancy is almost completely absent
- The Markov property (next char depends on previous) rarely produces identical consecutive chars
- In fact, English has constraints against long runs (e.g., "uu" is rare, "aaa" almost never occurs)

7.7.2 Sensitivity to Symbol Ordering

RLE's effectiveness depends critically on how symbols are arranged.

Definition

Order Sensitivity: The property that RLE compression ratio can change dramatically if the same data is reordered, even though the multiset of symbols remains identical.

Example

Dramatic Effect of Ordering:
Consider the multiset: 100 A's, 100 B's, 100 C's
Ordering 1: Interleaved (AB C AB C...) `ABCABCABC...` (300 bytes)

- Each run length = 1
- Number of runs = 300
- Encoded size: 600 values (assuming 1-byte counts)
- Result: 100% expansion!

Ordering 2: Grouped (AAA...BBB...CCC...) `AAA... (100) BBB... (100) CCC... (100)`

- 3 runs, each length 100
- Encoded size: $3 \times 2 = 6$ values
- Result: 98% compression!

Ordering 3: Alternating pairs (AA BB CC AA BB CC...) `AABBCCAABBCC...`

- Each run length = 2
- Number of runs = 150
- Encoded size: 300 values (still expansion)

Important

Implications of Order Sensitivity:

1. **Preprocessing matters:** Sorting or grouping data before RLE can dramatically improve compression
2. **Transform coding:** This is exactly what BWT does — reorders data to create runs
3. **File format design:** Some formats store data in plane-separated order (e.g., all red values, then all green, then all blue) to improve RLE

4. **Database optimization:** Column-oriented databases group similar values, enabling better compression

7.7.3 The Run-Length Distribution Problem

Even when runs exist, their distribution affects compression efficiency.

Definition

Run-Length Distribution Problem: The challenge of efficiently encoding run lengths when they vary widely in magnitude.

Important

The Problem in Detail:
Consider an image with:

- Many short runs (length 1-3) in detailed areas
- Some medium runs (length 10-50) in smooth areas
- Few very long runs (1000+) in background areas

Challenges:

1. **Fixed-width counts:** Waste bits on short runs, limit long runs
2. **Variable-width codes:** Need to design for the distribution
3. **Distribution mismatch:** Different parts of data may have different distributions
4. **Adaptivity:** Should encoding adapt to local statistics?

Example

Run-Length Distribution Examples:
Example 1: Mixed document

```
Background white: runs of 1000-5000 pixels
Text areas: runs of 1-10 pixels (alternating black/white)
Graphics: runs of 20-100 pixels
```

Example 2: Quantized image after DCT

```
DC coefficients: all length 1 (different each block)
AC coefficients: runs of zeros: 0,1,2,3,... (geometric distribution)
End-of-block: always runs of zeros to end
```

Example 3: Binary fax image

```
White runs: can be very long (up to full line width)
Black runs: typically short (text strokes)
Distribution highly skewed
```

Important

Solutions to Distribution Problems:

1. **Separate codes per context:** Fax uses different tables for black and white runs
2. **Hierarchical encoding:** Make-up + terminating codes (fax again)
3. **Adaptive Huffman:** Adjust codes based on recent history
4. **Golomb-Rice codes:** Good for geometric distributions
5. **Spline-based:** Encode runs as parameters to piecewise functions

7.8 RLE as a Preprocessing Technique

RLE's true power emerges when combined with other compression methods.

7.8.1 Combining RLE with Huffman Coding

The RLE+Huffman combination is a classic and effective approach.

Important

Why Combine RLE and Huffman:

- **RLE handles runs:** Converts repeated symbols into (symbol, count) pairs
- **Huffman handles frequencies:** Assigns shorter codes to frequent symbol-/count combinations
- **Synergy:** Each addresses a different type of redundancy

Example

RLE+Huffman Pipeline:

```
Original: AAAAABBBCCCCCDDDDD

Step 1: RLE
Output: (A,5)(B,3)(C,5)(D,5)
```

```
Step 2: Treat each pair as a symbol
Symbol set: A5, B3, C5, D5

Step 3: Build Huffman codes based on frequencies
If each appears once, all get same code length (2 bits each)

Step 4: Encode
A5(00) B3(01) C5(10) D5(11)

Final compressed: 4 pairs × (1 byte symbol + 1 byte count) → 8 bytes
But we encoded to 8 bits = 1 byte total? Wait, careful...

Actually: We need to encode both symbol and count.
Better approach: Have Huffman codes for (symbol, count) combinations.
Or: Separate Huffman for symbols and for counts.
```

Important

CCITT Group 3 Fax: The Classic Example

Group 3 fax does exactly this:

- Step 1: Identify runs of white and black
- Step 2: For each run length, use a Huffman code
- Step 3: Special make-up codes for long runs
- Step 4: Separate Huffman tables for white and black runs

Result: Achieves 10-20:1 compression on typical documents.

7.8.2 RLE After Other Transformations

RLE is often applied after transforms that create runs.

Important

Common Transform+RLE Pipelines:

1. Move-to-Front (MTF) + RLE:

- MTF converts local correlations into runs of small numbers
- RLE then compresses these runs
- Used in bzip2 after BWT

2. **Differential Coding + RLE:**
   - Encode differences between consecutive values
   - Constant regions become runs of zeros
   - RLE compresses the zero runs
   - Used in image compression, audio (DPCM)
3. **Discrete Cosine Transform + RLE:**
   - DCT concentrates energy
   - Quantization creates zeros
   - Zigzag scan creates long zero runs
   - RLE encodes zero runs
   - Used in JPEG, MPEG
4. **Wavelet Transform + RLE:**
   - Similar to DCT but multi-resolution
   - Zero-tree coding exploits zero runs across scales
   - Used in JPEG 2000

Example

MTF+RLE Example:
After BWT (which we'll study later), we might get: `mississippi` → BWT → `ipssm$pissii`
Apply MTF with initial list [a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z,?,!,$,...]
MTF output (simplified): 9,0,0,0,5,0,0,3,0,0,0,...
Now apply RLE to these MTF indices:

- Run of three 0's: (0,3)
- Then 5: literal 5
- Then two 0's: (0,2)
- Then 3: literal 3
- Then three 0's: (0,3)

Result: (0,3),5,(0,2),3,(0,3)
This is much more compressible than the original text!

7.8.3 The Crucial Role of RLE in the Burrows-Wheeler Transform (BWT) Pipeline

This is the key connection that motivates our chapter.

Important

The BWT Pipeline:
`Original Text → BWT → MTF → RLE → Huffman/Arithmetic`

Why RLE is essential after BWT+MTF:

1. **BWT groups similar contexts:** Characters with similar contexts end up near each other
2. **MTF converts to small numbers:** Recently seen characters get small indices (often 0)
3. **Result:** Long runs of zeros and other small numbers appear
4. **RLE compresses these runs:** Exactly what RLE is good at!

Example

BWT+MTF+RLE in Action:
Consider text: `the_cat_sat_on_the_mat`
After BWT (simplified, actual BWT would be different): `thheee___aaatt_mnoosstt??`
After MTF (with reasonable initial order): `2,0,0,1,0,0,0,0,3,0,0,0,0,4,0,0,0,0,0,5,6,0,0,0,`
After RLE on the MTF output:

- Notice the long runs of zeros!
- (0,3), (0,4), (0,4), (0,5), (0,5), ...
- These runs are highly compressible

Why bzip2 is effective: This pipeline turns complex text patterns into simple runs that RLE can handle efficiently.

Important

Key Insight for Students:
The BWT doesn't directly create runs of identical characters. Instead, it creates a sequence where identical characters appear near each other. MTF then converts this "nearness" into actual runs (of zeros). RLE then compresses those runs.
Without RLE, BWT+MTF would be much less effective. Without understanding RLE, you cannot fully understand BWT.
This is why we study RLE in detail before BWT!

7.9 Implementation Considerations

Practical RLE implementation requires attention to engineering details.

7.9.1 Streaming vs. Block-Based Encoding

The choice between streaming and block-based approaches affects memory use and functionality.

Important

Streaming RLE:
Characteristics:

- Process data as it arrives, produce output incrementally
- Minimal memory (O(1) beyond current run)
- Can handle arbitrarily large inputs
- No seeking backward possible
- Ideal for network transmission, real-time systems

Implementation sketch:

```
def encode_streaming(input_stream, output_stream):
    prev = None
    count = 0
    for symbol in input_stream:
        if symbol == prev:
            count += 1
        else:
            if prev is not None:
                output_stream.write((prev, count))
            prev = symbol
            count = 1
```

```
    if prev is not None:
        output_stream.write((prev, count))
```

Important

Block-Based RLE:
Characteristics:

- Process data in fixed-size blocks
- Can random-access within blocks
- Better for indexed file formats
- Block headers add overhead
- Easier to parallelize

Typical block header:

`[Block size][Number of runs][Optional checksum][RLE data...]`

Trade-offs:

Aspect	Streaming	Block-based
Memory	O(1)	O(block size)
Latency	Low (per symbol)	Higher (per block)
Random access	No	Yes
Error resilience	Loses sync	Can resync at blocks
Overhead	None	Block headers

7.9.2 Memory Usage and Buffering

Even simple RLE implementations need to consider memory carefully.

Important

Memory Considerations:
Encoder memory:

- Current symbol (1 value)
- Current run count (1 integer)
- Optional output buffer (tunable size)
- Total: typically < 1 KB

Decoder memory:

- Current (symbol, count) pair
- Current position in run
- Optional input buffer
- Total: similarly tiny

But watch out for:

- Very long runs: count must not overflow
- Output buffering: if decoding to memory, need space for result
- File I/O buffers: typically 4-64 KB for efficiency

Example

Memory-Efficient Decoding:

```
def decode_streaming(encoded_stream, output_callback):
    while True:
        try:
            symbol = next(encoded_stream)
            count = next(encoded_stream)
            for _ in range(count):
                output_callback(symbol)
        except StopIteration:
            break

# Usage: outputs character by character
decode_streaming(encoded_data, lambda c: print(c, end=''))
```

Listing 5: Streaming decoder with minimal memory

This uses O(1) memory regardless of file size!

7.9.3 Handling Large Run Counts Safely

When runs exceed the maximum representable count, we need strategies.

Important

Strategies for Large Runs:

1. Split runs:

- When count exceeds maximum, output multiple run chunks
- Example: max count = 255, actual run = 1000

- Output: (sym,255), (sym,255), (sym,255), (sym,235)

2. **Use larger count fields:**
   - Switch to 16-bit or 32-bit counts
   - But overhead increases for all runs
   - May still need splitting for very large runs
3. **Escape sequence for overflow:**
   - Special marker means "count continues in next field"
   - Like UTF-8 for integers
   - More complex but flexible
4. **Variable-length encoding:**
   - Use Elias gamma, Fibonacci, etc.
   - No maximum, efficient for all sizes
   - But slower, bit-level I/O needed

Example

Splitting Runs Example:
Assume 8-bit counts (max = 255)
Input run: 1000 'A's
Approach 1: Split into max-sized chunks

- (A,255), (A,255), (A,255), (A,235)
- Overhead: 4 pairs instead of 1
- Extra overhead: 3 extra count bytes + 3 extra symbol bytes

Approach 2: Use escape for overflow

- Define 255 as "overflow, next byte is additional count"
- (A,255,235) where 255 means "add next byte"
- Total count = 255 + 235 = 490 — still not enough!
- Need cascading: (A,255,255,235) for 255+255+235 = 745

- (A,255,255,255,235) for 1000
- This is essentially variable-length encoding

7.10 Worked Case Studies

Detailed examples help solidify understanding.

7.10.1 Blue Sky Example (Grayscale Image)

Consider a simple 10×10 grayscale image (8-bit pixels) with a blue sky gradient.

Example

Blue Sky Image Data:
Pixel values (simplified):

```
Row 1: 200,200,200,200,200,200,200,200,200,200
Row 2: 200,200,200,200,200,200,200,200,200,200
Row 3: 200,200,200,200,200,200,200,200,200,200
Row 4: 180,180,180,180,180,180,180,180,180,180
Row 5: 180,180,180,180,180,180,180,180,180,180
Row 6: 160,160,160,160,160,160,160,160,160,160
Row 7: 160,160,160,160,160,160,160,160,160,160
Row 8: 140,140,140,140,140,140,140,140,140,140
Row 9: 140,140,140,140,140,140,140,140,140,140
Row10: 120,120,120,120,120,120,120,120,120,120
```

RLE Encoding (row-major order):

- Run 1: 30 pixels of 200 (rows 1-3) → (200,30)
- Run 2: 20 pixels of 180 (rows 4-5) → (180,20)
- Run 3: 20 pixels of 160 (rows 6-7) → (160,20)
- Run 4: 20 pixels of 140 (rows 8-9) → (140,20)
- Run 5: 10 pixels of 120 (row 10) → (120,10)

Compression analysis:

- Original: 100 bytes
- Encoded: 5 pairs = 10 bytes
- Ratio: 10:1 (90% compression!)

With 2D structure: If we encode each row separately:

- 10 rows $\times$ 1 run each = 10 runs = 20 bytes
- Worse than treating whole image as 1D stream
- Shows importance of scan order

7.10.2 Binary Document Scan

A typical business letter scanned at 300 DPI.

Example

Document Scan Analysis:
Consider a letter with:

- 1-inch margins all around
- 10-point text (about 1/8 inch tall)
- Single-spaced text
- Page size: 8.5" $\times$ 11"

Dimensions at 300 DPI:

- Width: $8.5 \times 300 = 2550$ pixels
- Height: $11 \times 300 = 3300$ pixels
- Total pixels: 8.4 million
- Raw size (bi-level): 1 MB

Run characteristics:

- White space (margins): runs of 2550 pixels per line
- Between lines: 1/8 inch = 37.5 pixels of white
- Text lines: alternating black/white runs of 1-5 pixels
- Paragraph breaks: larger white runs

RLE encoding (per line): For a typical text line:

```
[White margin: 200][Text pattern...][White margin: 200]
```

White margin: 200 pixels (approx 0.67 inches at 300 DPI) Text pattern: alternating black/white runs: 2,1,2,1,3,1,2,1,... (character strokes)

Compression estimate:

- Average run length in text area: 2-3 pixels
- But long white runs dominate
- Overall average run length: 50-100 pixels
- Runs per line: 20-40
- Total runs per page: 3300 lines $\times$ 30 runs $\approx$ 100,000 runs
- With 2 bytes per run (1 symbol + 1 count): 200 KB
- Ratio: 5:1 (80% compression)

With Group 4 fax: Typically 20-50 KB per page (20-50:1 compression)

7.10.3 Why RLE Expands Shakespeare

A quantitative look at why RLE fails on literary text.

Example

Shakespeare's Sonnet 18:

> Shall I compare thee to a summer's day? Thou art more lovely and more temperate: Rough winds do shake the darling buds of May, And summer's lease hath all too short a date:

Statistical analysis:

- Total characters (including spaces, punctuation): 500
- Count runs of length 1: 490
- Runs of length 2: 5 (double letters: "ll" in "shall", "mm" in "summer", etc.)
- Runs of length 3: 0
- Average run length: 1.01

RLE encoding:

- Original: 500 bytes
- Encoded: 500 runs $\times$ 2 = 1000 values

- At 1 byte per value: 1000 bytes (100% expansion)

Why the expansion?

- Each run of length 1 requires a count of 1
- The count field adds 100% overhead to each character
- The few runs of length 2 save only 1 byte each (2 chars $\rightarrow$ 2 bytes vs 4 bytes for two singles)
- Savings from runs: 5 runs $\times$ (2 bytes saved per run) = 10 bytes saved
- But overhead from counts: 500 extra bytes
- Net loss: 490 bytes

Mathematically: Let n = original length, r = number of runs Basic RLE size $= 2r$ (if counts and symbols same size) Expansion when $2r > n$, i.e., $r > n/2$ For random text, $r \approx n$ (since avg run ≈ 1) Thus expansion $\approx 2n - n = n$ (100%)

7.11 Summary and Conceptual Bridge to Transform Coding

7.11.1 Equality Redundancy: A Foundational Compression Primitive

Important

Key Takeaways:

1. **Definition:** Equality redundancy is the simplest form of data redundancy—consecutive identical symbols.
2. **RLE basics:** Replace runs with (symbol, count) pairs. Simple, fast, optimal for certain data.
3. **Performance:**
   - Best case: near-infinite compression (single run)
   - Worst case: 100% expansion (no runs)
   - Break-even: depends on count field size and run length distribution
4. **Variants:** Escape symbols, thresholds, PackBits, binary RLE—each addresses specific limitations.
5. **Applications:** Fax, image formats, transform coding pipelines—RLE is everywhere.

6. **Limitations:** Fails on natural language, sensitive to ordering, distribution-dependent.

7.11.2 The Fundamental Limitation of RLE: Passivity

Important

RLE is Passive:
RLE can only exploit runs that already exist in the data. It cannot:

- Create runs where none exist
- Reorder data to make it more compressible
- Find higher-order patterns
- Adapt to the structure of the data

This is the key insight: RLE is a *reactive* compressor, not a *proactive* one.

7.11.3 From Passive Exploitation to Active Transformation

Important

The Central Question:
What if we could *transform* the data to make it more run-prone?

- Can we reorder data so that identical symbols end up together?
- Can we convert other types of redundancy into equality redundancy?
- Can we design algorithms that *create* the very patterns that RLE excels at compressing?

This is exactly what transform-based compression does!

7.11.4 Roadmap to the Burrows-Wheeler Transform

Important

Preview of Chapter 8: The Burrows-Wheeler Transform
The BWT answers our central question by:

1. **Reordering data:** It sorts all rotations of the input, bringing characters

with similar contexts together.

2. **Creating local similarity:** After BWT, identical characters appear near each other (though not necessarily in long runs yet).

3. **MTF conversion:** Move-to-Front coding converts this "nearness" into actual runs of zeros and small numbers.

4. **RLE application:** Now RLE can finally work its magic on the MTF output.

The pipeline:

```
Original → BWT (reorder) → MTF (create runs) → RLE (compress runs) → Entropy coding
```

Why this works: BWT doesn't compress directly. Instead, it *transforms* the data into a form that plays to RLE's strengths. RLE then does what it does best—compress runs.

Conceptual Bridge Summary

From RLE to BWT: The Evolution of Compression Thinking

Level 1: "I see runs, I compress runs." (Basic RLE)

Level 2: "I adapt RLE to handle different situations." (RLE variants)

Level 3: "I transform data to create the runs RLE needs." (BWT)

The BWT chapter will show you how to achieve Level 3 thinking.

7.12 Exercises and Projects

Exercise 7.0

Exercise 7.1: Basic RLE Implementation
Implement basic RLE encoding and decoding in your favorite programming language. Test on:

1. A file with long runs (e.g., 1000 'A's)

2. A file with no runs (e.g., "abcdefghijklmnopqrstuvwxyz")

3. A binary image file (you can create a simple PBM file)

Measure compression ratio for each case.

Exercise 7.1

Exercise 7.2: Run Length Analysis

Write a program that analyzes a file and reports:

1. Number of runs
2. Average run length
3. Distribution of run lengths (histogram)
4. Best and worst case compression ratios for this file

Test on different file types: text, image, executable, random data.

Exercise 7.2

Exercise 7.3: PackBits Implementation

Implement PackBits encoding and decoding. Compare its performance with basic RLE on:

1. Files with mixed long runs and literals
2. Files with no runs
3. Files with very long runs (¿128)

When does PackBits outperform basic RLE? When does it underperform?

Exercise 7.3

Exercise 7.4: Threshold Optimization

For a given file, find the optimal minimum run length threshold for RLE encoding.

1. Try thresholds from 2 to 10
2. For each threshold, compute compressed size
3. Plot threshold vs. compressed size
4. Explain why the optimal threshold is what it is

Exercise 7.4

Exercise 7.5: Binary RLE for Fax Simulation

Create a simple fax-like document (black text on white background) as a PBM file.

1. Implement binary RLE (store first bit, then run lengths)
2. Try different encodings for run lengths: fixed 8-bit, variable-length (gamma

code)

3. Compare compression ratios
4. Research actual Group 3 fax tables and implement them

Exercise 7.5

Exercise 7.6: RLE + Huffman

Combine your RLE implementation with Huffman coding:

1. After RLE, treat each (symbol,count) pair as a symbol
2. Build Huffman codes for these meta-symbols
3. Compress a file using this two-stage approach
4. Compare with RLE alone and Huffman alone

Exercise 7.6

Exercise 7.7: Preprocessing for RLE

Take a file that compresses poorly with RLE (e.g., English text).

1. Sort the entire file (destroying all order)
2. Apply RLE to the sorted file
3. Compare compression ratios
4. Explain why sorting helps (and why we can't actually do this in practice)

This exercise demonstrates why BWT is clever—it approximates sorting while preserving reversibility.

Exercise 7.7

Exercise 7.8: Project - Simple Image Compressor

Build a complete image compressor for simple graphics:

1. Input: PPM or BMP file
2. Convert to run-length encoded format of your design
3. Include options for:
   - Row-major vs. column-major scanning
   - Plane separation (RRR...GGG...BBB...)
   - Threshold-based RLE

- PackBits-style encoding

4. Compare compression for different image types:
   - Computer-generated graphics
   - Photographs
   - Screenshots
5. Write a report on which techniques work best for which image types

Exercise 7.8

Exercise 7.9: Research and Analysis

Research one of these topics and write a brief report:

1. The history and evolution of fax compression standards
2. How JPEG uses zero run-length encoding for AC coefficients
3. RLE in game console graphics (e.g., NES, Game Boy)
4. Comparison of RLE implementations in different file formats (BMP, PCX, TIFF)

Exercise 7.9

Exercise 7.10: Bridging to BWT

Take a small text file (e.g., "abracadabra").

1. Apply BWT (you can look ahead or use an existing implementation)
2. Apply MTF to the BWT output
3. Apply RLE to the MTF output
4. Compare with RLE applied directly to the original text
5. Explain why the BWT+MTF+RLE pipeline compresses better

This exercise prepares you for the next chapter!

Chapter Challenge Problem

Design a Better RLE:
Your task is to design an RLE variant optimized for a specific application:

- **Scenario A:** Real-time sensor data with occasional long runs of identical readings, but mostly varying data. Must be extremely fast.
- **Scenario B:** Archival storage of scanned documents. Compression ratio is critical; speed secondary.
- **Scenario C:** Embedded system with 2KB RAM, compressing data for radio transmission. Must handle runs up to 65535.

For your chosen scenario:

1. Specify your encoding format in detail
2. Justify design choices (count field size, escape handling, etc.)
3. Implement and test on representative data
4. Compare with at least two other RLE variants
5. Write a technical memo explaining your design

8: Lecture 8: Transform-Based Compression: The Burrows–Wheeler Transform

8.1 The Three Paradigms of Lossless Compression

Before diving into the Burrows-Wheeler Transform (BWT), let us step back and consider the intellectual journey we have taken so far. Lossless compression algorithms generally fall into three broad paradigms:

1. **Statistical/Entropy Coding (Lectures 1-4):** These methods build a probability model of the source and assign shorter codes to more probable symbols. Huffman coding and arithmetic coding are the quintessential examples. The fundamental limit is the source entropy $H(S)$.

2. **Dictionary Methods (Lectures 5-6):** Instead of modeling probabilities, these methods identify repeated phrases and replace subsequent occurrences with references to earlier ones. LZ77, LZ78, and their descendant DEFLATE dominate practical compression due to their speed and simplicity.

3. **Transform-Based Methods (Lecture 7):** These methods do not compress directly. Instead, they *reorganize* the data to make it more amenable to compression by subsequent stages. The Burrows-Wheeler Transform is the most famous example, but this paradigm also includes the discrete cosine transform (DCT) in JPEG and wavelets in JPEG2000 (though those are lossy).

This lecture introduces the third paradigm through the lens of the Burrows-Wheeler Transform, showing how a reversible sorting operation can dramatically improve compressibility.

8.2 Motivation: Beyond Local Matching

8.2.1 The Limits of Sliding Windows

Recall from Lecture 5 that LZ77-based compressors (like DEFLATE) use a sliding window to find repetitions. While effective, this approach has fundamental limitations:

- **Local vs. Global Redundancy:** LZ77 can only detect repetitions that occur within the window (typically 32KB in DEFLATE). Repetitions separated by larger distances are missed entirely.

- **The "Shakespeare Problem":** Consider the complete works of Shakespeare. The phrase "to be or not to be" appears multiple times throughout the text, but these occurrences might be hundreds of kilobytes apart. An LZ77 compressor with a

32KB window will treat each occurrence as novel text, missing this global repetition entirely.

- **Computational Cost of Large Windows:** While we could increase the window size (LZMA uses up to 4GB), the cost of searching grows significantly. More importantly, the *pointer* to reference distant matches requires more bits to encode, eating into compression gains.

Important

The fundamental insight of the Burrows-Wheeler Transform is this: instead of searching for repetitions, why not *rearrange* the data so that similar contexts—and hence similar symbols—are brought together? This rearrangement is reversible, so the original data can be recovered exactly.

8.2.2 The Global Reordering Idea

Imagine you have a long string of text. If you could somehow gather all characters that appear in similar contexts (e.g., all characters that follow "th"), they would cluster together. Within these clusters, the data would exhibit strong local correlations that simple compression methods could exploit.

This is precisely what the Burrows-Wheeler Transform achieves. It does not compress data; it *transforms* data into a form where:

- Identical characters tend to appear in runs
- The local structure is significantly enhanced
- Simple coding methods (move-to-front, run-length encoding, Huffman) become extraordinarily effective

Definition

[The Transform-Then-Compress Principle] Many modern compression algorithms follow a two-stage process:

1. **Transform:** Apply a reversible transformation to the data that reveals its structure (redundancy becomes explicit).
2. **Compress:** Apply statistical or dictionary coding to the transformed data.

The Burrows-Wheeler Transform is the canonical example of this principle in lossless compression.

8.3 The Burrows–Wheeler Transform

8.3.1 Historical Background

The Burrows-Wheeler Transform was developed by Michael Burrows and David Wheeler at DEC Systems Research Center in 1994. Their technical report, "A Block-sorting Lossless Data Compression Algorithm," introduced a completely new approach to compression that combined:

- The global structure exploitation of statistical methods
- The speed and simplicity of dictionary methods
- A clever use of sorting—a fundamental computer science operation

The algorithm was immediately recognized as revolutionary. Within years, it became the basis for `bzip2`, which offered significantly better compression than `gzip` on many file types, particularly text.

8.3.2 Intuition: Grouping Similar Contexts

The core idea of BWT is deceptively simple:

1. Generate all *cyclic rotations* of the input string (with a special end-of-string marker).
2. Sort these rotations lexicographically (dictionary order).
3. Extract the last column of the sorted rotation matrix.

The remarkable property is that the last column tends to contain long runs of identical characters. Why? Because rows that start with similar prefixes (contexts) are adjacent in the sorted order, and their last characters—which are the characters *preceding* those contexts in the original string—tend to be similar.

Example

[Intuition Through a Simple Example] Consider the word "BANANA". In English text, the letter 'N' often follows 'A'. In the BWT output, all the characters that precede 'N' in the original string will be grouped together because all rotations starting with 'N' are adjacent in the sorted list. This clustering effect is what makes the transform powerful.

8.3.3 Step-by-Step Example: `banana$`

Let us work through a complete example to understand the mechanics. We will use the string "banana" with a special end-of-file marker '$' that is lexicographically smaller than any other character.

Example

[Computing the BWT of "banana$"] Step 1: Generate all cyclic rotations
Original string (with marker): `banana$`
All rotations (starting at each character and wrapping around):

1. banana$
2. anana$b
3. nana$ba
4. ana$ban
5. na$bana
6. a$banan
7. $banana

Step 2: Sort rotations lexicographically
We sort these strings as you would in a dictionary:

Index	Sorted Rotation	Original Index
1	$banana	7
2	a$banan	6
3	ana$ban	4
4	anana$b	2
5	banana$	1
6	na$bana	5
7	nana$ba	3

Step 3: Extract the last column
The BWT output is the last character of each sorted rotation:

Sorted Rotation	Last Character
$banana	a
a$banan	n
ana$ban	n
anana$b	b
banana$	$
na$bana	a
nana$ba	a

Thus, **BWT("banana$") = "annb$aa"**
Notice the runs: "nn" appears, and "aa" appears at the end. The original "banana" had no consecutive identical characters!

8.3.4 Visualizing the Effect: Before and After

The transformation becomes clearer when we visualize the effect:

Original: banana$** → **BWT Output: annb$aa

b	a	n	a	n	a	$

a	n	n	b	$	a	a
	run				run	

The BWT output has:

- A run of two 'n's (positions 2-3)
- A run of two 'a's at the end (positions 6-7)
- The original had *no* runs at all!

8.4 Formal Definition of the BWT

8.4.1 Cyclic Rotations and Lexicographic Sorting

Let us formalize what we have done. Given an input string S of length n with a unique end-of-string marker $ that is lexicographically smaller than all other characters:

Definition

[Burrows-Wheeler Transform] The Burrows-Wheeler Transform of S, denoted BWT(S), is computed as:

1. Form a matrix M of size $n \times n$ where row i (0-indexed) is the cyclic rotation of S starting at position i:

$$M[i] = S[i:] + S[:i]$$

2. Sort the rows of M in lexicographic order (dictionary order, with '$' ¡ all other characters).

3. Let I be the index of the row containing the original string S in the sorted matrix.

4. The BWT output L is the last column of the sorted matrix:

$$L[j] = M_{\text{sorted}}[j][n-1]$$

5. The transform also outputs I, the index of the original row, which is needed for inversion.

Thus, $\text{BWT}(S) = (L, I)$.

In our "banana$" example:

- $L =$ "annb$aa"
- $I = 4$ (the original string "banana$" appears at row 5, which is index 4 in 0-based indexing)

8.4.2 The Last Column Construction

Why do we take the last column? Consider any two rows that are adjacent in the sorted order. They share a common prefix (the longer the prefix, the more similar their contexts). The last characters of these rows are the characters that *precede* these contexts in the original string. When contexts are similar, the preceding characters often come from a limited set, creating runs.

Important

The BWT is a *reversible* transformation. Given only L (the last column) and I (the index of the original row), we can reconstruct the original string S exactly. This reversibility is what makes the transform useful for compression—we can recover the original data after decompression.

8.4.3 Properties of the Transform

The BWT has several remarkable properties:

1. **Reversibility:** The transform is bijective—every input string maps to a unique (L, I) pair, and the mapping is invertible.
2. **Run Formation:** The output L tends to contain long runs of identical characters, especially in structured data like text.
3. **Context Grouping:** Characters that appear in similar contexts in the original string are brought together in the output.
4. **Structure Revealing:** Higher-order dependencies in the original string become lower-order dependencies in the transformed string.
5. **Block Structure:** The transform operates on fixed-size blocks (typically 100KB–900KB), allowing a trade-off between compression ratio and memory usage.

8.5 Inverting the BWT

The inversion algorithm is where the true cleverness of the BWT reveals itself. Given only L (the last column) and I (the index of the original row), we can reconstruct the original string without ever storing the full rotation matrix.

8.5.1 The First–Last (LF) Mapping

The key insight is that the first column F of the sorted rotation matrix can be derived from L alone:

1. F is simply L sorted lexicographically.
2. Why? Because the rows are sorted lexicographically, the first column is just all characters of the original string in sorted order.

In our example:

- $L = [a, n, n, b, \$, a, a]$
- Sorting L gives: $[\$, a, a, a, b, n, n] = F$

Now we have a crucial observation: For any character in the last column, we can find its corresponding character in the first column through a mapping that preserves the relative order of identical characters.

Definition

[LF-Mapping] The LF-mapping (Last-to-First mapping) is a function LF(i) that returns the row index in the first column that corresponds to the same character occurrence as position i in the last column.
For a character c that appears k times in the string, the k occurrences appear in the same order in both F and L. Therefore, if $L[i]$ is the r-th occurrence of character c in L (counting from 1), then LF(i) is the position of the r-th occurrence of c in F.

8.5.2 Computing the LF-Mapping Correctly

Let us compute the LF-mapping for our example step by step.

First, determine the rank of each character in L (the occurrence number):

Index i	$L[i]$	**Rank in** L	**Explanation**
0	a	1	First 'a' in L
1	n	1	First 'n' in L
2	n	2	Second 'n' in L
3	b	1	First 'b' in L
4	$	1	First '$' in L
5	a	2	Second 'a' in L
6	a	3	Third 'a' in L

Now, determine the rank of each character in F:

Index j	$F[j]$	**Rank in** F	**Explanation**
0	\$	1	First '\$' in F
1	a	1	First 'a' in F
2	a	2	Second 'a' in F
3	a	3	Third 'a' in F
4	b	1	First 'b' in F
5	n	1	First 'n' in F
6	n	2	Second 'n' in F

The LF-mapping matches each position i in L to the position j in F that has the same character and the same rank:

i	$L[i]$	**Rank in** L	**Matches in** F	$LF[i]$
0	a	1	First 'a' in F at index 1	1
1	n	1	First 'n' in F at index 5	5
2	n	2	Second 'n' in F at index 6	6
3	b	1	First 'b' in F at index 4	4
4	\$	1	First '\$' in F at index 0	0
5	a	2	Second 'a' in F at index 2	2
6	a	3	Third 'a' in F at index 3	3

Thus, the correct LF-mapping array is:

$$LF = [1, 5, 6, 4, 0, 2, 3]$$

8.5.3 Reconstruction Algorithm

With the correct LF-mapping, we can now reconstruct the original string by walking backwards from the known original row:

Algorithm 8 BWT Inversion

Require: $L[0..n-1]$ (last column), I (index of original row)
Ensure: Original string S

```
n ← length(L)
Compute F ← sort(L)
Build LF-mapping array LF[0..n − 1] as described above
Initialize S as empty string
row ← I
for k ← 0 to n − 1 do
    Prepend L[row] to S                ▷ Add character to front
    row ← LF[row]                      ▷ Move to previous row
end for
return S
```

Example

[Correct Inversion of "annb$aa" with I=4] Let us trace through the inversion using our correct LF-mapping:
$LF = [1, 5, 6, 4, 0, 2, 3]$, $I = 4$
We will walk backwards, prepending each character we encounter:

Step	**Current Row**	**Character to Prepend**	**New Row** ($LF[row]$)
Start	4		
1	4	$L[4] = \$$	$LF[4] = 0$
2	0	$L[0] = a$	$LF[0] = 1$
3	1	$L[1] = n$	$LF[1] = 5$
4	5	$L[5] = a$	$LF[5] = 2$
5	2	$L[2] = n$	$LF[2] = 6$
6	6	$L[6] = a$	$LF[6] = 3$
7	3	$L[3] = b$	$LF[3] = 4$

Now, let's track the string being built by prepending each character:

Step	**Character Prepended**	**Current String** S
1	$	"$"
2	a	"a$"
3	n	"na$"
4	a	"ana$"
5	n	"nana$"
6	a	"anana$"
7	b	"banana$"

Perfect! We have successfully reconstructed "banana$". The key was having the correct LF-mapping that preserves the order of identical characters.

8.5.4 Proof of Correctness (Sketch)

The inversion algorithm works because of a fundamental property of the sorted rotation matrix:

Theorem 8.1 (LF-Mapping Property)**.** *For any row i in the sorted rotation matrix, the character $L[i]$ (last column) is the character that precedes $F[i]$ (first column) in the original string. Moreover, the row $LF[i]$ in the sorted matrix corresponds to the rotation that starts with $L[i]$.*

This creates a chain that cycles through all rows exactly once. Starting from the row containing the original string (index I) and following the LF-mapping backwards, we

visit each row exactly once in reverse cyclic order, allowing us to reconstruct the original string by collecting the last column characters in reverse order.

8.6 Why BWT Improves Compressibility

8.6.1 Run Formation and Structure Revelation

The BWT's power for compression comes from its tendency to create runs of identical characters. To understand why, consider the following:

1. In the sorted rotation matrix, rows with similar prefixes are adjacent.
2. The last column contains the characters that *precede* these prefixes in the original string.
3. If the original text has statistical regularities (e.g., 'q' is almost always followed by 'u'), then all rows starting with 'u' will be adjacent, and their last characters will all be 'q' (or characters that commonly precede 'u').

Important

[Important Clarification on Entropy] It is crucial to understand that the BWT does **not** reduce the zero-order entropy of the data. In fact, the zero-order entropy of L equals that of S because the transform is a permutation of the original characters. What the BWT achieves is something more subtle: it transforms **higher-order dependencies** in the original string into **lower-order dependencies** in the transformed string. After BWT, patterns that were previously spread across the text become localized, making them exploitable by simple zero-order coding methods like Move-To-Front and Run-Length Encoding.
In other words, the BWT rearranges the data so that the *conditional probabilities* that characterize higher-order structure become visible as *empirical frequencies* that zero-order coders can exploit.

Example

[Visualizing the Effect] Consider English text. The character 'u' frequently follows 'q'. In the original text, these 'u's are scattered. In the BWT output:

- All occurrences of 'u' (as first column entries) will be grouped together because rotations starting with 'u' are adjacent in sorted order.
- Their corresponding last column entries will be the characters that preceded these 'u's—often 'q'.
- Result: A run of 'q's appears in the output, which can be efficiently compressed by RLE.

8.6.2 Connection to Context Modeling

Recall from Lecture 4 that higher-order Markov models capture dependencies by considering the previous k characters. The BWT achieves something analogous without explicit probability modeling:

- In Lecture 4, we built explicit models: $P(\text{next symbol} \mid \text{previous } k \text{ symbols})$
- In BWT, sorting rotations groups symbols that share the same context (the following characters)
- The result is that conditional probabilities become visible as local patterns in the output

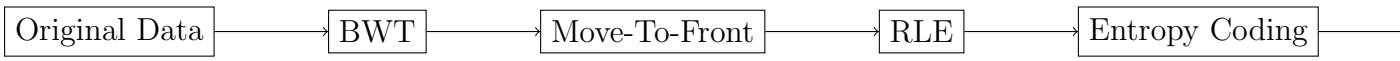
This is why BWT can be seen as an *implicit context modeling* transform—it reorganizes data so that context-dependent structure becomes spatially local.

Important

The BWT doesn't compress; it reveals structure. The actual compression is done by subsequent stages that exploit the local correlations the BWT creates.

8.7 The Full BWT Compression Pipeline

A complete BWT-based compressor typically consists of several stages:



8.7.1 Move-To-Front (MTF) Encoding

After the BWT, we have a string with local correlations but not yet in a form optimal for entropy coding. Move-To-Front encoding transforms this string into a sequence of small integers that are highly compressible.

Definition

[Move-To-Front Encoding] The MTF encoder maintains a list of symbols. For each input symbol:

1. Output its current index in the list (0-based)
2. Move that symbol to the front of the list

Example

[MTF of BWT Output] For pedagogical clarity, we use a simplified alphabet containing only the characters that appear in our example: $[a, b, n, \$]$. In practice, MTF operates on the full byte alphabet (0–255), but the principle remains identical.

Apply MTF to "annb$aa" with initial list [a, b, n, $] (sorted order for predictability):

Input	Current List	Index	Output
a	[a, b, n, $]	0	0
	Move a to front → [a, b, n, $] (already at front)		
n	[a, b, n, $]	2	2
	Move n to front → [n, a, b, $]		
n	[n, a, b, $]	0	0
	Move n to front → [n, a, b, $] (already at front)		
b	[n, a, b, $]	2	2
	Move b to front → [b, n, a, $]		
$	[b, n, a, $]	3	3
	Move $ to front → [$, b, n, a]		
a	[$, b, n, a]	3	3
	Move a to front → [a, $, b, n]		
a	[a, $, b, n]	0	0
	Move a to front → [a, $, b, n]		

MTF output: [0, 2, 0, 2, 3, 3, 0]

Notice how runs of identical symbols in the BWT output become runs of zeros in the MTF output! This is the key to BWT's effectiveness—local structure in the BWT output translates to runs of small numbers after MTF.

8.7.2 Run-Length Encoding (RLE)

After MTF, we have a sequence dominated by small numbers, often with long runs of zeros. Run-length encoding compresses these runs efficiently.

Example

[RLE of MTF Output] MTF output: [0, 2, 0, 2, 3, 3, 0]

After RLE (encoding runs of identical values):

- 0 appears alone (position 1) → encode as (0,1)
- 2 appears alone → (2,1)

- 0 appears alone → (0,1)
- 2 appears alone → (2,1)
- 3 appears as a run of length 2 → (3,2)
- 0 appears alone → (0,1)

RLE output: (0,1), (2,1), (0,1), (2,1), (3,2), (0,1)
This representation has even lower entropy than the MTF output.

8.7.3 Final Entropy Coding (Huffman or Arithmetic)

The final stage applies statistical coding (Huffman or arithmetic coding) to the RLE output. Because the data now consists of small integers with skewed distributions, entropy coding achieves excellent compression ratios.

8.8 Efficient Implementation

8.8.1 Naive Rotation Matrix: Why It Is Impractical

The naive BWT implementation described earlier—constructing an $n \times n$ matrix of rotations—is completely impractical for real data:

- For a 1MB file ($n \approx 10^6$), storing all rotations would require 10^{12} characters (1TB) of memory!
- Even storing indices rather than full strings, we need n rotations of length n, which is $O(n^2)$ space.

Clearly, we need a more efficient approach.

8.8.2 Suffix Arrays for Efficient Construction

The key insight is that we don't need to store all rotations explicitly. For a string with a unique end marker, the cyclic rotations correspond exactly to the suffixes of the string when we consider the end marker as a special character.

Definition

[Suffix Array] A suffix array for a string T of length n is an array of integers $SA[0..n-1]$ that gives the starting positions of the suffixes of T in lexicographic order.

For BWT construction, we only need the suffix array of $S\$$ (the original string with the unique end marker). The relationship is:

- The suffix starting at position i corresponds to the rotation that begins at position i in the original string.
- The lexicographic order of suffixes is identical to the lexicographic order of rotations because the end marker ensures no suffix is a prefix of another.

Algorithm 9 Efficient BWT Construction Using Suffix Arrays

Require: Input string S of length n with unique end marker '\$'
Ensure: BWT output $L[0..n-1]$ and primary index I

```
T ← S                                          ▷ T includes the end marker
Build suffix array SA for T                    ▷ O(n log n) or O(n) time
for i ← 0 to n − 1 do
    pos ← SA[i]
    if pos = 0 then
        L[i] ← $                               ▷ Last character when suffix starts at beginning
        I ← i                                  ▷ Record position of original string
    else
        L[i] ← T[pos − 1]                      ▷ Character before this suffix
    end if
end for
return (L, I)
```

8.8.3 Time and Space Complexity

Modern BWT implementations achieve excellent efficiency:

- **Time Complexity:** $O(n \log n)$ for sorting-based suffix array construction, or $O(n)$ for advanced algorithms (SA-IS, DC3)
- **Space Complexity:** $O(n)$ for the suffix array plus input/output buffers
- **Practical Performance:** `bzip2` processes data at several MB/s, with block sizes typically 100KB–900KB

Important

The development of practical linear-time suffix array construction algorithms and efficient in-place implementations was crucial for making BWT feasible for real-world compression.

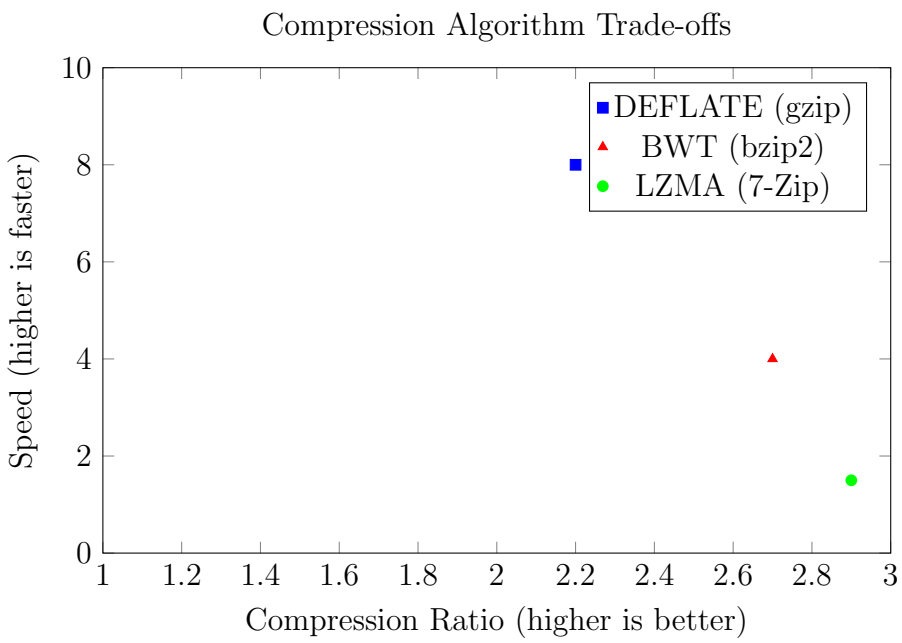
8.9 Comparison with Lempel–Ziv Methods

8.9.1 Local vs. Global Redundancy

The fundamental difference between LZ77 and BWT lies in how they exploit redundancy:

Aspect	LZ77 (DEFLATE)	BWT (bzip2)
Redundancy Type	Local (within sliding window)	Global (across entire block)
Detection Method	String matching in window	Sorting all rotations
Memory Reference	Explicit (distance, length)	Implicit (context grouping)
Adaptivity	Online (streaming)	Block-based

8.9.2 Compression Ratio vs. Speed Trade-offs



8.10 Case Study: `bzip2`

8.10.1 Architecture Overview

`bzip2` is the canonical BWT-based compressor. Its pipeline is:

1. **Run-Length Encoding Prelim:** A preliminary RLE stage to compress repeated characters (simplifies later stages)
2. **Burrows-Wheeler Transform:** Applied to blocks (default 900KB)
3. **Move-To-Front Transform:** Converts local correlations into runs of zeros
4. **Run-Length Encoding:** Compresses runs from MTF
5. **Huffman Coding:** Multiple Huffman tables for different parts of the data

8.10.2 Why BWT Beats DEFLATE on Text

On natural language text, `bzip2` typically achieves 15-30% better compression than `gzip` because:

- Text has global structure (frequent words, common phrases) that spans entire documents
- LZ77's window limits miss these long-range dependencies
- BWT's global reordering captures these dependencies implicitly

Example

[Compression Comparison] On the Canterbury Corpus (standard compression benchmark):

- File: `alice29.txt` (Alice in Wonderland, 152KB)
- `gzip -9`: 56KB (compression ratio 2.71)
- `bzip2 -9`: 44KB (compression ratio 3.45)
- Improvement: 21% better compression

8.11 Limitations and Modern Extensions

8.11.1 Block Size Constraints

BWT operates on fixed-size blocks. The choice of block size involves trade-offs:

- **Larger blocks:** Better compression (more context), but more memory, slower processing
- **Smaller blocks:** Faster, less memory, but poorer compression

`bzip2` allows block sizes from 100KB to 900KB. Beyond 900KB, compression gains diminish while memory usage grows linearly.

8.11.2 Memory Usage Considerations

BWT requires:

- $O(n)$ memory for the suffix array (4-8 bytes per character)
- Additional memory for MTF and RLE stages
- For 900KB blocks, `bzip2` uses about 7MB of RAM—modest by modern standards but significant for embedded systems

8.11.3 When BWT May Not Improve Compression

While BWT is reversible and theoretically cannot reduce compressibility, in practice several factors can lead to worse compression than simpler methods:

- **Block overhead:** The transform operates on fixed-size blocks. Very small blocks (¡ 1KB) may have overhead that outweighs any compression benefit.
- **Primary index storage:** The index I must be stored with each block, adding a small constant overhead.
- **MTF inefficiency on random data:** If the input has no structure, sorting creates no useful patterns, and MTF may actually increase the entropy by spreading out values.
- **Algorithmic overhead:** The multi-stage pipeline (BWT + MTF + RLE + entropy coding) adds computational cost that may not be justified for data that compresses well with simpler methods.

For these reasons, BWT-based compressors like `bzip2` excel on structured data like text but may be outperformed by simpler LZ77 variants on binary data or already-compressed files.

8.11.4 Relation to FM-Index and Text Indexing

The BWT's impact extends far beyond compression. In 2000, Paolo Ferragina and Giovanni Manzini introduced the FM-index (Full-text index in Minute space), which uses the BWT to create a compressed data structure that supports:

- **Counting queries:** Find the number of occurrences of a pattern P in $O(|P|)$ time, regardless of the text size.
- **Locating queries:** Find the positions of matches in $O(|P| + \text{occ} \cdot \log^{\epsilon} n)$ time, where occ is the number of occurrences and the log factor depends on sampling rates.
- **Extract queries:** Retrieve arbitrary substrings of the original text without full decompression.

The FM-index revolutionized bioinformatics:

- **Bowtie and BWA:** DNA sequence aligners used in countless genomics projects, aligning millions of short reads against reference genomes.
- **Pattern matching in compressed data:** Search without full decompression, enabling efficient querying of large genomic databases.

- **Compressed suffix arrays:** Building block for many string algorithms that operate directly on compressed representations.

Important

The BWT demonstrates a profound principle: compression and indexing are two sides of the same coin. A good compressed representation often enables fast search directly on the compressed data, with time complexity that depends only on pattern size, not on text size.

8.12 Summary and Key Insights

8.12.1 The Transform-Then-Compress Principle

The BWT embodies a powerful design philosophy:

1. **Separate concerns:** First transform the data to reveal its structure, then compress the transformed representation
2. **Reversibility is key:** The transform must be exactly invertible
3. **Structure exploitation:** Different transforms reveal different kinds of structure

8.12.2 When to Use BWT-Based Compression

Scenario	Recommendation	Rationale
Text files, source code	BWT (bzip2)	Global repetition matters
Web content (HTTP)	DEFLATE/gzip	Streaming, compatibility
Archival storage	LZMA, Zstandard	Maximum compression
Real-time systems	LZ77 variants	Low latency, streaming
Embedded systems	Custom/simple RLE	Memory constraints
DNA sequences	BWT-based + FM-index	Search+compress combined

8.12.3 Beyond Compression: The FM-Index Revolution

The BWT's legacy extends far beyond file compression:

- **Compressed data structures:** Store data in compressed form while supporting queries
- **Bioinformatics:** Billions of DNA reads aligned daily using BWT-based tools
- **Information retrieval:** Compressed full-text search engines
- **Theory:** Deep connections between sorting, compression, and indexing

8.13 Hands-On Exercises

Exercise 8.0

Apply the Burrows-Wheeler Transform to the string "mississippi$" by hand. Show:

1. All cyclic rotations
2. The sorted rotations
3. The BWT output L and primary index I
4. Identify any runs in the output

Exercise 8.1

Implement a simple BWT encoder in Python:

```python
def bwt_encode(s):
    """Return BWT(s) and primary index."""
    # Add end marker
    s = s + '$'
    # Generate rotations (hint: use list comprehension)
    # Sort rotations
    # Extract last column and find original row
    # Return (L, I)
    pass
```

Test your implementation on "banana" and "mississippi".

Exercise 8.2

Compare compression performance on a text file:

1. Find a text file ¿ 1MB (e.g., a book from Project Gutenberg)
2. Compress with `gzip -9` and `bzip2 -9`
3. Compare sizes and compression times
4. Use `time` command to measure speed
5. Explain the results in terms of the algorithms' characteristics

Exercise 8.3

Challenge: Why does BWT perform poorly on random data? Connect your answer to:

1. The sorting process in BWT
2. The subsequent MTF and RLE stages

3. The concept of structure versus randomness

Exercise 8.4

Research question: The FM-index enables substring search on BWT-compressed data. Explain:

1. How the LF-mapping enables backward search
2. Why counting queries are $O(|P|)$ time
3. Applications in bioinformatics (BWA, Bowtie)

Exercise 8.5

Compare the three compression paradigms we've studied:

Aspect	**Statistical**	**Dictionary**	**Transform**
Core idea	Model probabilities	Find repetitions	Reorganize data
Example	Huffman, Arithmetic	LZ77, LZ78, DEFLATE	BWT, bzip2
Strengths	Optimal for known model	Fast, adaptive	Global structure
Weaknesses	Needs good model	Limited window	Block-based, slower
Best for	Well-characterized sources	General purpose	Text, structured data

Fill in this table based on Lectures 1-7.

8.14 Looking Forward

The Burrows-Wheeler Transform concludes our exploration of lossless compression paradigms. In subsequent courses, you may encounter:

- **Transform coding for lossy compression:** DCT in JPEG, wavelets in JPEG2000
- **Context mixing:** PAQ and its descendants (top-tier compression)
- **Asymmetric numeral systems (ANS):** Modern entropy coding used in Facebook's Zstandard
- **Learned compression:** Using machine learning to predict and compress

The principles you've learned—entropy as a fundamental limit, the value of modeling, the power of transforms, the trade-offs between speed and ratio—will serve you well in all these advanced topics.

Important

[Key Takeaway] The Burrows-Wheeler Transform teaches us that sometimes the best way to solve a problem is to transform it into a different problem. By reordering data to reveal its inherent structure, we make the subsequent compression task dramatically easier—a lesson that applies far beyond data compression.